



BASE24[®]

Release 6.0 Version 3 Implementation Guide

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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one version to another are indicated with the symbol “|” in the outside margin.

Version 3.0 (E. Allen, D. Barton, D. Becker, L. Culp, J. Dargy, G. Gruber, R. Krogstrand, M. Lovejoy, C. McGeorge, C. Rink, A. Rittenbach, R. Schieuer, P. Spanke, J. Urbanec)-02/2003

Third Version

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Section 1

Introduction

The ***BASE24 Release 6.0 Version 3 Implementation Guide*** is the primary reference document for information needed to upgrade existing BASE24 Release 6.0 Version 2 systems to BASE24 Release 6.0 Version 3.

This section provides an overview of the planning requirements for BASE24 users migrating to release 6.0 version 3. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for release 6.0 version 3, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for migrating their BASE24 systems to release 6.0 version 3. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

ACI has provided BASE24 users with the **BASE24 Release 6.0 Version 3 Implementation Guide** to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to release 6.0 version 3.

BASE24 Release 6.0 products include the following:

- BASE24-atm Release 6.0
- BASE24-pos Release 6.0
- BASE24-teller Release 6.0
- BASE24 Kernel/IEP Release 6.0
- BASE24-telebanking Release 6.0
- BASE24-billpay Release 6.0
- BASE24-customer service Release 6.0
- BASE24 Interchange Interfaces Release 6.0
- BASE24 Value-Added Products:
 - BASE24-Infobase
 - BASE24 Retail Products
 - BASE24-card
 - ACI Host Link
 - ACI Plastic Card Control System
 - ACI Claims Manager
 - ACI Proactive Risk Manager
 - ACI Settlement Manager
 - ACI Payments Manager

Who should use this guide

There are two audiences for this guide: those who plan and manage the migration process and those who implement the migration to release 6.0 version 3. In either case, the reader should be experienced with BASE24 software, HP NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in Section 3.

To migrate to release 6.0 version 3, users must be on release 6.0 version 2. Refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)*** to assist in planning the tasks necessary to get to release 6.0 version 2.

Using this guide, you can do the following

- Determine what ACI services are available to assist in the migration to release 6.0 version 3.
- Develop a project plan to migrate to release 6.0 version 3.
- Determine staffing and training needs.
- Gain a detailed understanding of the release 6.0 version 3 enhancements.

Software Requirements

HP NSK Software Requirements

ACI recommends that the BASE24 user be on a current HP NSK-supported operating system to obtain maximum performance and support. However, BASE24 release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 release 6.0 requires TMF for several files.

XPNET

ACI requires that the BASE24 user be on the XPNET 3.0 system.

Conversion Planning

BASE24 users must develop their own project plans for upgrading their systems to release 6.0 version 3. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

In general, all changes for release 6.0 version 3 can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the user's release 6.0 version 2 modules.

Conversion Program Guidelines

Users of previous releases of BASE24 and previous versions of BASE24 Release 6.0 must run file conversion programs in order to migrate to the latest version of release 6.0. Information about specific conversion programs can be found in subsequent sections of this document under the applicable enhancements. Following are general conversion program guidelines.

The *UC04 subvolumes (e.g., AT60UC04, BA60UC04, PS60UC04) contain the conversion programs for users migrating from release 5.3 to the latest version of release 6.0. The conversion programs on these subvolumes are kept up-to-date with each new version of release 6.0.

The *UC05 subvolumes (e.g., AT60UC05, BA60UC05, PS60UC05) contain the conversion programs for users migrating from release 6.0 version 1 to release 6.0 version 2. No changes are made to these programs other than necessary fixes. Also residing on these subvolumes are the Transaction Security Services (TSS) conversion programs, which are used to move key data to the new TSS database. The TSS conversion programs are required only if the user is implementing TSS.

The *UC06 subvolumes (e.g., AT60UC06, BA60UC06, PS60UC06) contain the conversion program for users migrating from release 6.0 version 2 to release 6.0 version 3. No changes are made to these programs other than necessary fixes.

If a release 5.3 user wants to convert to the latest version of release 6.0, the conversion programs on *UC04 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 2, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 2 user wants to convert to release 6.0 version 3, the conversion programs on *UC06 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 3, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run. Following this, the user should run the conversion programs on *UC06.

Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the release 6.0 version 3 software. Such manuals include the formal publications mentioned under the Publications heading in Section 3, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 Classic 6.0 Installation Guide (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24 release 6.0 system from scratch. This document is included as a Microsoft Word document and is shipped on the BASE24 6.0 CD-ROM.

The BASE24 Classic 6.0 Installation guide is also provided on the software tape with the code. The document is located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to move the install document to your hard drive and open it with Microsoft Word.

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Switch 6.0 interfaces. It explains how customers can implement Switch 6.0 interfaces in their existing networks as well as in a BASE24 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a 6.0 network. This document is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 release 6.0 network. This document is included as a HP NSK TGAL document and is shipped with the release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

User Bulletin for BASE24 Release 6.0 Version 3 (Informal)

A primary reference for information about all changes made to each BASE24 Release 6.0 Version 3 source module. This documentation is included as a set of HP NSK TGAL documents and is shipped with BASE24 Release 6.0 Version 3 software. This documentation consists of a number of files located on the BA60UB06 subvolume. All files ending in FH contain fix history sections for all changes made. All files ending in FP contain a list of all fixed Procedures for each change made to all modules, except those files that have been resequenced.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade to release 6.0 version 3. This document is included as a HP NSK TGAL document and is shipped with the release 6.0 version 3 software. This document is named AAREAD06 and is located on the BA60UC06 subvolume.

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A utility that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file (also requires an operator to manually rename the file that has been converted), move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects.

This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Dynamic Key Management (DKM)

The automated distribution of encryption keys between BASE24 and an outside entity.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the **BASE24-atm Transaction Processing Manual**, **BASE24-pos Transaction Processing Manual**, or the **BASE24 ITS Kernel Configuration Manual** for additional information about this utility.

Online (Inline) Conversion

A utility that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with event messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

Section 2

Functionality

This section contains an overview of the BASE24 Release 6.0 Version 3 enhancements.

Functionality has been organized by product, and within each product, according to whether the enhancement is a general enhancement or an optional enhancement.

- General enhancements are those enhancements that are inherent in the release 6.0 version 3 offerings. These enhancements affect all BASE24 users. BASE24 users must review the features that are part of these enhancements to ensure an understanding of the new processing and ensure proper configuration within the BASE24 environment.
- Optional enhancements are those enhancements that can be enabled or disabled in release 6.0 version 3. BASE24 users can implement these enhancements based on their own BASE24 processing and marketing requirements.

Enhancements are described in this section by product as follows:

- BASE24 Base (applicable to all products)
- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24 Kernel/IEP (applicable to all ITS based products)
- BASE24-telebanking
- BASE24-billpay
- BASE24 Interchange Interfaces
- BASE24 Value-Added Products
 - BASE24-InfoBase
 - BASE24 Retail Products
 - BASE24-check auth
 - BASE24-frequent shopper
 - BASE24-refunds
 - BASE24-card
 - ACI Host Link
 - ACI Plastic Card Control System

- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Settlement Manager
- ACI Payments Manager

BASE24 users should acquaint themselves with the BASE24 Base section (Section 4) as well as the applicable product-specific sections in order to obtain an understanding of the overall release 6.0 version 3 offering.

BASE24 Base Enhancements

The BASE24 Base product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Base product, refer to Section 4.

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General Enhancements

The following enhancements are common to all products or provide the base for product-specific enhancements. As these enhancements are inherent in release 6.0 version 3, they must be reviewed by all BASE24 users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

1. New data fields were added to the Key File (KEYF) DDL for Canadian-specific processing. User field USER-FLD9A was depleted for the definition of the fields. No source code in BASE24 product references the new fields.
2. The Cardholder Authorization File (CAF) requester and server have been modified with this version to perform all data validation for the screen fields in the server code. This was done because of code space limitations in the requester code. Although functionally the same as the release 6.0 version 2 CAF requester/server (except for any new enhancements added for release 6.0 version 3) the code is significantly different due to the data validation moves.

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Optional Enhancements

The following BASE24 Base Release 6.0 Version 3 enhancements are optional. BASE24 users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Eracom Mark II Security Device Support

The Transaction Security Services (TSS) environment has been enhanced to support the Eracom Mark II security devices. Originally supported only in the Asia/Pacific versions of BASE24, support for these devices has been uplifted into global product and is available worldwide. Also known as the Protecthost White (ESM 2000), these devices provide high-speed cryptographic processing and secure key management. PIN translation, PIN verification, card verification, key generation, MACing and triple DES are supported, as well as IBM 3624 DES and Visa PVV PIN verification methods. Currently supported protocols include Async and Ethernet.

MFK/LMK/AKB Key Translate Utilities

A new utility has been developed for the Transaction Security Services (TSS) environment to allow users to change their security device's master key (Atalla MFK, or Thales/Racal LMKs). As a replacement to the RSMCOM and ASMCOM tool, this utility translates all keys stored on the TSS database from encryption under the 'old' MFK/LMK to encryption under a 'new' MFK/LMK. This feature is important to customers that need to change their security device's master key in the event of a key compromise, or if the master key must be lengthened to support triple DES. The utility also facilitates a conversion of Atalla keys to the new Atalla AKB keyblock format.

Persistent UAF Support

The Usage Accumulation File (UAF) is normally purged daily at logical network cutover or at midnight cutover depending on the field cutover settings for each institution. Functionality has been added with this enhancement to allow the UAF to remain on the system and be cleaned up by a new UAF Cleanup process. This process is bound into the Super Extract process and can be initiated via Settlement at logical network cutover or at midnight cutover, automatically based on a timer set in the Extract Configuration File (ECF), or manually by an operator.

Racal Access Module (RAM)

An enhanced version of RAM, called 'URAM', is available to support IP connections to Thales (Racal) security devices. URAM includes two performance enhancements

over RAM: the implementation of IP support directly from the URAM process, and the use of UDP/IP, a connectionless protocol, rather than TCP/IP. By placing UDP/IP programming directly in the URAM process, I/O to the XPNET process is eliminated, along with the additional check-pointing overhead associated with XPNET. By using UDP/IP, the overhead from establishing and breaking down sessions is eliminated. Because of these enhancements, the implementation of URAM with Transaction Security Services can result in a net performance improvement. URAM can be implemented with or without Transaction Security Services.

Transaction Security Services Diebold and Identikey PIN Verification Support

Diebold PIN Verification and Identikey PIN Verification is supported in Transaction Security Services (TSS). TSS stores the data required for Diebold PIN Verification and Identikey PIN Verification in new files, the DNTD and IDNTD. These files are maintained using the ACI Desktop Graphical User Interface (GUI). When using the Transaction Security Services process, the KEYA is replaced by Transaction Security Services files and is no longer accessed when performing PIN verification, card verification or chip authentication in hardware security modules.

Transaction Security Services Multiple Logical Network Support

This enhancement allows customers who run multiple BASE24 logical networks (LNs) to use a separate Transaction Security Services (TSS) database for each LN. This functionality is required by customers who run their LNs at separate physical sites but do not want to partition a single set of TSS data sources across the physical sites. The enhancement also provides ease of migration to customers who want to convert multiple LNs to TSS by allowing them to convert one at a time.

The enhancement performs an extra PIN translation for transactions that travel from one LN to another. This operation translates the PIN from encryption under the acquiring endpoint's PIN key to encryption under a common key known to all LNs in the BASE24 system.

This common key approach requires only one extra key record to be added to each TSS database. Were this approach not used with separate TSS databases, the acquiring endpoint key records in each LN would have to be replicated in the other LNs in order for the TSS processes in each to have access to the required PIN keys. Replicating all these key records would not work. For one, it would be wasteful, since there may be hundreds of records to replicate. More importantly, the replicated records cannot be kept current as devices and interchanges may dynamically update the original records with new keys.

Although the enhancement was done to support transactions passed between LNs using TSS, it can also be used by LNs that have not migrated to TSS. Thus, in

addition to supporting TSS-to-TSS transactions, it can also support TSS-to-non-TSS, non-TSS-to-TSS, and non-TSS-to-non-TSS transactions. It can also support traffic between Atalla AKB and Atalla non-AKB LNs. This permits customers to perform a systematic migration of their LNs to TSS.

BASE24-atm Authorization and BASE24-pos Router/Authorization perform the special PIN translation on transactions that are SPROUTE LGNT-routed from one LN to another because of a match on the card prefix in the second LN's Card Prefix File (CPF). The authorization processes use logic introduced by the Multiple Logical Network Routing Indicator enhancement (release 6.0 version 2) to detect this routing condition. Transactions that are routed using any other SPROUTE method (LGNT with file type IP, AMEX, BASE or VISA, DSCV, INAS or BNET, DFLT, etc.) are not supported at this time.

The enhancement is enabled by the presence of a new optional LCONF param. For LNs that use TSS, the param contains a sixteen-digit key locator that points to a TSS Interchange Security Data Source (ISECD) record. The ISECD record contains the encrypted common PIN encryption key. For LNs that do not use TSS, the param contains the *encrypted* single-length common key itself. For BASE24 systems that contain both TSS and non-TSS LNs, the key locator used for TSS LNs should equal the encrypted key used by the non-TSS LNs.

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BASE24-atm Enhancements

The BASE24-atm product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-atm product, refer to Section 5.

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General Enhancements

The following BASE24-atm enhancements are inherent in the release 6.0 version 3 offerings. These enhancements must be reviewed by all BASE24-atm users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

ATD Infrastructure

In order to support future enhancements, the BASE24-atm Terminal Data file (ATD) has been modified to contain offsets to additional data structures in the data area of the record. In addition, some fields for release 6.0 version 3 enhancements have been added outside the ACI User Field. To implement the new data record, logic has been added to the required modules to perform an inline conversion of the ATD records as they are used.

Because this enhancement changes the format of the ATD, it is required that this enhancement be implemented prior to several other enhancements in release 6.0 version 3. These enhancements contain a note that the ATD Infrastructure change is required. In addition, going forward, it will be assumed that this enhancement has been implemented and the additional offsets are available for new data.

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Optional Enhancements

The following BASE24-atm Release 6.0 Version 3 enhancements are optional. BASE24-atm users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Automated Key Distribution System

The NCR 5XXX device handler has been enhanced to support the Automated Key Distribution System (AKDS) enhancement. AKDS provides a means of securely downloading a unique master key (A-key) from BASE24 to an ATM using the Rivest-Shamir-Adleman (RSA) public key method.

RSA is a highly secure cryptography method that utilizes public/private key pairs. The public key is transmitted between sender and recipient. The private key is kept secret by its owner and is never transmitted. There is a mathematical relationship between the public and private key, but it is virtually impossible to derive one from the other. These key pairs are used for data encryption. Data is encrypted under the recipient's public key, which can only be decrypted using the recipient's private key. These key pairs are also used for mutual authentication. A digital signature is created for the encrypted data using the sender's private key. The signature can only be authenticated using the sender's public key. (A signature is an encrypted hash of the data, used by the recipient to authenticate the identity of the sender and to verify that the original content of the data has not been altered.)

A BASE24 system has one public/private key pair, which is generated by an attached Host Security Module (HSM) during the initial setup of AKDS and stored in the Transaction Security Services database. BASE24's public key is sent to the Certificate Authority (CA) to be signed. (NCR acts as its own CA.) The Certificate Authority returns BASE24's public key signature, as well as the CA's own public key. These values are entered into the Transaction Security Services database.

Each ATM has its own public/private key pair, which is generated and stored in the ATM's Encrypting PIN Pad (EPP) at the time of manufacture. Also stored in the EPP is the ATM's public key signature (provided by the Certificate Authority) and the CA's own public key.

Once the initial keys and signatures are established in BASE24 and the ATM, an online dialogue can proceed. An operator at the Device Control Terminal (DCT) executes a LOAD PUBLIC KEY command, initiating an Authentication and Public Key Exchange sequence (alternatively, this command can be executed via NCS/NPCOM). During this load sequence, BASE24 and the specified ATM authenticate each other and exchange public keys. These public keys are used for

encryption and mutual authentication in all future master key downloads. This command is typically performed only once – when a new ATM with an AKDS-enabled EPP is initially deployed, or when an ATM already in the field is fitted with a new AKDS-enabled EPP.

After BASE24 and the ATM have authenticated each other and exchanged their public keys, the master key can be downloaded to the ATM. An operator at the DCT executes a LOAD MASTER KEY command, initiating a Master Key Download sequence (alternatively, this command can be executed via NCS/NCPCOM). During this load sequence, BASE24 generates a new master key and downloads it to the specified ATM. This command can be performed as often as is necessary to preserve the integrity of the ATM's keys. In most cases, this is done relatively infrequently. After this command completes successfully, the DES encryption keys (e.g. PIN Encryption Key, MAC Key) may be generated and downloaded to the ATM, as per normal procedures prior to this enhancement.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Automated Key Distribution System. For more information, refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Diebold Merchant Banking Center

The Diebold Merchant Banking Center (MBC) was initially implemented in BASE24-atm as an Americas Channel CSE. With release 6.0 version 3, this functionality is available in global product. The MBC self-service terminal is one of the latest hardware offerings by Diebold. It allows institutions to provide automated services to their customers such as purchase cash, make change, and perform after-hour bag deposits at locations where bank branch offices are not conveniently located. The MBC eliminates the need for financial institutions to have brick-and-mortar branch locations (and associated overhead expenses) in high priced retail space locations yet allows them to meet their retail customers ongoing night depository and coin/currency needs.

MBC also provides functional, detailed transaction receipts to customers, as well as detailed reports on terminal activities to the supporting financial institution.

The BASE24-atm Diebold MBC application is built as a bindable extension to the standard Diebold 10XX/478X Device Handler. This allows full Diebold functionality to be performed in addition to the MBC-specific transactions with the MBC device handler extension.

There are two new transaction options supported at the Diebold ATM supporting the Merchant Banking Center. These are Money Exchange transactions and Bag Deposit transactions. The Money Exchange transaction allows merchants to obtain change (bills and rolled coins) of various denominations. The consumer may pay for the change either from an online funding account or by inserting cash into the currency

acceptor. The Bag Deposit transaction allows merchants to make bag deposits and obtain multiple copies of receipts. If the consumer entered a deposit bag ID as part of the transaction, the deposit bag ID can be included on the receipt copies. Both of these transactions require the ability to print multiple copies of the receipt.

Refer to 'Diebold Merchant Banking Center' in the BASE24-atm Optional Enhancements in Section 5 for more information about this enhancement.

Dynamic Key Management

Dynamic Key Management was initially implemented in BASE24-atm as an Americas Channel CSE for the Canadian market. With release 6.0 version 3, this functionality is available in global product. In Canada, this enhancement was known as Application Interface Layer (AIL). Generally speaking, this enhancement enables BASE24-atm to dynamically change the Personal Identification Number (PIN) and Message Authentication Code (MAC) keys based on transaction count, security error threshold, or by a key change interval.

In BASE24-atm a PIN or MAC key is changed when:

- The number of PIN verify or MAC verify/generate transactions exceeds a predetermined limit;
- The number of PIN verify or MAC verify/generate failures exceeds a predetermined limit;
- The number of MAC consecutive failures exceeds a predetermined limit.
- Twenty-four hours has passed since the last MAC or PIN key change.

Once any of the limits has been reached, the process to load new keys to the ATM is initiated. To minimize the time an ATM is down, if one key reload is required, both the PIN and the MAC key are reloaded, all of the counters are reset, and the key change timestamp is updated.

In BASE24-atm Release 6.0 Version 3, this enhancement is being added to the NCR 5XXX Device Handler, Diebold 10XX/478X Device Handler, and the Triton Device Handler.

Prerequisite: The ATD Infrastructure enhancement must be installed prior to implementing the Dynamic Key Management enhancement in the Diebold 10XX/478X and NCR 5XXX Device Handlers.

Refer to 'Dynamic Key Management' in the BASE24-atm Optional Enhancements in Section 5 for more information about these enhancements.

Interactive Response Notification Fee Processing and Negative Surcharge Support

BASE24-atm fee processing has been enhanced to support interactive response notification of fees at an ATM. This differs from the original interactive **request** notification surcharge processing, in that the original interactive request notification surcharge processing performed an interactive request prior to the transaction going to the issuer. The cardholder account was impacted after the cardholder was notified of, and accepted, the fee.

The interactive **response** notification of fees, however, routes the transactions for authorization without notifying the cardholder first. The cardholder account is impacted before the cardholder accepts the fee, and the transaction may need to be reversed if the cardholder does not accept the fee displayed at the ATM.

The interactive response notification of fees can include both issuer fees and acquirer fees (i.e., surcharges). BASE24 maintains information in the Surcharge File (SURF) for only acquirer fees, and does not maintain information in its database on issuer fees. The setting of issuer fees must be done at a host system or an interchange whose interchange interface on BASE24 has been enhanced to carry the issuer fee information in a new BASE24 Issuer Fee/Rebate token.

Standard product interactive request notification surcharge processing has been enhanced to allow a negative fee amount to provide a cardholder with an incentive to use the ATM device. This type of incentive is also available with the interactive response notification fee processing.

This enhancement is compliant with the UK LINK Switch Interchange Standard (LIS5), Version 5.1, and can be used by the following UK product module:

- UK LINK ISO Interchange (LIS5) Interface

Refer to 'Interactive Response Notification Fee Processing and Negative Surcharge Support' in the BASE24-atm Optional Enhancements in Section 5 for more information about this enhancement.

Multiple Currency Dispense

Multiple Currency Dispense was initially implemented in BASE24-atm as an Americas Channel CSE for the Mexican market. With release 6.0 version 3, this functionality is available in global product. In Mexico, this enhancement was known as Dollar and Peso Dispense. As product, this enhancement enables any combination of currencies to be dispensed at an ATM. It is supported in release 6.0 version 3 on the Diebold 10XX/478X and NCR 5XXX devices. Features include:

- Supports dispensing a single currency or up to four currencies (dependent upon device support) for any currency that is configured in the BASE24 network.

- Provides Multiple Currency Dispense support in the NCR 5XXX and Diebold 10XX/478X (912 emulation mode) device handlers only.
- Ensures compatibility of the Multiple Currency Dispense enhancement if the Multiple Currency Authorization add-on (i.e. the multiple currency account enhancement) is present. However, the multiple currency dispense enhancement does not require that a customer purchase the Multiple Currency Authorization add-on in order to function correctly.
- Provides display screens at the ATM that allow cardholders to select the currency to be dispensed.

Prerequisite: The ATD Infrastructure enhancement must be installed prior to implementing the Multiple Currency Dispense enhancement.

Refer to 'Multiple Currency Dispense' in the BASE24-atm Optional Enhancements in Section 5 for more information about this enhancement.

NCR Bunch Note Acceptor / Check Processing Module Support

NCR has developed two new modules, the Bunch Note Acceptor (BNA) and the Check Processing Module (CPM), for their ATMs to enhance their deposit capabilities. The Check Processing Module is a replacement of NCR's existing Document Processing Module (DPM). The Check Processing Module supports a subset of the functionality provided with the DPM and can exist either as a sidecar with a Cash Dispense ATM or integrated into a full function ATM. The Bunch Note Acceptor allows a bunch of notes (up to 50 bills) to be deposited into a secured part of the ATM. As part of the deposit process, the BNA accepts, verifies, and count the notes, rejecting any that are considered invalid. Once this process is complete, the customer is asked to confirm the amount. If the customer agrees the transaction request is sent to the processing host. If the customer does not confirm the amount, the notes, which were kept in escrow, are returned to the customer. The bills can be of different denominations, as well as different currencies, if supported by the ATM. At this time, a particular transaction must contain only a single currency. If multiple currencies are entered into the ATM, BASE24 rejects the transaction. The BNA can exist either as a p73 Personal Cash Deposit terminal or in a sidecar with or without the CPM.

In order to increase ACI's marketability and keep current with technology, changes were made to BASE24-atm Release 6.0 Version 3 to support these new NCR modules and the associated deposit transaction processing.

Prerequisite: The ATD Infrastructure enhancement must be installed prior to implementing the NCR Bunch Note Acceptor / Check Processing Module Support enhancement.

Refer to 'NCR Bunch Note Acceptor / Check Processing Module Support' in the BASE24-atm Optional Enhancements in Section 5 for more information about these enhancements.

Offline PIN Management Support

The United Kingdom is in the process of implementing offline PIN validation as the primary Cardholder Verification Method (CVM) for electronic transactions performed at the Point-of-Sale (POS). With EMV cards becoming increasingly common, offline PIN (i.e. where the PIN is verified by the EMV card) has been selected as the preferred CVM option for POS transactions in the U.K. Online PIN continues to be used as the CVM for ATM transactions in the U.K.

Currently, the cardholder's signature is used as the primary CVM option for POS transactions in the U.K. However, increasing levels of fraud have shown that signature verification is ineffective in proving the identity of the cardholder. The project to implement 'PIN@POS' in the U.K. has been launched by the Association for Payment Clearing Services (APACS).

In order to ensure that card issuers can successfully manage the offline PIN function, e.g. ensuring that the offline PIN is synchronized with the online PIN known to the issuer, there is a requirement to implement an offline PIN Management Service. The management of the offline PIN function on the card introduces two new administrative functions for card issuers:

- Support for offline PIN change, where the cardholder wants to change their PIN to a new value.
- Support for offline PIN unblocking, where the PIN Try Counter on the card has reached zero and needs to be reset to the normal value (the PIN Try Limit).

Both of these tasks are technically achieved by delivering a script message to the card during an authorization response message, although it is impractical for this to take place during a routine POS transaction. Instead, ATMs are felt to be the ideal secure terminal at which to offer this service.

The basic business requirement is to provide a widely available, reciprocal, PIN Management Service (encompassing both online PINs and offline PINs) for EMV cards only at ATMs. PIN Management comprises both PIN change and PIN unblock (offline PIN try counter reset). BASE24-atm Release 6.0 Version 3 has been enhanced to support this functionality. Specifically, in order to satisfy the U.K. offline PIN CVM requirements, BASE24-atm processing has been enhanced to support the following functions:

- Unblock Offline PIN
- Synchronized Change of Online and Offline PIN
- Script Failure Notification

Transaction Security Services Multiple Logical Network Support

This enhancement allows customers who run multiple BASE24 logical networks (LNs) to use a separate Transaction Security Services (TSS) database for each LN if their environment requires it. It also provides ease of migration to customers who want to move multiple LNs to TSS by allowing them to migrate one LN at a time. BASE24-atm Authorization and BASE24-atm ISO Host Interface were modified for this enhancement. Refer to the BASE24 Base Optional Enhancements for more information.

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BASE24-pos Enhancements

The BASE24-pos product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-pos product, refer to Section 6.

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General Enhancements

The following BASE24-pos enhancements are inherent in the release 6.0 version 3 offerings. These enhancements must be reviewed by all BASE24-pos users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

Balance Inquiry Reversals for SPDH and ISO Host Interface

The BASE24-pos ACI Standard POS Device Handler (SPDH) and the BASE24-pos ISO Host Interface have been enhanced to allow balance inquiry transactions to be reversed if they do not complete as intended. This is controlled via an LCONF param and is documented in the LCONF Param Changes section.

Canadian Processing Options

Fields were added to the BASE24-pos Terminal Data files (PTD) DDL and AFT screens, which are used for Canadian-specific processing. These fields fall under the heading of Pass-Through Parameters. Use of these options requires the use of specific Canadian interface processes.

Enhanced Chargeback Processing

BASE24-pos has been enhanced to support generating chargeback transactions for pre-authorization completion transactions in which a hold was not found, the hold has expired, or the completion amount is greater than the hold amount. This is controlled via a new flag on the BASE24-pos Terminal Data files (PTD) and the existence of a new token.

If the PTD flag is set to generate chargebacks for pre-authorization completion transactions, the SPDH adds the POS Merchant token to the internal request message with a Y in the PRE-AUTH-CHRGBK field. When RTAU processes a pre-authorization completion transaction, if the PRE-AUTH-CHRGBK field in the POS Merchant token is a Y and the hold cannot be found, the hold has expired or the completion amount is greater than the hold amount, a chargeback transaction is generated.

Enhanced ISO Host Interface SAF Processing

The BASE24-pos ISO Host Interface process has been enhanced to place all transactions in the Store-and-Forward File (SAF) prior to sending them to the host. This is controlled via an LCONF param and is documented in the LCONF Param Changes section.

ISO Response Code Usage

The BASE24-pos ACI Standard POS Device Handler (SPDH) has been enhanced to map BASE24 response codes to ISO response codes when BASE24 is the issuer. A new FID has been added to send the ISO response code to the terminal. This is controlled via a table in a new file that is compiled into the executable object.

The ISO response code is placed in the interchange response value field in the PSTM if it is not already populated. Until the table is updated with valid ISO response codes, the PSTM.ICHG-RESP is always set to spaces. The table is called B24-POS-RESP-TBL and in is the file SPDHNAMS.

Preauthorization Purchase Transactions with Cash Back

Visa mandates require support for a cash back amount in preauthorization purchase and preauthorization purchase completion transactions for debit accounts/cards. In addition to modifications previously made in the VisaNet Interface, BASE24-pos HISO and BASE24-pos BIC ISO, modifications to the BASE24-pos Release 6.0 Router/Authorization were required.

The additional processing requires that the entire transaction amount (purchase plus cash back) be carried in the PSTM.TRAN.AMT-1 field. The cash back amount is then carried in the PSTM.TRAN.AMT-2 field.

This preauthorization processing of a purchase request with cash back is only allowed on debit accounts/cards. Preauthorization processing for a lesser amount is not supported with the preauthorization purchase with cash back transaction. Preauthorization purchase with cash back and preauthorization purchase completion with cash back transactions use the same transaction codes as a preauthorization purchase (transaction code '11') and preauthorization purchase completion (transaction code '12'), respectively. No new transaction codes are added with this enhancement.

This enhancement includes the following impacts:

- The cash back amount in field 2 is the same currency as field 1.
- Maximum cash back amount/limit is not included.
- Card use limit checking is not changed since the cash back amount is included with the amount contained in AMT-1 field.
- BASE24-pos CRT Auth was not modified to support the preauthorization with cash back transactions.

Response Code Description File (RCDF)

A new BASE24-pos file has been added called the Response Code Description File (RCDF). RCDF records define the response code description for each terminal group, FIID, language, record type (ISO or BASE24) and response code combination. The BASE24-pos ACI Standard POS Device Handler (SPDH) is the only process that uses the information in the file. The response code description can be configured to be sent to the terminal on a transaction response.

The information in the RCDF is verified against the data in the Account Type Table File (ATT). The ATT contains the account types for the ISO account types displayed in the ACCOUNT TYPE field.

SPDH Return Available Balance to Terminal

The BASE24-pos ACI Standard POS Device Handler (SPDH) can be configured to return the available balance to the terminal for all transactions when it is available. This is controlled via a new flag on the BASE24-pos Terminal Data files (PTD) and by the existence of the POS Balance token. This token was added with the Multiple Currency Support enhancement. Because of the use of the token, the functionality of returning the balance to the terminal cannot be processed with the Multiple Currency add-on product.

If the PTD flag is set to return the available balance to the terminal, the SPDH adds the POS Balance token to the internal request message. The issuer updates the token's TXN-AMT field with the available balance. When formatting the response message to the terminal, the SPDH populates FID J with TXN-AMT-1 from the POS Balance token if the token exists, the FID is configured to be sent on the transaction response, and the PTD Balance Returned flag is set to Y.

SPDH New Key Transaction

A new administrative transaction has been added to the BASE24-pos ACI Standard POS Device Handler (SPDH), so the terminal can request new keys. This transaction is called the New Key transaction.

Suppress Chargeback Response Message

The BASE24-pos Router/Authorization process has been enhanced to not send a response message to the originator when it receives a chargeback request (0402) message. This is controlled via a new LCONF param.

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Optional Enhancements

The following BASE24-pos Release 6.0 Version 3 enhancements are optional. BASE24-pos users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

ACI Standard POS Device Handler Data Encryption

The ACI Standard POS Device Handler Data Encryption module supports field encryption of the account balance field returned to the ACI Standard POS Device Handler (SPDH) terminal. This involves the use of a new encryption key – the field encryption key. This enhancement only works with the SPDH. If the balance to be returned to the terminal is encrypted for any transaction type other than a balance inquiry, additional configuration is required. Refer to the General Changes section for more information on configuring available balances to be returned to the SPDH terminal.

The product requires that the balance always be returned to the terminal. The balance returned to the terminal may be encrypted using either the Electronic Code Block (ECB) or Cipher Block Chaining (CBC) algorithms.

The Transaction Security Services (TSS) enhancement must be installed prior to implementation of the ACI Standard POS Device Handler Data Encryption enhancement. For more information about TSS, refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

ACI Standard POS Device Handler Dynamic Key Management

The ACI Standard POS Device Handler Dynamic Key Management enhancement provides the capability to control the exchanging of the PIN, MAC and Field Encryption keys. The keys are exchanged based on a threshold of number of transactions or number of key errors with a minimum of one key exchange per day per key. This feature is only supported with the ACI Standard POS Device Handler (SPDH).

Use of ACI Standard POS Device Handler Dynamic Key Management does *not* require the Transaction Security Services enhancement; however, it works either with or without TSS present in the BASE24-pos environment.

CRT Authorization TQF Token Buffer Space Expansion

CRT Authorization is a product that allows manual entry of POS transactions by customer service representatives (CSR) for online authorization. The On-Line transaction screen (OLT) in the BASE24 AFT subsystem is used to enter and display

the last seven POS transactions that have been entered by a CSR from a particular terminal. The last seven transactions are stored in a key-sequenced Enscribe file on the HP that is called the Transactions Queue File (TQF). Each of the seven transactions in the TQF has a buffer to hold tokens.

Prior to release 6.0 version 3, the amount of token buffer space was hard-coded at 283 bytes in the TQF. Because customers can configure the specific tokens through the TKN file that they use for processing, each customer can have a different number of tokens. Some customers may have encountered insufficient token buffer space in the TQF.

With release 6.0 version 3, customers have the ability to configure how they want token data handled in CRT Authorization. They have the option of not logging any token information in the TQF, logging token information in the TQF (if under 283 bytes) and an additional file (if over 283 bytes), or retrieving token information from the PTLF if required for a reversal. Customers can use these options to configure the system in a manner most suited for their environment.

Discretionary Data Suppression

BASE24-pos Router/Authorization has been modified to suppress the logging of discretionary data, address verification data, and CVD2 data in PTLF records for transactions routed outbound for authorization.

Discretionary data suppression takes place after transaction data has been moved to the PTLF record structure, and before it is written to disk. Router/Authorization determines if a transaction is eligible for discretionary data suppression using the PSTM.RTE.ISS. Records are sometimes added to the CPF file for not-on-us prefixes, and those prefixes have an issuer code (CPF.POSCPF.ISS) between '30' and '99'. All cards identified by algorithm (Visa, MasterCard, American Express, and Discover) are also assigned an issuer code of '99'. When the PSTM.RTE.ISS is not between '00' and '29', a transaction is considered eligible for discretionary data suppression.

Router/Authorization uses three new LCONF params to control discretionary data suppression.

- POS-ACQ-DISCR-DATA-SUPPRS is used to control suppression of discretionary data (any data past the expiration date) in TRACK1 or TRACK2.
- POS-ACQ-CVD2-DATA-SUPPRS is used to control the suppression of manual card verification data.
- POS-ACQ-AVS-DATA-SUPPRS is used to control the suppression of address verification data.

If the param exists and its value has been set to 'Y', the associated data is suppressed in PTLF records eligible for discretionary data suppression.

Enhanced Pre-Authorization Support

Because of the incorporation of some processing previously supported for Canadian customers only, enhancements to pre-authorization have been added. The pre-authorization segment for the CAF and UAF has been modified to contain the additional fields of completion timestamp, host flag, approval code and terminal ID.

The pre-authorization completion amount contains the amount of the original pre-authorization request transaction. The approval code from the pre-authorization request is used to match the pre-authorization with its corresponding completion.

Pre-authorization completion transactions can be declined if the completion amount is greater than the original request, the pre-authorization hold is not found, or the pre-authorization hold has expired. Pre-authorization holds can be kept over a number of business days (standard BASE24 processing allows holds only for the current business day on the UAF).

With this functionality, the UAF is able to keep pre-authorization holds for several days. A flag in the IDF controls this functionality by indicating that the UAF for this FIID is a persistent UAF. A UAF cleanup program purges the UAF records that do not have outstanding holds.

Functionality also exists to allow holds to be stored on the UAF/CAF for FIIDs that have an Authorization Level of Online only.

MHI Device Handler Compatibility with Release 6.0

The BASE24-pos Merchant Host Interface (MHI) has been made compatible with BASE24-pos Release 6.0. The MHI device handler acquires authorization and draft capture transactions originating from terminals controlled by a host or controller. It accepts and optionally accumulates totals by terminal. Using ACI's Standard POS message formats, both financial and administrative transactions are supported.

The uplift of MHI to release 6.0 compatibility does not include any new features/functions, such as DUKPT, triple DES or EMV. The device handler retains all functionality present in the release 5.3 version.

Secure Internet Validation

The BASE24-pos Secure Internet Validation module provides product support for today's most current e-commerce consumer payment validation schemes. It allows issuers to seamlessly integrate BASE24 with internet authentication product offerings from the ACI Secure Commerce Suite. Alternatively, the Secure Internet Validation module also provides a BASE24 message interface for third party e-commerce authentication vendors. The Secure Internet Validation module supports the following authentication schemes:

- Visa's 3-D Secure CAVV
- MasterCard's SecureCode UCAF
- Virtual Primary Account Numbers (VPANs)

MasterCard and Visa have instituted new consumer authentication programs to secure public network (Internet and mobile) electronic commerce transactions. Visa's 3-D Secure technology, marketed to consumers and merchants as Verified by Visa, and MasterCard's three SecureCode Universal Cardholder Authentication Field (UCAF) payment schemes are approved authentication methods for validating the identity of cardholders during the online purchase process.

The ACI Secure Commerce Suite provides issuers with solutions for consumer enrollment and registration, Verified by Visa and/or SecureCode cardholder authentication, and the generation of Cardholder Authentication Validation Value (CAVV) and UCAF authentication values. The Secure Internet Validation module provides a message interface to obtain CAVV, UCAF, and additional merchant and consumer information from the Secure Commerce Suite. The Secure Internet Validation module also provides the ability to validate this authentication data prior to authorizing the financial transaction. Note that the preauthorization processing for a lesser amount is not supported in the Secure Internet Validation module.

Virtual Primary Account Numbers (VPANs) protect both issuers and their cardholders by using a proxy credit card number that is linked to the real card number. The ACI Secure Commerce Suite contains products, which generate and provide VPANs to authenticated consumers. Using the same message interface and processing logic developed for support of SecureCode and Verified by Visa, the Secure Internet Validation module provides the ability to validate the virtual card number while maintaining a single set of limits for the real PAN.

SecureCode, Verified by Visa, and VPANs have been introduced to reduce risk/fraud, decrease chargebacks and increase consumer confidence with Internet purchases. The following is additional information on Secure Internet Validation support of Visa 3-D Secure, MasterCard SecureCode, and VPANs.

Visa 3-D Secure

BASE24-pos can participate in the initial consumer enrollment/registration authentication via a card verification transaction and in the following financial transactions via a purchase, preauthorization request, or preauthorization completion transaction. The ACI Secure Commerce Suite generates the CAVV and optional TransStain number during the consumer authentication request from the consumer. These new data elements are sent to BASE24-pos immediately via the ISO Host acquiring interface using a card verification message. The necessary card specific and 3-D Secure data will be placed in a new file called the E-Commerce Authentication File (EAF) for later validation during authorization. 3-D Secure's CAVV

can optionally be calculated from the information included in the authorization message and does not have to be provided to BASE24-pos for comparison of the e-commerce data using the EAF.

If the customer is making a 3-D Secure Internet purchase, the merchant will have captured all of the data necessary to complete the transaction during the authentication transaction. The merchant then sends the authorization request with the necessary e-commerce data to the acquirer. The acquirer sends the purchase/pre-authorization purchase transaction through the VisaNet network using the VisaNet ISO or Visa DPS ISO Interface for issuer validation and authorization.

BASE24-pos Router/Authorization interrogates the authorization message to determine if the transaction originated as an e-commerce transaction (PS51TKN.E-COM-FLG), the CAVV is present (TRANS-STAIN-XID-TKN, token ID equals 'C6') and the CAVV should be verified based on the CPF.POS.CAVV-AVV-DISP-FLD and CPF.POS.CAVV-AVV-CHK fields. If the data indicates an e-commerce transaction, Visa has not already verified the CAVV, and the CAVV should be verified by BASE24, Router/Authorization must either verify the CAVV using the CVV keys from the KEYA and data from the message by making a call to the Security Module. If the CAVV is validated, Router/Authorization will set the CAVV status field in the PS51token accordingly. Note that a new Secure Internet Validation (SIV) KEYA group has been added to the CPF for this authentication method. If the CAVV cannot be validated, Router/Authorization determines whether to deny the transaction or set the CAVV status field and continue processing based on a new field set in the CPF.

MasterCard SecureCode

If the customer is making a SecureCode Internet purchase, the ACI Secure Commerce Suite assigns the Accountholder Authentication Value (AAV) during the purchase request from the consumer. This e-commerce data is sent to BASE24-pos immediately via the acquiring ISO Host Interface using a card verification message. The necessary card specific and SecureCode data is placed in the EAF for later validation during authorization. The merchant sends the authorization request with the necessary e-commerce data to their acquirer. The acquirer sends the purchase/pre-authorization purchase transaction through the MasterCard network using the Banknet ISO or MMDS ISO Interface for issuer validation and authorization.

The BASE24-pos Router/Authorization process interrogates the authorization message to determine if the transaction originated as an e-commerce transaction (i.e., PS51TKN.E-COM-FLG), the AAV is present (i.e., AUTHN-DATA-TKN, token ID equals 'CE'), and the AVV should be verified. Validation of the AAV is performed based on the CPF.POS.CAVV-AVV-DISP-FLD and CPF.POS.CAVV-AVV-CHK fields. If the data indicates an e-commerce transaction and the AVV should be verified by BASE24, the Router/Authorization process attempts to locate the transaction in the EAF file. If the AAV is validated (using a compare against the EAF

data), BASE24 sets the CAVV status field in the PS51Token. The AAV result is not passed back to MasterCard, but for trouble shooting the issuer could use it. The transaction is marked as used in the EAF provided it is not a split payment transaction. If the transaction is a split payment, the transaction is marked in the EAF to allow further authorizations to complete with the same UCAF. If the AAV cannot be validated, BASE24 determines whether to deny the transaction or set the CAVV/AAV status field and continue processing based on a new field set in the CPF.

Virtual Primary Account Numbers

If the customer is making a Virtual PAN (VPAN) internet purchase, it functions similar to MasterCard SecureCode. The consumer requests a VPAN from the ACI Secure Commerce Suite, which is then used by the consumer in a subsequent Internet purchase. The ACI Secure Commerce Suite product sends a card verification request containing the normal verification information with the real and VPAN and its expiration date to BASE24-pos for later validation during authorization. The merchant passes the authorization request using the Virtual Card Number with the necessary e-commerce data to their acquirer. The acquirer sends the transaction through the appropriate network for issuer validation and authorization. This data is passed in the purchase/pre-authorization purchase transaction to be verified by the issuer institution.

During normal BASE24-pos Router/Authorization processing, the Router/Authorization process reads the CPF using the VPAN in the PSTM's TRACK2 field to determine the card type. If the card type indicates a VPAN, Router/Authorization then reads the EAF using the VPAN to determine the real PAN. Router/Authorization places the real PAN and its expiration date in the Track2 field and places the Virtual PAN and its expiration date in a new Pseudo Card Number token. It uses the real PAN and real expiration date for authorizing the transaction and logging to the PTLF. However, prior to responding to the acquiring interface, e.g. Host ISO, Banknet ISO, Router/Authorization will replace the Track2 data with the VPAN and its expiration date, so the acquiring interface will be able to find the suspended transaction. The Pseudo Card Number token is logged to the PTLF for subsequent extract. The Pseudo Card Number token data should be saved by the settlement processing system for subsequent settlement and adjustment processing. VPANs can optionally support 3-D Secure or SecureCode. The Router/Authorization process determines whether a CAVV or AVV should be validated based on the presence or absence of the transaction stain XID transaction token (token ID 'C6') and the authentication data token (token ID 'CE').

Transaction Security Services Multiple Logical Network Support

This enhancement allows customers who run multiple BASE24 logical networks (LNs) to use a separate Transaction Security Services (TSS) database for each LN if their environment requires it. It also provides ease of migration to customers who want to move multiple LNs to TSS by allowing them to migrate one LN at a time. BASE24-pos Router/Authorization and BASE24-pos ISO Host Interface were modified for this enhancement. Refer to the BASE24 Base Optional Enhancements for more information.

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BASE24-teller Enhancements

The BASE24-teller product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-teller product, refer to Section 7.

ACI Worldwide Inc.

General Enhancements

The following BASE24-teller enhancements are inherent in the release 6.0 version 3 offerings. These enhancements must be reviewed by all BASE24-teller users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

BASE24-teller Uplift to Release 6.0

BASE24-teller, BASE24-teller ISO Host Interface, and BASE24-teller From Host Maintenance (FHM) have been uplifted to release 6.0.

Extract Configuration File enhancement

The Extract Configuration File (ECF) has been modified to support the configuration of reading past the initially determined end-of-file of the BASE24-teller Transaction Log File (TTLF) or reading up to the initially determined end-of-file of the TTLF. Refer to Section 7 for more information about this enhancement.

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Optional Enhancements

The following BASE24-teller Release 6.0 Version 3 enhancements are optional. BASE24-teller users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Multiple Currency Support

BASE24-teller has not been enhanced to support multiple currency transactions. Changes have been made to BASE24-teller to ensure that BASE24-teller runs in a multiple currency system, including OLTP checks for unsupported currency codes and PBF Refresh file format changes. Refer to the **BASE24 Release 6.0 Implementation Guide (Version 2)** for more information about this enhancement.

Transaction Security Services

The Transaction Security Services (TSS) is available in BASE24 Release 6.0 for managing all cryptographic functions such as PIN verification, message authentication, chip authentication and card verification. Transaction Security Services needs to be implemented to support full end-to-end triple DES and/or to support Atalla's new AKB key formats.

This requires all keys be stored in a new database for all BASE24 modules that call hardware security modules. The benefit is that BASE24-teller can be installed within the same logical network as BASE24-atm and BASE24-pos, whether or not the teller terminals support Triple DES. Refer to the **BASE24 Release 6.0 Implementation Guide (Version 2)** for more information about this enhancement.

Note:

- Not all PIN verification methods are supported in TSS and not all BASE24-teller Device Handlers support triple DES encryption.
- Automated Key Distribution System (AKDS) functionality is not available for teller terminals.

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BASE24 Kernel/IEP Enhancements

The BASE24 ITS Kernel/IEP product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 ITS Kernel/IEP product, refer to Section 8.

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General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24-telebanking Enhancements

The BASE24-telebanking product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-telebanking product, refer to Section 9.

ACI Worldwide Inc.

General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24-billpay Enhancements

The BASE24-billpay product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-billpay product, refer to Section 10.

ACI Worldwide Inc.

General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24-customer service Enhancements

The BASE24-customer service product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-customer service product, refer to Section 11.

ACI Worldwide Inc.

General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24 Interchange Interface Enhancements

All BASE24 Interchange Interface product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Interchange Interfaces, refer to Section 12.

ACI Worldwide Inc.

General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

The following BASE24 Interchange Interface enhancements are optional. BASE24 Interchange Interface users converting to release 6.0 version 3 should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

- Discretionary Data Suppression

BASE24 Interchange Interface users should refer to the documentation provided for each Interchange Interface for implementation instructions. This documentation can be found on the release 6.0 Interchange Interface subvolume and the Interchange Interface add-on product subvolume with a file name of 'XXXXMAN', where 'XXXX' is the Interchange Interface acronym name. All BASE24 Release 6.0 Interchange Interfaces that have been enhanced with the described general or optional enhancements are delivered on 'SW60XXXX' subvolumes. A formal manual is available on the ACI CD-ROM for the BIC ISO Interchange Interface, Europay EPS-Net Interchange Interface, LINK (LIS5) Interchange Interface and Switch Authorization Network Interface (SWANI).

Discretionary Data Suppression Support

Several BASE24 Release 6.0 Interchange Interfaces (Switch 6.0) have been modified to suppress the logging of discretionary data, address verification data, and CVD2 data in ILF records for transactions routed outbound for authorization.

The interchange interfaces enhanced to support discretionary data suppression are:

- Banknet ISO interchange interface
- BASE24 Interface ISO (BIC ISO) interchange interface
- EPS-Net ISO interchange interface
- MDS/Cirrus interchange interface
- VisaNet ISO interchange interface

Discretionary data suppression takes place after transaction data has been moved to the ILF record structure, and before it is written to disk. Interchange Interfaces determine if a transaction is eligible for discretionary data suppression using the ORIG-B24 field in the primary key. This field is set by an interface based on the message type and direction (inbound or outbound). If the ORIG-B24 field indicates that a transaction originated on BASE24, the transaction is considered eligible for discretionary data suppression.

The BIC ISO Interface does not maintain an ORIG-B24 field in the primary key. The terminal logical network from the internal message and the logical network from the PCT (from ICFE screen 1 Interchange Logical Net) are compared to determine if a transaction is eligible for discretionary data suppression. If the terminal logical network does not match the logical network in the PCT (from ICFE screen 1 Interchange Logical Net), the transaction is considered an acquired transaction and is eligible for discretionary data suppression. If the terminal logical network matches the logical network in the PCT, the transaction is considered an issuer transaction and is not eligible for discretionary data suppression.

Interchange Interfaces use three new LCONF params to control discretionary data suppression.

- SWI-ACQ-DISCR-DATA-SUPPRS is used to control suppression of discretionary data (any data past the expiration date) in TRACK1 or TRACK2.
- SWI-ACQ-CVD2-DATA-SUPPRS is used to control the suppression of manual card verification data.
- SWI-ACQ-AVS-DATA-SUPPRS is used to control the suppression of address verification data.

If the param exists and its value has been set to Y, the associated data is suppressed in the ILF records eligible for discretionary data suppression.

For more information about Discretionary Data Suppression, contact your ACI Customer Manager.

BASE24 Value-Added Products Enhancements

There are no enhancements inherent to these products.

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Section 3

Support Services

This section summarizes the various support services available from ACI. These support services are described as follows:

- Product consulting and project management
- Publications
- Education
- HELP24
- ASTech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned; please contact your ACI Customer Manager for more details of the services ACI can offer.

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Product Consulting and Project Management

No formal service programs have been created by ACI for upgrading customers to BASE24 Release 6.0 Version 3. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system.

After reviewing, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading, after reviewing all materials.

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Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system to release 6.0 version 3. Information contained in this guide provides insight into required planning and control of the entire project.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user.

1. IDENTIFY PROJECT TEAM

One of the initial tasks for the BASE24 upgrade is for the user to assign a project team who will be responsible for the success of the project. A project manager and team members will need to be assigned and specific responsibilities established.

2. DETERMINE HARDWARE REQUIREMENTS

An ACI sales representative can assist in determining whether additional hardware will be required for the BASE24 upgrade. It is the BASE24 user's responsibility to order the hardware and establish the delivery date with HP NSK. HP NSK will perform diagnostic testing on the hardware when it is installed at the site. Should hardware encryption be required, it will be necessary to determine the hardware requirements. Additional hardware should be ordered for installation before testing can occur.

If applicable, the BASE24 user should order any new device firmware/software required to implement the enhancements they have chosen to support. The delivery date for the device firmware/software should be agreed upon with the vendor, as this time frame will affect the entire project.

3. IDENTIFY COMMUNICATION REQUIREMENTS

If the BASE24 user has identified additional protocol requirements, the communications lines and modems should be ordered. The customer must coordinate this effort with the local telephone company to ensure that the lines are configured in the least costly manner. The lines and modems must be installed and tested prior to the scheduled software installation date.

4. UPDATE HP NSK SOFTWARE REQUIREMENTS

ACI recommends that the BASE24 user be on a current HP NSK supported operating system to obtain maximum performance and support. BASE24 Release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a

minimum Operating System level of D46. In addition, BASE24 Release 6.0 requires TMF for several files.

5. UPGRADE TO XPNET 3.0

Customers should note that XPNET 3.0 is the minimum requirement for BASE24 Release 6.0.

6. PROJECT SCOPE MEETING

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 Version 3.

7. MIGRATION PLANNING

Users will need to use this document, and any references within it, for detailed analysis when creating their migration plan.

8. DEVELOP PROJECT SCOPE DOCUMENT

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 Version 3.

9. DEFINE REQUIREMENTS DEFINITION

The customer will be required to review existing CSMs to determine which ones will require retrofitting to the new version. With ACI's Fix Release methodology, BASE24 users should evaluate their CSMs to determine if they are handled as standard product in release 6.0 version 3 or if they are specific to the customer's environment and will need to be upgraded to the new version.

10. DEVELOP PROJECT PLAN

The customer's project manager is responsible for developing an internal project plan. A review of the ACI project guidelines will help this person identify job responsibilities and establish time frames. A tracking mechanism should be included in the plan to ensure that the project is proceeding on schedule.

11. IDENTIFY CUSTOMER SPECIFIC MODIFICATION DEVELOPMENT

If the customer elects to have ACI develop modifications identified as required for the new version, a request will be submitted to ACI Software Development for a cost estimate. The functional descriptions and estimates will be sent to the customer for review and approval. Resources will be assigned and the final delivery dates established upon receipt of the signed contract.

Should the customer decide to develop any modifications in-house, resources will need to be assigned and scheduled. Once delivery dates are established, impacts to system testing can be determined.

12. DEFINE CERTIFICATION REQUIREMENTS

Customers currently on Fortify.3 will need to verify with each of the interchanges they connect to, to determine if they need to re-certify with the Interchanges with release 6.0 version 3 code.

If re-certification is required, preliminary test times should be determined and a copy of the interchange test script secured.

The customer needs to set up test time with the interchange well in advance in order to insure time frames are met.

13. ESTABLISH TEST FACILITY

It is recommended that an area be designated specifically for testing the BASE24 Release 6.0 Version 3 software. The following items are recommended for the test area:

- 3 PCs
- Test devices
- Hard copy printer
- Test cards

14. INSTALL PRODUCTION HARDWARE

Installation of any additional computer hardware or peripherals is generally performed by the local hardware service organization or customer staff.

15. ATTEND BASE24 EDUCATION CLASSES

ACI offers a full suite of instructor led courses for BASE24. An enhancement course is also available upon request. The course descriptions can be found in the BASE24 Education Catalog available in hardcopy through your ACI Customer Manager or viewed on the ACI Worldwide web site at www.aciworldwide.com.

16. DEVELOP INTERNAL PROCEDURES

There are no new internal procedures that ACI is aware of for BASE24 Release 6.0 Version 3 enhancements. User/Operational impacts are noted in the detailed sections for each enhancement described within this document.

17. DEVELOP SYSTEM TEST PLAN

The customer is responsible for developing a system test plan. Included in the test plan should be a description of each step specifying the test card to be used, the account to be used, the transaction to be executed, and the expected results. A procedure for retaining the results of the transaction processing should also be specified. Controlled testing and result verification procedures are key elements for successful system testing.

The execution of the system test plan will occur over a period of weeks to fully test all aspects of the system. BASE24 product functionality should be the first application tested. Any system modifications should then be thoroughly tested. The interface between the host and BASE24 should also be tested. This includes testing on-line authorization and the refresh and extract programs.

18. DEVELOP HOST SOFTWARE

During the course of a business day, BASE24 logs all completed and declined transactions to a transaction log file. After network settlement, the transaction log file can be extracted to tape and used as input into the customer's nightly processing cycle on the host. If the customer would like to take advantage of the new features in release 6.0 version 3, a host program will need to be developed to interpret the new token data that can be extracted with release 6.0 version 3.

After the nightly processing cycle on the host or periodically during the business day the host may need to update card and account files used by BASE24 for pre-authorization or stand-in processing. If the customer would like to take advantage of the new features in release 6.0 version 3 in these files, a host program will need to be developed to initialize the new data fields in BASE24 files.

19. INSTALL BASE24

Release 6.0 version 3 software will need to be installed at the customer site. A tape containing the licensed release 6.0 version 3 software modules will be sent to the customer upon request for installation on their system.

20. CONVERT DATA FILES

ACI provides database conversion programs for selected BASE24 files. These files are discussed in the detail sections for each enhancement described within this document.

21. CERTIFY INTERCHANGE

This task involves implementing the interchange test script and receiving notice of certification from the interchange. Current BASE24 users should not have to re-

certify their Interchange Interfaces, unless they are implementing new functionality not previously tested.

22. PERFORM SYSTEM TESTING

Full business day cycle testing of the system as determined by the customer testing plan and procedures.

23. DEVELOP CONVERSION PLAN

The customer should develop an implementation plan for converting existing devices and interchange links to release 6.0 version 3. The installation of new devices and interchanges should also be included in this plan. A method of informing users of any changes to existing procedures should also be developed. The conversion schedule should be routed to all departments.

24. GO-LIVE

At the customer's request, an operational support analyst will visit the site one day prior to the "go-live" date to review the current software status and the BASE24 facilities. This individual will remain onsite to assist the customer with production cutover. The migration plan developed by the customer will be utilized in production cutover. On-site go-live support is a billable service.

25. COMPLETE POST IMPLEMENTATION REVIEW

In general, the purpose of this meeting is to ensure overall satisfaction of the customer's project.

26. TURN OVER TO CUSTOMER MANAGEMENT

If an ACI Project Manager has handled the BASE24 upgrade, a formal turnover meeting is held between the ACI Project Manager and the ACI Customer Manager to ensure a smooth transition in turnover of responsibilities for on-going customer support.

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Publications

The BASE24 formal publications set has been updated to include the release 6.0 version 3 enhancements. The updated BASE24 documentation is available via CD-ROM.

The BASE24 6.0 CD-ROM also contains the previous release (5.3/1.3) documentation. The 5.3/1.3 version of a manual can be considered the most current if the manual is not listed under the 6.0 documentation menus. The XPNET 3.0 documentation is available on a separate CD-ROM titled NET24-XPNET.

BASE24 customers can order a copy of the BASE24 6.0 CD-ROM and/or XPNET 3.0 CD-ROM from their ACI Customer Manager.

ACI has discontinued hard copy distribution of manuals that are available on CD-ROM. Hard copies, however, can be printed at your site from the CD-ROM as necessary and in accordance with your license agreement.

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Education

BASE24 Technical Education offers a complete set of one- to four-day modules covering the components of the BASE24 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Enhancement course is also available upon request for existing customers upgrading to BASE24 6.0 version 3.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

For a current education schedule, see the BASE24 Education Catalog available in hardcopy through your ACI Customer Manager or on the ACI Worldwide web site at [\\www.aciworldwide.com](http://www.aciworldwide.com).

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HELP24

Users migrating to BASE24 Release 6.0 Version 3 should use their existing procedures to report problems or ask questions. HELP24 will continue to use the same response and tracking methodology as they currently do today.

All HELP24 staff have received training in all methods of release upgrades. In addition, there is a complete library of migration documentation available for reference in quickly resolving upgrade issues.

HELP24 has implemented Remote Diagnostics or Access capabilities for assistance in troubleshooting complex problems. The service has grown because of the obvious benefits it offers. Information about Remote Diagnostics or Access is available from HELP24 or your ACI Customer Manager. HELP24 strongly recommends that customers start planning to take advantage of this service now, so it will be available and ready in a crisis situation.

Access to HELP24's bulletin board (in the EMEA this is provided through a web site) is still available as an additional method for communication of important support information. BASE24 users are assigned individual logons to use this facility and are able to peruse the information 24 hours a day. The system can be used to transfer files such as fix files, log files, audit files, and dump files to help expedite problem resolution.

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ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are example descriptions of the various services ASTech could undertake to assist you during the BASE24 upgrade. The number of staff weeks identified to accomplish these tasks are estimates and will vary depending on the amount of customer specific code to be developed and delivered by ACI.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Effort: Two to four staff weeks

ASTech personnel will perform a complete review of the customer's BASE24 software system and determine what current modifications (CSM code) exist at the customer location. ASTech personnel will perform a detailed comparison of existing customer specific software modifications to release 6.0 version 3 functions. Upon completion of the review, the customer will be provided with a document that identifies all modules that contain code unique to their BASE24 system (file names, custom functions, custom function applicability, etc.).

Conversion Planning

Effort: One and one-half to three staff weeks depending on the size and current system configuration.

This service option provides the customer with comprehensive support during the system migration to the new version. ASTech assigns the customer a Technical Project Lead, or Tech Lead from ACI. The Tech Lead will be responsible for ensuring the success of the BASE24 upgrade project.

The Tech Lead will:

- Review the migration options and develop a detailed migration plan for the conversions project
- Review the Project Scope Document on behalf of the customer (if applicable)
- Review the customer's internal procedures to accommodate release 6.0 version 3 functions.

Upgrade Custom Code

Effort: The amount of time necessary for this task varies depending on the number of modules identified as having modifications unique to the customer. Systems with a great number of customizations may take up to sixteen staff weeks. Actual estimates would be provided upon completion of the Customer Software Analysis.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

With this service, ASTech will upgrade all custom code to release 6.0 version 3. This includes unit testing of the code. Customizations that are no longer necessary will be removed. Customizations that must be carried forward can either be upgraded in their current form or be reengineered to take advantage of new functionality in release 6.0 version 3. Reengineering customizations will add time to the estimate.

System Test Planning

Effort: One to four staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products to be tested. Some of this time will be on-site to perform the System Test Plan.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

ASTech personnel will develop and execute a BASE24 System test plan designed for the customer's specific needs. This service also includes the performance and documentation of the system acceptance testing.

Switch and Host Certification Support

Effort: Two to five staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products and interfaces to be certified. All of this time will be on-site time.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

This service option allows ASTech personnel to provide complete support of the various switch and/or host communication certification testing. This includes reviewing and refining the test scripts, performance of certification testing functions, evaluation of the testing results, and documentation and presentation of the results obtained during the certification testing.

Go-Live Support

Effort: One week (on-site).

ASTech personnel will be at the customer site assisting in all conversion tasks and being present for go-live support. This includes remaining on-site for a couple of days to ensure system stability.

Post Go-Live Support

Effort: Varies.

After a successful go-live, a Tech Lead will provide on-going support which may be required by the customer. Tech Lead support can be provided at the customer location and/or remotely through ACI.

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Section 4

BASE24 Base

Section 4 discusses the changes and enhancements made to the BASE24 Base modules in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing BASE24 users must consider when upgrading from release 6.0 version 2.

Changes and enhancements to the BASE24 Base modules impact all users. Consequently, all users should review the material in this section to determine how it applies to them.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the Base modules changes that affect all BASE24 users, along with the associated conversion and implementation requirements that all BASE24 users must address.
- Optional enhancements describe the Base modules changes that affect only those BASE24 users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 4 must be used in conjunction with Sections 5 through 13 to identify the changes that BASE24 users need to consider when upgrading to release 6.0 version 3.

Standard Information Format

A standard format is used in Sections 4 through 13 to highlight and describe the various changes, conversion requirements, and implementation requirements that BASE24 users need to consider when upgrading to release 6.0 version 3. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

CUSTMACS Changes

Describes the CUSTMACS changes required to implement a given change or enhancement. This information includes Make defines that have been added, deleted, or modified.

LCONF Assign Changes

Describes the LCONF assign changes required to implement a given change or enhancement. This information includes assigns that have been added, deleted, or modified, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes the LCONF param changes required to implement a given change or enhancement. This information includes params that have been added, deleted, or modified, along with an explanation of how these params can or must be set.

Database Configuration

Describes the data file changes required to implement a given change or enhancement. This information includes file fields that have been added, deleted, or modified, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given change or enhancement.

Template Changes

Identifies file templates that must be created or modified in order to implement a given change or enhancement. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given change or enhancement.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given change or enhancement.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given change or enhancement.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given change or enhancement.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given change or enhancement.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given change or enhancement.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given change or enhancement. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given change or enhancement. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given change or enhancement. This includes NEF changes.

File Conversions

Identifies the files that need to be converted in order to implement a given change or enhancement. This implies that a given file could be impacted by multiple changes and/or enhancements. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user's host or co-network as a result of a given change or enhancement. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract tape formats.

User/Operational Impacts

Identifies potential changes to a user's internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given change or enhancement. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given change or enhancement.

Reference Documents

Lists any outside sources used in the design of a given change or enhancement. BASE24 users can refer to these documents for additional information.

General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 users must address in order to upgrade to release 6.0 version 3.

- General Changes

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

1. New data fields were added to the Key File (KEYF) DDL for Canadian-specific processing. User field USER-FLD9A was depleted for the definition of the fields. No source code in BASE24 product references the new fields.
2. The Cardholder Authorization File (CAF) requester and server have been modified with this version to perform all data validation for the screen fields in the server code. This was done because of code space limitations in the requester code. Although functionally the same as the release 6.0 version 2 CAF requester/server (except for any new enhancements added for release 6.0 version 3) the code is significantly different due to the data validation moves.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the Base modules enhancements that can be enabled or disabled with release 6.0 version 3, along with the changes, conversion requirements, and implementation requirements. These enhancements are the following:

- Offline PIN Management Support
- Persistent UAF Support
- Transaction Security Services Diebold and Identikey PIN Verification Support
- Transaction Security Services Multiple Logical Network Support

BASE24 users can choose whether or not to implement the above enhancements. Users who plan to implement these particular enhancements must address the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements. Users who do not plan to implement these particular enhancements can review the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements for future reference.

ACI Worldwide Inc.

Offline PIN Management Support

EMV cards have the capability to hold the cardholder's PIN within the ICC. This allows "offline PIN verification" (i.e. where the PIN entered by the cardholder is verified by the ICC) to be offered as a Cardholder Verification Method (CVM) for electronic transactions. Many payment terminals will not be enhanced to offer offline PIN verification, and so "online PIN verification" (i.e., where the PIN entered by the cardholder is verified by the issuer's system) will continue to be widely used.

Where both online and offline PIN verification can be used as the CVM for an EMV card, cardholder confusion would result if the offline PIN value (i.e., the PIN held on the ICC) differed from the online PIN value (i.e. the PIN held on the issuer's system). Also, if a cardholder exceeds the allowable bad PIN tries when entering the offline PIN, a mechanism is required to allow the bad PIN try counter to be reset. Therefore, it is critical that card issuers provide a management function for the offline PIN.

The management of the offline PIN function on the card introduces two new administrative functions for card issuers:

- Support for offline PIN change, where the cardholder wants to change their PIN to a new value;
- Support for offline PIN unblocking, where the PIN Try Counter on the card has reached zero and needs to be reset to the normal value (the PIN Try Limit).

Both of these tasks are technically achieved by delivering a script message to the card during an authorization response message, although it is impractical for this to take place during a routine POS transaction. Instead, an ATM is the ideal secure terminal at which to offer this service.

In order to ensure that card issuers can successfully manage the offline PIN function, e.g. ensuring that the offline PIN is synchronized with the online PIN known to the issuer, the BASE24-atm EMV product provides an offline PIN Management Service.

BASE24-atm support for PIN Management comprises both PIN change and PIN unblock (offline PIN try counter reset). BASE24-atm also ensures that the issuer system is notified if the card cannot be updated with the PIN management information in the script message.

Refer to the **BASE24 Release 6.0 Implementation Guide (Version 2)** for information on how to implement EMV Support.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Offline PIN Management Support enhancement. For more information refer to the **BASE24 Release 6.0 Implementation Guide (Version 2)**.

Unblock Offline PIN

An offline PIN may be unblocked either implicitly or explicitly.

- An EMV transaction request initiated at an ATM contains data retrieved from the EMV chip card. This data includes the Card Verification Results (CVR), which contains an indicator (byte 3, bit 7) of whether the cardholder has exceeded the maximum number of PIN tries allowed for offline transactions. The checking of this indicator on all EMV transactions allows BASE24 to provide an “implicit” mechanism to unblock the offline PIN.
- A PIN Unblock transaction request may be initiated by an EMV card at an EMV-capable ATM, providing an “explicit” mechanism to unblock the offline PIN.

A Host system may reset the bad PIN try counter on an EMV card by including a ‘PIN Unblock’ script command in any EMV transaction response message. In BASE24, issuers are able to specify whether offline PIN unblocking is permitted.

- A new parameter, configurable by card prefix, allows an issuer to specify whether “implicit” offline PIN unblocking is permitted. One setting of this parameter will result in the transaction request being declined, meaning that the cardholder must perform an “explicit” PIN Unblock transaction or PIN Change transaction to reset the bad PIN try counter on the card.
- A new parameter, configurable by card prefix, allows an issuer to specify whether “explicit” offline PIN unblocking is permitted.

BASE24 is capable of generating a ‘PIN Unblock’ script command, either “implicitly” (if byte 3, bit 7 of the CVR is set to ‘1’) or “explicitly” (in response to a ‘PIN Unblock’ transaction request). A new parameter in BASE24, configurable by card, allows an issuer to specify the conditions under which a ‘PIN Unblock’ script should be generated. A ‘PIN Unblock’ script command may be returned to the card only if the cardholder’s online PIN is verified successfully and the transaction’s Authorization Request Cryptogram (ARQC) is verified successfully.

BASE24 is also capable of generating a ‘PIN Change’ script command in response to a ‘PIN Unblock’ transaction request. Having forgotten the PIN, a cardholder may be issued with a new PIN (but not a new card) through the post, for example. This processing will then allow a cardholder to resynchronize the online and offline PINs without actually changing the online PIN. (Note that a ‘PIN Change’ script command will unblock the offline PIN in addition to changing the offline PIN.) A new parameter in BASE24, configurable by card, allows an issuer to specify the conditions under which a ‘PIN Change’ script should be generated. A ‘PIN Change’ script command may be returned to the card only if the cardholder’s online PIN is verified successfully and the transaction’s Authorization Request Cryptogram (ARQC) is verified successfully.

Synchronized Change of Online and Offline PIN

An offline PIN may be changed using an “explicit” mechanism only. A PIN Change transaction request (and optionally a PIN Unblock transaction request), which may be initiated by an EMV-card at an EMV-capable ATM, provides the required functionality.

A Host system may change the PIN on an EMV card by including a ‘PIN Change’ script command in a transaction response message. A new parameter in BASE24, configurable by card prefix, allows an issuer to specify whether an offline PIN change is permitted.

The new parameter also indicates whether the new PIN data should be logged to the BASE24 database, as this information may be required if the card subsequently has to be re-issued with the new PIN. Current PIN Change transaction logic allows the new PIN verification digits to be written to the Transaction Log File (TLF). This information can then be extracted and provided to a Card Management System, which may require the new PIN verification digits if the card subsequently has to be re-issued. For Visa PVV PIN verification, however, the new PIN cannot be derived from the new PVKI/PVV; therefore, additional information (e.g. the new PIN data itself) would be required.

BASE24 is capable of generating a ‘PIN Change’ script command, if the PIN Change request is processed successfully (i.e. the cardholder’s PIN verification digits are updated successfully) in the BASE24 database. A new parameter in BASE24, configurable by card, allows an issuer to specify the conditions under which a ‘PIN Change’ script should be generated. A ‘PIN Change’ script command may be returned to the card only if the cardholder’s online PIN is verified successfully and the transaction’s Authorization Request Cryptogram (ARQC) is verified successfully.

New BASE24 processing supports the HSM functionality required to encrypt the PIN block returned in the script command. For Thales (Racal) HSMs, the existing secure messaging command (‘KU’) has been enhanced to support this function. For Atalla HSMs, a new command (‘351’) has been implemented to support this function (which also provides secure messaging for the script command).

For an EMV card that does not contain an active offline PIN, it would be inappropriate to return a ‘PIN Change’ script command, as the card may not be able to process this type of script command. Conversely, for an EMV card that does contain an active offline PIN, it would be inappropriate to authorize non-EMV PIN Change transactions, as a ‘PIN Change’ script command could not be returned in the transaction response (resulting in non-synchronization of the online PIN and the offline PIN). Another new parameter in BASE24, configurable by card, allows an issuer to specify whether synchronization of the online and offline PIN is required for the card. The parameter specifies the action to take when a PIN Change request (either EMV or non-EMV) is performed on the EMV card.

Script Failure Notification

Once authorized by BASE24 or the issuer Host system, it is possible for script processing to fail at the ATM.

If BASE24 is notified that the card failed to unblock the offline PIN, following “implicit” PIN Unblock processing, then the BASE24 system records this information, so that appropriate personnel can be notified.

If BASE24 is notified that the card failed to change or unblock the offline PIN (following a PIN Change transaction or an “explicit” PIN Unblock transaction), then the PIN Change/Unblock transaction is reversed. This has the effect of either restoring the cardholder’s original PIN verification digits to the BASE24 database, or notifying the issuer that the offline PIN has not been unblocked. Similarly, the reversal is forwarded to an issuer Host system (if external to BASE24), if the BASE24 system did not authorize the transaction.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Note that these modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

EMV-PIN-UNBLOCK-SCRIPT

Specifies the hexadecimal representation of binary data that is used to reset the offline bad PIN tries accumulator on a chip card. It includes both the Script header and the Script command. This param also provides the “template” used by BASE24-atm when generating a script to change the offline PIN on a chip card. This param is required if the EMV extensions are present and initialization fails if the param is missing.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field will depend on the configuration desired by the institution.

Card Prefix File (CPF)

CPF screen 11 has been modified to include the following new fields:

- The BAD OFFLINE PIN ACTION field indicates the EMV processing required when a PIN-based EMV transaction contains data indicating that the offline bad PIN try limit has been exceeded.
- The OFFLINE PIN MANAGEMENT ACTION field indicates the EMV processing required when an EMV PIN Unblock or a PIN Change transaction is received.

Set the new fields on this screen according to the information in the **BASE24-atm EMV Support Manual**.

Cardholder Authorization File (CAF)

CAF screen 13 has been modified to include the following new fields:

- The ONLINE/OFFLINE PIN SYNC field indicates whether synchronization of the online and offline PIN is required for the card.
- The SEND PIN UNBLOCK field indicates the circumstances under which a 'PIN Unblock' script command will be generated and returned by BASE24-atm.
- The SEND PIN CHANGE field indicates the circumstances under which a 'PIN Change' script command will be generated and returned by BASE24-atm.

Set the fields on this screen according to the information in the **BASE24-atm EMV Support Manual**.

Token File (TKN)

The following new BASE24 Base token has been added for this enhancement:

- The EMV Issuer Script Results Token (Token ID "BJ") has been added and is used by the NCR 5XXX ATM Device Handler and the VisaNet and LINK Interchange Interfaces. The EMV Issuer Scripts Results Token contains information about the processing of EMV Script data. This token will be created by the acquirer process (e.g. device handler or interchange). It will contain information about the results of EMV script processing.

If the new token for this enhancement is not to be logged to the TLF or ILF, the appropriate token records must be configured.

If the new token for this enhancement is to be extracted with the TLF or ILF, sent to the host (via HISO), or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

Not applicable

Template Changes

COBNAMES

The following is an excerpt of the ATM-TRAN-DESCRIPT-TABLE from the COBNAMES file on the BA60SRC subvolume that has been modified for the Offline PIN Management enhancement. The table requires updating to include a new entry for the PIN Unblock transactions, together with its corresponding screen text:

```
05      FILLER      PIC X(20)  VALUE "8200000*EMV PIN UBLK".
```

After the modification of this table, the BASE24 Pathway requesters and servers must be recompiled.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

In order to implement this enhancement as an issuer, BASE24 must have access to a host security module. The enhancement will support the use of an Atalla or Thales e-Security (Racal) security module. Refer to the Vendor Impacts subsection of this enhancement for further information.

File Conversions

A file conversion for the CPF is supplied with the Offline PIN Management enhancement.

Refer to the ***BASE24 Classic 6.0 Installation Guide*** and the ***BASE24 Release 6.0 Implementation Guide (Version 2)*** for further information on file conversions.

Card Prefix File (CPF)

The CPF conversion program must be run to convert CPF records from the release 5.3 format to the latest release 6.0 format (or from the release 6.0 version 2 format to the release 6.0 version 3 format).

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF from the release 5.3 format to the latest release 6.0 format. The CPF conversion program on the BA60UC06 subvolume can be used to convert the CPF from the release 6.0 version 2 format to the release 6.0 version 3 format (users running a release 6.0 version 1 system will also have to use the CPF conversion program on the BA60UC05 subvolume to convert from the release 6.0 version 1 format to the release 6.0 version 2 format first). If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPFS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cnvcpf.t,name $cpfm,nowait/  
cnvcpfm -force -list -listreplace
```

Users must update the CNVCPFR obey file on the BA60UC04 or BA60UC06 subvolume with the appropriate information. CNVCPFR must then be obeyed to execute the conversion program.

For more information on this file conversion, refer to the AAREAD04 and CNVCPFR files on the BA60UC04 subvolume or the AAREAD06 and CNVCPFR files on the BA60UC06 subvolume.

Cardholder Authorization File (CAF)

A CAF file conversion can be performed via the Refresh process. BASE24 users can perform a CAF refresh to initialize the new fields in the CAF EMV segment to a value that they want.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interchange Interface, ISO Host Interface, Super Extract, Refresh and From Host Maintenance processes.

BIC ISO Interchange Interface Process

PIN Unblock transactions are supported with a new ISO processing code of “970000” in BASE24 external messages. Hosts should be analyzed for any impacts that may result from receiving the new transaction codes.

In approved PIN management transaction responses, the EMV Response Token and EMV Script token (containing a ‘PIN Change’ or ‘PIN Unblock’ script) may be included in the BASE24 external message. When the HISO and BICISO modules send a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either ‘C’ or ‘U’ to indicate the type of script present in the EMV Script token. When the HISO and BICISO modules receive a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either ‘C’ or ‘U’ to indicate the type of script present in the EMV Script token.

If the new EMV Issuer Script Results Token is configured to be included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

ISO Host Interface Process

PIN Unblock transactions are supported with a new ISO processing code of “970000” in BASE24 external messages. Hosts should be analyzed for any impacts that may result from receiving the new transaction codes.

In approved PIN management transaction responses, the EMV Response Token and EMV Script token (containing a ‘PIN Change’ or ‘PIN Unblock’ script) may be included in the BASE24 external message. When the HISO and BICISO modules

send a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token. When the HISO and BICISO modules receive a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token.

If the new EMV Issuer Script Results Token is configured to be included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Super Extract Process

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to the CAF and CPF records.

Host code changes are required for all BASE24 users that use ULF Extract to accept and process information for the changes to the CAF records.

The PTLF Extract record format will change if additional EMV transaction data is extracted from BASE24-pos to provide Management Information regarding offline PIN verification.

The TLF Extract record format will change if additional EMV transaction data is extracted from BASE24-atm to provide Management Information regarding offline PIN verification.

The new EMV PIN Unblock transactions are supported with an internal transaction code of "820000". Hosts should be analyzed for any impacts that may result from receiving TLF Extract records containing the new transaction code.

If the PIN Change Token (containing new PIN data) and/or new EMV Issuer Script Results Token is configured to be included in the TLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

Refresh Process

Host code changes are required for all BASE24 users that utilize the release 6.0 CAF Refresh file message format to accept and process the new fields in the CAF file. Users utilizing the release 5.x CAF Refresh file message format will not be able to refresh the EMV segment.

From Host Maintenance Process

Host code changes are required for all BASE24 users that utilize the release 6.0 FHM ISO message format to accept and process the new fields in the CAF file. Users

utilizing the release 5.x CAF FHM ISO message format will not be able to perform file maintenance on the EMV segment.

User/Operational Impacts

Disk Impacts

Additional EMV transaction data logged by BASE24-atm to provide Management Information regarding offline PIN verification will cause the length of TLF records to increase.

Additional EMV transaction data logged by BASE24-pos to provide Management Information regarding offline PIN verification will cause the length of PTLF records to increase.

The TLF and ILF require additional disk space if the PIN Change Token (containing new PIN data) and/or new EMV Issuer Script Results Token is configured to be included in the transaction records logged to these files. The total impact varies for each BASE24 user depending on the transaction volume in the BASE24 system.

User Impacts

The following configuration requirements apply when the PIN Change transaction is authorized on BASE24-atm:

- The PIN verification method must be IBM 3624-format or Visa PVV.
- The BASE24-atm authorization method must be Positive or Positive Balance, so the cardholder has a CAF record.
- PIN verification digits must be carried in the CAF.

Both FHM reports have been amended to show headings for the new EMV fields and also the values obtained from the new fields.

Operational Impacts

New event (log) messages were created as result of this enhancement, e.g. to provide notification of a script processing failure following an "implicit" PIN unblock. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

Performance Impacts

If BASE24 is configured to generate the 'PIN Change' and/or 'PIN Unblock' script command, then the additional HSM command will have an impact on system performance.

Vendor Impacts

To implement this enhancement, BASE24 must have access to a host security module. The enhancement supports the use of Atalla and Thales e-Security (Racal) security modules.

If BASE24 is configured to generate the 'PIN Change' script command, the security module must have a version of software that is capable of supporting the Transaction Security Services enhancement. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for more details.

If BASE24 is configured to generate the 'PIN Change' script command, then a Thales HSM with version 6 software or higher (supporting the enhanced 'KU' command) or an Atalla HSM (supporting the new '351' command) is required.

Reference Documents

- Visa International: Visa Integrated Circuit Card Specification (VIS), version 1.4.0, dated 31 October 2001.
- MasterCard International: M/Chip 4 Card Application Specification, draft version 0.7, dated 16 May 2002.
- APACS: Chip CVM Architecture, version 0.4, dated April 2001.
- Europay/LINK/Visa: Reciprocal PIN Management Services Technical Solution Proposal, version 1.0, dated 31 October 2001.
- Visa International: PIN Management for IC Cards, Member Implementation Guide, version 1.1, dated 30 August 2002.
- MasterCard Europe: PIN Management Services, Service Description (no version information), dated July 2002.
- LINK Interchange Network Ltd.: LINK Switch Service Interchange Standard (LIS5), 2003-1, version 1.1.
- Switch Card Services Ltd.: Inter-Member System Specification, Part 5, Authorization Specification, Phase 12, version 2.0, dated 18 June 2002.
- Thales e-security: HSM 6.0 Firmware Maintenance Release 1233HS0C HSM 6.0 Release, dated 22 February 2002.
- Compaq/Atalla Security Products: Functional Specification for EMV PIN Change (Command 351), version 1.9, dated 04 December 2002.

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Persistent UAF Support

The Usage Accumulation File (UAF) is normally purged daily at logical network cutover or at midnight cutover depending on the field cutover settings for each institution. Functionality has been added with this enhancement to allow the UAF to remain on the system and be cleaned up by a new UAF Cleanup process. This process is bound into the Super Extract process and is initiated via Settlement at logical network cutover or at midnight cutover, automatically based on a timer set in the Extract Configuration File (ECF), or manually by an operator.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field depend on the configuration desired by the institution.

Extract Configuration File (ECF)

Field Name: UAF CLEANUP
Screen: ECF Screen 2
DDL Name: ECF.ECFBASE.UAF-CLEANUP.FLG

A code indicating whether UAF cleanup should be performed. Valid values are Y (Yes, perform the UAF cleanup) and N (No, do not perform UAF cleanup). The default is N.

Field Name: UAF CLEANUP GROUP
Screen: ECF Screen 2
DDL Name: ECF.ECFBASE.UAF-CLEANUP.GRP

A value indicating the refresh group against which UAF cleanup is performed. Valid values include the value ALL which indicates the UAF cleanup is performed on all UAFs, and a user-identified group name

which indicates the UAF cleanup is performed on the UAF associated with this specified group. The user-identified group name value must correspond to a value in the IDF REFRESH GROUP field on IDF screen 1.

Institution Definition File (IDF)

An FIID Auth File Segment Indicator on screen 5 of the IDF has been added for the Online Record Maintenance segment added to the UAF for this enhancement. The file segment indicator must be updated to a "Y" for any institution that utilizes the persistent UAF functionality with the UAF Cleanup process.

Field Name: PERSISTENT UAF
Screen: IDF Screen 3
DDL Name: IDF.IDFBASE.PERSISTENT-UAF

A value indicating whether this institution utilizes a UAF and if so, whether the UAF is to be maintained as persistent. Valid values are:

0 – No, the UAF is not utilized or the Persistent UAF functionality is not required. (Default)

1 – Yes, Persistent UAF is utilized along with Settlement support. The UAF cleanup function is initiated automatically by the Settlement process at logical network cutover or at midnight cutover, or is initiated automatically based on a timer set in the Extract Configuration File (ECF), or is initiated manually by an operator.

2 – Yes, Persistent UAF is utilized without Settlement support. The UAF cleanup function is initiated automatically based on a timer set in the Extract Configuration File (ECF), or is initiated manually by an operator.

Usage Accumulation File (UAF)

Field Name: LAST RESET DATE
Screen: UAF Screen 1
DDL Name: UAF.ONL-REC-MAINT.LAST-RESET-DAT

The date (YYMMDD) that the usage accumulation totals on screen 1 of the UAF were last cleared.

Field Name: LAST USED DATE
Screen: UAF Screen 4
DDL Name: UAF.ONL-REC-MAINT. ATM-LAST-USED-DAT

The date (YYMMDD) that the BASE24-atm usage accumulation fields on screen 4 of the UAF were last cleared.

Field Name: LAST RESET DATE
Screen: UAF Screen 6
DDL Name: UAF.ONL-REC-MAINT. POS-LAST-USED-DAT

The date (YYMMDD) that the BASE24-pos usage accumulation fields on screen 6 of the UAF were last cleared.

For additional information on the usage of these fields, refer to the **BASE24 Base Files Maintenance Manual**.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

An entry was added to the SEG-TABLE and the SEG-DESCR-TABLE for the online record maintenance segment.

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Extract Configuration File (ECF)

An inline conversion of the ECF will be performed to initialize the new data fields defined for this enhancement to default values. If a release 5.3 file is being converted to release 6.0 version 3, the conversion programs on the BA60UC04 subvolume will initialize the new data fields in the ECF.

If the BASE24 user wishes to enable this enhancement, the user must update the new UAF Cleanup field with a 'Y' and the UAF Cleanup Group field with a valid value. Otherwise, the new fields in the ECF indicate that the UAF Cleanup will not be performed.

Institution Definition File (IDF)

An inline conversion of the IDF will be performed to initialize the new data fields defined for this enhancement to default values. If a release 5.3 file is being converted to release 6.0 version 3, the conversion programs on the BA60UC04 subvolume will initialize the new data fields in the IDF.

If the BASE24 user wishes to enable this enhancement, the user must update the new Persistent UAF field with a '1' or '2'. Otherwise, the new field in the IDF indicates that the UAF is not utilized or is not persistent.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24 users that use IDF Extract to accept and process information for the changes to the IDF records.

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to the ECF, IDF and UAF records.

User/Operational Impacts

Disk Impacts

The UAF requires additional disk space for the new online record maintenance segment added for this enhancement. This segment will always be present whether persistent UAF functionality is configured or not. The total impact varies for each BASE24 user depending on the number of records defined in the BASE24 system.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide Inc.

Transaction Security Services Diebold and Identikey PIN Verification Support

Diebold PIN Verification and Identikey PIN Verification are now supported via Transaction Security Services (TSS). TSS will store the data required for Diebold PIN Verification and Identikey PIN Verification in new TSS files, the DNTD and IDNTD. These files are maintained using the ACI Desktop Graphical User Interface (GUI). When using the Transaction Security Services process, the KEYA is replaced by Transaction Security Services files and is no longer accessed in processing for PIN verification, card verification or chip authentication in host security modules. Refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)*** for information on how to implement TSS.

Support for Diebold PIN Verification in TSS has been added for Atalla and Thales e-Security (Racal) HSMs. TSS will not support Diebold PIN Verification for non-AKB Atalla HSMs.

Support for Identikey PIN Verification in TSS has been added for Atalla HSMs. TSS will not support Identikey PIN Verification for Thales HSMs.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

KEYA

With the implementation of Transaction Security Services the KEYA is no longer used by BASE24 for Diebold Number Table PIN verification or Identikey PIN Verification.

LCONF Param Changes

Not applicable

New Data Files

New TSS data files, the DNTD and IDNTD, are required to replace the BASE24 KEYA records no longer accessed in processing for verification in host security modules. These files can be created by running a conversion program or records can be added, deleted or modified manually using the ACI Desktop Graphical User Interface.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

If all PINS are encrypted in hardware, access to the KEYA should be disabled using the BASE24 Security screen.

Access to the Transaction Security Services equivalents of this file will be through the new ACI Desktop Graphical User Interface that is set up separately from the BASE24 Pathway CRT access system. Refer to the ***BASE24 Classic 6.0 Installation Guide*** and the ***BASE24 Release 6.0 Implementation Guide (Version 2)*** for more information.

Routing Changes

Not applicable

Network Configuration

To implement this enhancement, BASE24 must have access to a host security module. Diebold PIN Verification supports the use of the Atalla AKB and Thales

e-Security (Racal) host security modules. Identikey PIN Verification supports the use of the Atalla host security module (AKB and non-AKB).

File Conversions

A file conversion for the KEYA is supplied with the Diebold PIN Verification and Identikey PIN Verification enhancement. There are no structural changes to the KEYA, but the Transaction Security Services DNTD and IDNTD files can be created and populated from BASE24 files. DNTD and IDNTD data can also be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs.

Refer to the ***BASE24 Classic 6.0 Installation Guide*** and the ***BASE24 Release 6.0 Implementation Guide (Version 2)*** for further information on file conversions.

Key Authorization File (KEYA)

The KEYA conversion program can be run to create DNTD and IDNTD records and a new KEYA. Encrypted keys in the new KEYA will be set to zeroes. The original KEYA is not modified. Running the conversion program is not required. DNTD and IDNTD records can be entered manually using the ACI Desktop Graphical User Interface instead.

The KEYA conversion program is on the BA60UC05 subvolume. If modifications to the conversion program are necessary, the program can be modified by editing the TSSKEYAS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#keyatss.t,name $keyam,nowait/  
tsskeyam -force -list -listreplace
```

Users must update the TSSKEYAR obey file on the BA60UC05 subvolume with the appropriate information. TSSKEYAR must then be obeyed to execute the conversion program.

For more information on this file conversion, refer to the AAREAD05 and TSSKEYAR files on the BA60UC05 subvolume.

Host/Co-Network Impacts

Not applicable.

User/Operational Impacts

Disk Impacts

The KEYA file is not required if Transaction Security Services is implemented and all encryption/decryption is performed by an HSM.

The Transaction Security Services process requires several files. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for more details.

Operational Impacts

Event messages generated by Transaction Security Services components running on the HP NonStop system are sent to an EMS collector process and are displayed on the EMS log. The ACI Desktop Graphical User Interface can be used to view or modify Transaction Security Services event messages. Operators must be trained how to view and modify these messages.

Vendor Impacts

To implement this enhancement, BASE24 must have access to a host security module. The enhancement supports the use of Atalla and Thales e-Security (Racal) security modules. The security module must have a version of software that is capable of supporting the Transaction Security Services enhancement.

- Atalla AKB version 1.0 is supported for Diebold PIN Verification.
- Thales e-Security (Racal) version 5.05 is supported for Diebold PIN Verification.
- Atalla version 2.9 software and AKB version 1.0 are supported for Identikey PIN Verification.

Refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)*** for information on Atalla AKB HSMs.

Reference Documents

- Banking Command Reference Manual (424645-002 May 2001)
- Atalla Key Block Banking Command Reference Manual (TP522901-001, October 2001)
- Host Security Module RG7000 Programmers Manual (1270A514 Issue 4)

Transaction Security Services Multiple Logical Network Support

This enhancement allows customers who run logical networks (LNs) at separate physical sites to run separate TSS databases for each LN. It also provides ease of migration to those customers who don't want to implement TSS on all LNs at one time.

To serve these purposes, the BASE24-atm Authorization and BASE24-pos Router/Authorization processes perform an extra PIN translation just prior to SPROUTE-routing transactions to another LN. This operation re-encrypts the PIN under a common PIN encryption key known to all LNs in the BASE24 system.

The common PIN key is specified by a new LCONF param named MULT-LN-PIN-KEY. This param controls whether the enhancement is enabled or disabled – the special PIN translation is only performed if it is present and contains valid data. If the param is absent or contains invalid data, no extra PIN processing is performed.

The param should only be configured by customers with multiple LNs and then only in the following situations: (1) The customer is migrating their LNs to TSS and a single TSS database is to be used for all of them. In this case the param can be deleted as soon as migration is complete for all LNs. (2) The customer is migrating their LNs to TSS where a separate TSS database is to be used for each LN. Here the param must be used permanently, even after migration is complete. It should be noted that configuring the param under any other circumstances will result in unneeded calls to the security device.

For LNs using TSS, MULT-LN-PIN-KEY should contain a key locator pointing to a TSS Interchange Security Data Source (ISECD) record that in turn holds the common PIN key. For LNs that don't use TSS, the param should hold the actual *encrypted* PIN key. For BASE24 systems that contain both TSS and non-TSS LNs, the key locator used for TSS LNs should equal the *encrypted* key used by the non-TSS LNs.

If the enhancement is enabled, the authorization processes on the acquiring LN only perform the extra translation on a certain category of SPROUTE LGNT-routed transactions: The trans have to be destined for another LN because of a match on the card prefix in the second LN's Card Prefix File (CPF). The authorization processes use logic introduced by the Multiple Logical Network Routing Indicator enhancement to detect this routing condition. That enhancement modified the XPNET SPROUTE^LOOKUP call to return a status 2 if the card prefix is found in another LN's CPF. If true, BASE24-atm Authorization and BASE24-pos Router/Authorization add a MULT-LN Token to the internal message.

If MULT-LN-PIN-KEY is present, the authorization processes now check the SPROUTE^LOOKUP return status to determine whether to translate the PIN. If the

call returns a 2, the processes translate the PIN from encryption under the acquiring endpoint's key to encryption under the common PIN key. The PIN is re-encrypted as an ANSI PIN block as part of the translation. Once the operation is complete, the authorization processes move the new PIN and the common PIN key into the internal message for shipment to the issuing LN. On PIN Change transactions, BASE24-atm Authorization also performs this translation on new PINs carried in the PIN Change Token.

BASE24-atm Authorization and BASE24-pos Router/Authorization were also modified to support their PIN verification roles on the issuing LN. During PIN verification these processes pass an ORIGIN flag to the BASE24 Security Utilities (SECUTILS) and, if TSS is used, the TSS process. The flag indicates whether the transaction originated at an interchange or a device. This information is important for two reasons. First, when TSS is used, it tells whether the TSS process should retrieve a Channel Security Data Source (CSECD) record or an ISECD record for the PIN key. Second, when a Thales (Racal) security device is used *without* TSS, it indicates whether the encryption key is a Terminal PIN Key (TPK) or a Zone PIN Key (ZPK).

The authorization processes normally use the internal message's ORIGINATOR field to determine the value of ORIGIN. With this enhancement enabled, ORIGIN is now hard-coded to 'interchange originated' if the PIN has been previously re-encrypted under the common key. For issuing LNs using TSS, this value ensures that the TSS process always looks for the common key on an ISECD record. For issuing LNs using a Thales security device *without* TSS, the value causes SECUTILS to format the proper verify command (i.e., one that uses a ZPK). BASE24-atm Authorization and BASE24-pos Router/Authorization hard-code the ORIGIN flag only if the following are true: (1) The MULT-LN-PIN-KEY param is present and contains a valid PIN key/locator. (2) The Multiple LN Token is present in the internal message.

In addition to the changes to BASE24-atm Authorization and BASE24-pos Router/Authorization, modifications were also made to the ISO Host Interface (HISO). The authorization processes can direct transactions to a host without first processing the PIN, so HISO has to perform its own ORIGIN logic when translating the PIN to encryption under the external host key. It does this only if the new LCONF param is present and the Multiple LN Token is present in the message.

The only BASE module modified for this enhancement is the BASE portion of HISO. It was changed to read the MULT-LN-PIN-KEY param on initialization and warmboots.

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement.

MULT-LN-PIN-KEY

This param points to the common PIN encryption key used to re-encrypt PINs on transactions passed between LNs in a multiple-LN BASE24 system. It must be set at the logical network level (i.e., with a 'used by' process name of all asterisks).

For LNs that use TSS, the param must contain a sixteen-digit hexadecimal key locator that points to a TSS ISECD (Interchange Security Data Source) record that in turn contains the actual encrypted PIN key.

For LNs that don't use TSS, the param must contain the actual sixteen-digit hexadecimal *encrypted* PIN key. The key has to be of single length because the BASE24 Security Utilities (SECUTILS) don't support longer keys for the non-TSS security device types.

This restriction means that, for BASE24 systems that contain one or more non-TSS LNs, the common key used by all LNs has to be of single length. If all LNs use TSS, however, either a single-, double-, or triple-length common key can be used.

Finally, if a Thales security device is in use, the common PIN key must be a Zone PIN Key (ZPK). A Terminal PIN Key (TPK) can't be used since no Thales PIN translation commands translate to TPKs.

Database Configuration

Interchange Security Data Source (ISECD)

A new TSS ISECD record must be added for each LN that uses TSS in a multiple-LN system. This record, which is pointed to by a corresponding MULT-LN-PIN-KEY LCONF param, should contain the common PIN key shared by all LNs.

If the system contains at least one LN that doesn't use TSS, this record should be configured with a single-length PIN encryption key. If all LNs use TSS, however, either a single-, double-, or triple-length key can be used.

If a Thales security device is in use, the common PIN key must be a Zone PIN Key (ZPK). A Terminal PIN Key (TPK) can't be used since no Thales PIN translation commands translate to TPKs.

When adding the ISECD common key record, only the following elements need to be entered. All other fields can be left empty or set to their defaults.

GUI Label: Interface Name
Element: INTF_NAM

Must contain the same value as the key locator specified in the MULT-LN-PIN-KEY LCONF param.

GUI Label: PIN Block Format
Element: PIN_BLK_TYP

Must be set to "ANSI PIN". The special PIN translation always re-encrypts the PIN as an ANSI PIN block.

GUI Label: Outbound Keys (Slot 1) Parts 1, 2, 3
Element: OUTBND_KPT1_KEYX

Must contain the common PIN encryption key. If an Atalla AKB security device is used, all three key parts must be filled. If a non-AKB Atalla security device is used, only the first part should be filled. If a Thales device is used, one, two, or three parts can be filled.

GUI Label: Outbound Current Index
Element: OUTBND_KPT1_IDX

Must contain a value of 1.

GUI Label: Inbound Keys (Slot 1) Parts 1, 2, 3
Element: INBND_KPT1_KEYX

Must contain the common PIN encryption key. If an Atalla AKB security device is used, all three key parts must be filled. If a non-AKB Atalla security device is used, only the first part should be filled. If a Thales device is used, one, two, or three parts can be filled.

GUI Label: Inbound Current Index
Element: INBND_KPT1_IDX

Must contain a value of 1.

GUI Label: Outbound Keys (Slot 1) Header
Element: OUTBND_KPT1_HDR

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB header. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Outbound Keys (Slot 1) MAC
Element: OUTBND_KPT1_KAM

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB MAC value. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Outbound Key Length
Element: OUTBND_KPT_LGTH

Must contain "Single", "Double", or "Triple" depending on the size of the common key entered in OUTBND_KPT1_KEYX.

GUI Label: Inbound Keys (Slot 1) Header
Element: INBND_KPT1_HDR

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB header. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Inbound Keys (Slot 1) MAC
Element: INBND_KPT1_KAM

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB MAC value. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Inbound Key Length
Element: INBND_KPT_LGTH

Must contain "Single", "Double", or "Triple" depending on the size of the common key entered in INBND_KPT1_KEYX.

GUI Label: External Key Block Format
Element: EXTRN_KEY_BLK_FRMT

If a Thales security device is used, this field should be set to "Variant method". If an Atalla security device is used (AKB or otherwise), the field should be set to "Standard Atalla".

New Data Files

Not applicable.

Template Changes

Not applicable.

Device Configuration

Not applicable.

Mega Table (MEGATBL) Changes

Not applicable.

PI-Table Changes

Not applicable.

SEG-Table Changes

Not applicable.

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable.

CRT Access Security Changes

Not applicable.

Routing Changes

Not applicable.

Network Configuration

Not applicable.

File Conversions

Not applicable.

Host/Co-Network Impacts

Not applicable.

User/Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable.

Reference Documents

Not applicable.

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Section 5

BASE24-atm

Section 5 discusses the changes and enhancements made to BASE24-atm in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing BASE24-atm users must consider when upgrading from release 6.0 version 2.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-atm users, along with the associated conversion and implementation requirements that all BASE24-atm users must address.
- Optional enhancements describe the changes that affect only those BASE24-atm users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-atm users must address to implement the enhancements.

Section 5 must be used in conjunction with Section 4 to identify the changes that BASE24-atm users need to consider when upgrading to release 6.0 version 3.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-atm users must address in order to upgrade to release 6.0 version 3.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- ATD Infrastructure

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

When an optional enhancement is developed, it is usually developed as a bindable extension on its own subvolume (for example, the NCR 5XXX device handler's AKDS extension on AT60AN50). The main software module (AT60N50.N50S) contains calls (hooks) to procedures that reside in the extension. Dummy procedures are used to allow the main module to compile cleanly when the optional enhancement is not purchased or is otherwise deactivated. In the past, these dummy procedures resided in a separate dummy file for each enhancement, or within the main module itself. (Note that these older dummy procedures will not be relocated at this time.) Beginning with version 3 of release 6.0 and going forward, certain modules will now employ a single "common" dummy file that will contain all dummy procedures for all new enhancements. The table below lists each module and its new dummy file.

Module	New Dummy File
Diebold 10XX/478X Device Handler	at601000.c1000d
NCR 5XXX Device Handler	at60n50.n50d

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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ATD Infrastructure

In order to support future enhancements, the ATD has been modified to contain offsets to additional data structures in the data area of the record. These offsets have been added to both the ATD Static and Dynamic records. Also, some fields for release 6.0 version 3 enhancements have been added outside the ACI User Field. The inline conversion to implement these new fields is explained in this section.

To use the additional data structures, the device handler-specific DDLs (e.g., N50DDLs) were modified to contain offsets and lengths that point to ATDD1.ADNL-OFST, ATDD1.ADNL-LGTH, ATDS1.ADNL-OFST and ATDS1.ADNL-LGTH. This allows each device handler to individually define what the additional offsets mean for that device type. Following is an example of how this is done.

Definition ADNL-OFST.

Begin

02 BNA-CPM	Type binary 16.
02 Enhancement-2	Type binary 16.
02 Enhancement-3	Type binary 16.
02 Enhancement-4	Type binary 16.
02 Enhancement-5	Type binary 16.
02 Enhancement-6	Type binary 16.
02 Enhancement-7	Type binary 16.
02 Enhancement-8	Type binary 16.
02 Enhancement-9	Type binary 16.
02 Enhancement-10	Type binary 16.

End.

Definition ADNL-LGTH.

Begin

02 BNA-CPM	Type binary 16.
02 Enhancement-2	Type binary 16.
02 Enhancement-3	Type binary 16.
02 Enhancement-4	Type binary 16.
02 Enhancement-5	Type binary 16.
02 Enhancement-6	Type binary 16.
02 Enhancement-7	Type binary 16.
02 Enhancement-8	Type binary 16.
02 Enhancement-9	Type binary 16.
02 Enhancement-10	Type binary 16.

End.

The device handler calculates the location of each enhancement offset based on the other enhancements supported. For example, to determine the offset for the BNA/CPM enhancement, the device handler adds all of the ATDD1.ADNL-LGTH

fields together to determine where to place the BNA/CPM data within the DATA-AREA. All of the ATDD1.ADNL-LGTH fields are set to zero initially but as additional enhancements are added, the lengths and offsets will be populated. BNA/CPM may not be the first enhancement for the given ATM. For example, if enhancement-2 is added, the ADNL-OFST and ADNL-LGTH are populated based on the data for enhancement 2. The length for BNA-CPM is zero. If the BNA/CPM enhancement is added to the given ATM, the ADNL-OFST and ADNL-LGTH for the first enhancement are populated and the offset for BNA/CPM is after the data for enhancement-2. This means no conversion is required when the BNA/CPM enhancement is added.

The Diebold Merchant Banking Center and NCR Bunch Note Acceptor / Check Processing Module Support enhancements make use of the ADNL-OFST and ADNL-LGTH in the ATD.

Since the structure of the ATD was already being modified and converted, it was decided to add data fields for two of the release 6.0 version 3 enhancements (Dynamic Key Management and Multiple Currency Dispense) outside of USER-FLD-ACI. Each of these enhancements would have used a large portion of USER-FLD-ACI. By moving the existing data to make room for the data for these enhancements, USER-FLD-ACI is still available for future enhancements with no conversion required.

To implement the new data record, logic has been added to the required modules to perform an inline conversion of the ATD records as they are used. In addition to the device handler modules listed below, the ATD Server and ATM Settlement have been modified to perform the inline conversion of the ATD record. The logic for the inline conversion is described in File Conversions.

This subsection describes the changes and conversion/migration requirements associated with the ATD Infrastructure enhancement.

Section 2 contains a functional overview of the ATD Infrastructure enhancement.

Note: This enhancement applies to the following BASE24-atm modules:

- Diebold 10XX/478X Device Handler
- NCR 5XXX Device Handler

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

This section describes the new fields in the files that have been modified to support this enhancement. The Device Handler will perform ATD conversions during startup initialization.

BASE24-atm Terminal Data Dynamic file – general data (ATDD1)

The ATDD1 record length was made static (4062) and two arrays were added to the ATDD1 definition immediately following the DEV-DATA-LGTH:

ADNL-OFST[0:9]

ADNL-LGTH[0:9]

The DATA-AREA length was reduced to 3938. Data within ATDD1 DATA-AREA shifted to accommodate the two new arrays.

ATDD1 before conversion:

...	dev^data^lgth	data^area
-----	---------------	-----------

ATDD1 after conversion:

...	dev^data^lgth	adnl^ofst[0:9]	adnl^lgtht[0:9]	data^area
-----	---------------	----------------	-----------------	-----------

The ATDD1-CORE definition was modified to include a new DKM structure immediately following the USER-DEFINED-AREA.

The ATDD1-CORE length was increased to 540. Data following ATDD1-CORE was shifted to accommodate the change.

ATDD1-CORE before conversion:

...	user^defined^area	user^fld^aci	user^fld^regn	user^fld^cust	dev specific
-----	-------------------	--------------	---------------	---------------	--------------

ATDD1-CORE after conversion:

...	user^defined^area	DKM fields	user^fld^aci	user^fld^regn	user^fld^cust	dev specific
-----	-------------------	------------	--------------	---------------	---------------	--------------

BASE24-atm Terminal Data Static file – general data (ATDS1)

The ATDS1 definition was modified to include two arrays. The arrays are located immediately following the DEV-DATA-LGTH:

ADNL-OFST[0:4]

ADNL-LGTH[0:4]

The variable DATA-AREA buffer was reduced to a limit of 3960. Data within ATDS1 DATA-AREA was shifted to accommodate the two new arrays.

ATDS1before conversion:

...	dev^data^lgth	data^area
-----	---------------	-----------

ATDS1after conversion:

...	dev^data^lgth	adnl^ofst[0:4]	adnl^lgtht[0:4]	data^area
-----	---------------	----------------	-----------------	-----------

The ATDS1-CORE definition was modified to include two new fields immediately following the EMV-TERM-CAP:

TERM-LOC-IND

SELECT-CRNCY-CDE[0:3]

The ATDS1-CORE length was increased to 296. Data following ATDS1-CORE shifted to accommodate the change.

ATDS1-CORE before conversion:

...	EMV^term^ cap	user^fld^aci	user^fld^regn	user^fld^cust	dev specific
-----	------------------	--------------	---------------	---------------	--------------

ATDS1-CORE after conversion:

...	EMV^term^ cap	term^loc ^ind	selct^crncy ^cde	user^fld^aci	user^fld^regn	user^fld^cust	dev specific
-----	------------------	------------------	---------------------	--------------	---------------	---------------	--------------

Host/Co-Network Impacts

Host impacts will be incurred by those BASE24 users implementing the BASE24-atm ATD Infrastructure enhancement. Modifications to existing files and processing will result in impacts to the Super Extract process.

Super Extract Process

Host code changes are required for all BASE24-atm users that utilize the OMF Extract Process to accept and process information in the ATD files.

User/Operational Impacts

Disk Impacts

The records-per-block for the ATDD1 and ATDS1 will not be reduced with this enhancement.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the BASE24-atm enhancements that can be enabled or disabled with release 6.0 version 3, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Automated Key Distribution System
- Diebold Merchant Banking Center
- Dynamic Key Management
- Interactive Response Notification Fee Processing and Negative Surcharge Support
- Multiple Currency Dispense
- NCR Bunch Note Acceptor / Check Processing Module Support
- Offline PIN Management Support
- Transaction Security Services Multiple Logical Network Support

BASE24-atm users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements for future reference.

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Automated Key Distribution System

This section describes the implementation requirements associated with the Automated Key Distribution System (AKDS) enhancement.

Section 2 contains a functional overview of this enhancement.

A standard BASE24-atm module has been modified to support this enhancement. That module is the NCR 5XXX device handler.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Automated Key Distribution System. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

Set the following flag to true to bind the AKDS module to the NCR 5XXX device handler.

- atm_dh_n50_akd_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Note that these modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-AKDS-TSS-TIMER

This param is used by the NCR 5XXX device handler. It specifies the number of seconds the device handler waits for a reply from Transaction Security Services before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 600 is used.

ATM-DH-N50-AKDS-SERIAL-NUM-VRFY

This param is used by the NCR 5XXX device handler. It specifies whether or not Transaction Security Services will verify the Encrypting PIN Pad's (EPP's) serial number during the exchange of public keys between the EPP and BASE24. This is an optional param. Valid values are "Y" (yes, verify the serial number) and "N" (no,

do not verify the serial number). If the param is not found or is invalid, the default value of "N" is used.

Database Configuration

Not applicable

New Data Files

The following data sources have been added to the Transaction Security Services database for this enhancement. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for more information.

- Channel Public Key Security Data Source (CHNPKD)

This data source contains the public key information of the Encrypting PIN Pad (EPP) of the channel device (ATM).

- Host Public Key Security Primary Data Source (HSPKPD)

This data source contains the primary public key and secret (private) key information generated for the host (BASE24).

- Host Public Key Security Secondary Source (HSPKSD)

This data source contains the secondary public key and secret (private) key information generated for the host (BASE24).

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Initial Setup for Transaction Security Services

Initial setup procedures must be carried out for Transaction Security Services before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM. BASE24's public/private key pair must be created. The public key must be sent to the Certificate Authority (NCR) to be verified and signed. The Certificate Authority must return BASE24's public key signature, as well as the Certificate Authority's own public key. These values must be entered into the Transaction Security Services database.

For more information about these initial setup procedures, refer to the ***BASE24 Transaction Security Services Processing Guide***.

NCR 5XXX Device Handler – DCT Commands

The following commands have been added to the NCR 5XXX Load Commands Screen (screen 45) of the Device Control Terminal (DCT).

- **LOAD PUBLIC KEY**

When the operator places the cursor on this command and presses function key F6-LOAD, the DCT sends a command to the device handler to initiate an Authentication and Public Key Exchange sequence. During this load sequence, BASE24 and the specified ATM authenticate each other and exchange public keys. These public keys are used for encryption and mutual authentication in all future master key downloads.

The Authentication and Public Key Exchange sequence cannot be initiated at the ATM.

This command is typically performed only once – when a new ATM with an AKDS-enabled EPP is initially deployed, or when an ATM already in the field is fitted with a new AKDS-enabled EPP.

After this command completes successfully, the LOAD MASTER KEY command can be performed.

- **LOAD MASTER KEY**

When the operator places the cursor on this command and presses function key F6-LOAD, the DCT sends a command to the device handler to initiate a Master Key Download sequence. During this load sequence, BASE24 generates a new master key and downloads it to the specified ATM.

The Master Key Download sequence cannot be initiated at the ATM.

The Authentication and Public Key Exchange sequence (see LOAD PUBLIC KEY) must complete successfully at least once before this command can be performed.

This command can be performed as often as is necessary to preserve the integrity of the ATM's keys.

After this command completes successfully, the DES encryption keys (e.g. PIN encryption key, MAC key) may be generated and downloaded to the ATM.

- **LOAD KEY ENTRY MODE x**

Where 'x' is a single character specifying the key entry mode:

- '1' – Mode is single length without XOR
- '2' – Mode is single length with XOR
- '3' – Mode is double length with XOR
- '4' – Mode is double length restricted

When the operator types a valid value at 'x' and presses function key F6-LOAD, the DCT sends a command to the device handler to download the key entry mode to the specified ATM.

The key entry mode can also be set at the ATM when in supervisor mode.

Setting the key entry mode causes all keys to be deleted in the EPP, including BASE24's public key and the master key. The exceptions are the EPP's public and private (secret) keys and NCR's public key, which are not deleted. To reduce the chances of accidentally changing the mode and deleting the keys (thus rendering the ATM inoperable), the EPP will accept this command only if the Authentication and Public Key Exchange sequence (see LOAD PUBLIC KEY) has been performed since the last power-up, ensuring that BASE24 has the ability to restore the keys. If this sequence has not been performed since the last power-up the EPP will reject the command.

NCR allows one exception to this restriction. To assist in the migration from a previous release of NDC+ to release 7.00, this command will be accepted one time only without having to first perform the Authentication and Public Key Exchange sequence if the EPP is operating in BAPE (Basic Alpha PINpad and Encryptor) emulation mode.

The following describes one such migration scenario. An EPP can be installed on an ATM running a previous release of NDC+. Releases prior to 7.00 do not recognize the full functionality of the EPP. In this case the EPP is operating in BAPE emulation mode, which does not support AKDS. When NDC+ 7.00 is later installed, it will detect this condition and continue to run in BAPE emulation mode, rather than reprogram the encryptor as an EPP. Thus the encryptor's current keys are preserved and the ATM continues to function as normal. The encryptor's keys remain preserved until the key entry mode is changed, either remotely using the LOAD KEY ENTRY MODE command, or locally at the ATM when in supervisor mode. Setting the key entry mode by either method causes all keys to be deleted and the encryptor to be re-programmed, switching from BAPE emulation mode to full EPP mode. At this point, the EPP is AKDS-enabled.

After this command completes successfully, the Authentication and Public Key Exchange sequence must be performed (see LOAD PUBLIC KEY), followed by the Master Key Download sequence (see LOAD MASTER KEY).

NCR 5XXX Device Handler – NCS/NCPCOM Commands

The following 9500 commands are supported in the NCR 5XXX device handler. They are entered at the BASE24 AFT NCS screen or through NCPCOM.

- LPUBLIC

This 9500 command is an alternative to the DCT's LOAD PUBLIC KEY command. It causes the device handler to initiate an Authentication and Public Key Exchange sequence. For more information refer to the LOAD PUBLIC KEY command in the preceding subsection (NCR 5XXX Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LPUBLIC <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX device handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the BASE24-atm Terminal Data file (ATD).

- LMASTER

This 9500 command is an alternative to the DCT's LOAD MASTER KEY command. It causes the device handler to initiate a Master Key Download sequence. For more information refer to the LOAD MASTER KEY command in the preceding subsection (NCR 5XXX Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LMASTER <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX device handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the BASE24-atm Terminal Data file (ATD).

- LKEYMODE

This 9500 command is an alternative to the DCT's LOAD KEY ENTRY MODE command. It causes the device handler to download the key entry mode to the specified ATM. For more information refer to the LOAD KEY ENTRY MODE command in the preceding subsection (NCR 5XXX Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LKEYMODE <terminal ID>
<mode>"

<process name>:	The symbolic name assigned to the NCR 5XXX device handler process in the XPNET node.
<terminal ID>:	The terminal ID defined for the ATM in the BASE24-atm Terminal Data file (ATD).
<mode>:	The key entry mode. Valid values are: '1' – Mode is single length without XOR '2' – Mode is single length with XOR '3' – Mode is double length with XOR '4' – Mode is double length restricted

Power-Up Considerations

Prior to the Automated Key Distribution System enhancement, the NCR 5XXX device handler automatically performed a full load (or a load of just the PIN encryption and MAC keys, if a full load is not needed) after receiving a power-up message from an ATM. With this enhancement the device handler will continue to do so, but there is an exception: The load will **not** be automatically performed if the ATM has an AKDS-enabled EPP and the EPP does not yet contain a master key. Instead, the device handler will log an event message informing the operator that the public keys and/or master key must be loaded. The master key must be present before a PIN encryption or MAC key can be loaded.

Financial Institution Table (FIT) Considerations

The local PIN verification key (PEKEY), if used, is encrypted under the master key and stored in the FIT. If a new master key is downloaded to an ATM, every PEKEY in the FIT must be re-encrypted under the new master key and the FIT must be downloaded to the ATM.

Supported Keys

Only the master key (A-key) can be downloaded to an ATM using the AKDS enhancement. AKDS does not support the downloading of the power-up COMM key (B-key) because this key is not currently supported by the NCR 5XXX device handler. The DES keys (e.g. PIN encryption key, MAC key) will continue to be encrypted under the master key and downloaded to the ATM as per normal procedures prior to this enhancement.

Event Messages

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume or the BASE24 Documentation CD-ROM.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Automated Key Distribution System.

- NCR model 56XX ATMs running NDC+ 7.0.
- Atalla NSP HSM models A8100, A9100 and A10100. Software version AKB 1.20 is required. AKB products require the use of a Secure Configuration Terminal (SCT), release 4.0 or later. SCT 4.0 devices support the Atalla AKB products while remaining fully backward compatible with older Atalla releases.
- Thales e-Security (Racal) HSM models RG7100, RG7110, RG7300, RG7310 and RG7400. Software version 6.0 and optional DSP plug-in (for RSA functions) are required.

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement AKDS. An EPP is a tamper-proof ATM component comprised of a PIN pad, a security module to perform cryptographic processing, and the cryptographic keys. All of the ATM's cryptographic functions are performed by the EPP.

Protocol Considerations

Some ATM communication protocols may not support the increased message lengths required by the AKDS enhancement. Current device and protocol configurations should be discussed with the ATM vendors prior to implementing AKDS.

Reference Documents

- Atalla: Atalla Key Block Banking Command Reference Manual (524130-002, July 2002)
- NCR: Modifications to NDC+ to Support: EPP, RSA Initial Key Loading, ISO PIN Block Formats (July 17, 2001)
- Thales: Host Security Module RG7000 Programmer's Manual (1270A514 Issue 5, July 2002)

Diebold Merchant Banking Center

This section describes the changes, conversion requirements, and implementation requirements associated with the Diebold Merchant Banking Center enhancement.

Diebold's Merchant Banking Center (MBC) terminals permit customers to purchase cash and perform bag deposits at locations where bank branch offices may not be conveniently located. There are three new transactions supported by BASE24-atm for these MBC terminals. They are Money Exchange from account, Money Exchange from cash and Bag Deposit. The Money Exchange transaction allows merchants to obtain change (bills and rolled coins) of various denominations. The merchant may pay for the change either from an online funding account or by inserting cash into the currency acceptor. Multiple copies of the receipt can be printed for these transactions. The Bag Deposit transaction allows merchants to make bag deposits and obtain multiple copies of receipts. If the merchant entered a deposit bag ID as part of the transaction, the deposit bag ID can be included on the receipt copies.

Card initiated administrative functions were enhanced to include processing for the rolled coin cassettes on balance with standard cash, print subtotals, and balance with current cash transactions. Rolled coin increase/decrease transactions were added to assist in the maintenance of the rolled coin cassettes. The Device Control Terminal (DCT) was not modified as part of this enhancement to include the rolled coin cassette information for these administrative functions. The new rolled coin hopper information will be displayed in the Transaction Log File (TLF) on a transaction detail screen for the associated administrative transaction.

New Receipt Field Type Codes were added to allow customers to configure their MBC Money Exchange and Bag Deposit Receipts with the new MBC information.

These changes, conversion requirements, and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Diebold Merchant Banking Center enhancement.

CUSTMACS Changes

The following Make flag was added to the CUSTMACS file to indicate whether the Diebold 10XX/478X Device Handler supports the Diebold Merchant Banking Center enhancement:

- atm_dh_c1000_mbc_on

LCONF Assign Changes

Not applicable.

LCONF Param Changes

Not applicable.

Database Configuration

This section describes the fields in existing files that have been modified or added to support this enhancement. The setting of each field depends on the configuration desired by the institution.

BASE24-atm Terminal Data files (ATD)

A field has been added to screen 7 of the ATD. The Merchant Banking Center field is used to signify that the terminal is a Merchant Banking Center terminal.

Two screens were added to display the MBC rolled coin hoppers. Screen 14 contains MBC rolled coin hoppers one, two and three. Screen 15 contains MBC rolled coin hoppers four, five and six.

Screen 16 was added to display the Currency Acceptor Contents fields, Cash In Acceptor and Currency Code. The MBC Receipt Group allows the customer to indicate the receipt group used to retrieve receipt templates for MBC transactions. The MBC terminal's Money Exchange Activity (count and amount of Money Exchange transactions) is also displayed on this screen as well as the ineligible transaction override field.

Field Name: MERCHANT BANKING CENTER
Screen: ATD Screen 7
DDL Name: ATDS1.[atds1-dev-info-diebold-10xx].MBC

This field identifies whether the terminal is a Merchant Banking Center ATM.

Field Name: CONTENTS
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CONTENTS

This field identifies what is contained in each hopper.

Field Name: COIN VAL
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.COIN-VAL

This field identifies the denomination of the coin.

Field Name: COINS IN A ROLL
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.COINS-IN-A-ROLL

This field identifies the number of coins in one roll.

Field Name: STD BEG CASH
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.STD-BEG-CASH

This field identifies the standard amount used to replenish the RCD hopper.

Field Name: BEG CASH
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.BEG-CASH

This field identifies the amount at the beginning of the current balance period.

Field Name: CASH INCR
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CASH-INCR

This field identifies the amount added to the RCD hopper via mid-point adjustments.

Field Name: CASH DECR
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CASH-DECR

This field identifies the amount removed from the RCD hopper via mid-point adjustments.

Field Name: CASH OUT
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CASH-OUT

This field identifies the amount dispensed from the RCD hopper.

Field Name: END CASH
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.END-CASH

This field identifies the amount currently available in the RCD hopper.

Field Name: CURRENCY CODE
Screen: ATD Screen 14
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CRNCY-CDE

This field identifies the type of currency in the RCD hopper.

Field Name: CONTENTS
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CONTENTS

This field identifies what is contained in each hopper.

Field Name: COIN VAL
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.COIN-VAL

This field identifies the denomination of the coin.

Field Name: COINS IN A ROLL
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.COINS-IN-A-ROLL

This field identifies the number of coins in one roll.

Field Name: STD BEG CASH
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.STD-BEG-CASH

This field identifies the standard amount used to replenish the RCD hopper.

Field Name: BEG CASH
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.BEG-CASH

This field identifies the amount at the beginning of the current balance period.

Field Name: CASH INCR
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CASH-INCR

This field identifies the amount added to the RCD hopper via mid-point adjustments.

Field Name: CASH DECR
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CASH-DECR

This field identifies the amount removed from the RCD hopper via mid-point adjustments.

Field Name: CASH OUT
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CASH-OUT

This field identifies the amount dispensed from the RCD hopper.

Field Name: END CASH
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.END-CASH

This field identifies the amount currently available in the RCD hopper.

Field Name: CURRENCY CODE
Screen: ATD Screen 15
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].HOPR-RCD.CRNCY-CDE

This field identifies the type of currency in the RCD hopper.

Field Name: CASH IN ACCEPTOR
Screen: ATD Screen 16
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].CRNCY-ACCPT.CASH-IN-VAULT

This field identifies the amount of cash accepted by the currency acceptor during the current balance period.

Field Name: CURRENCY CODE
Screen: ATD Screen 16
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].CRNCY-ACCPT.CRNCY-CDE

This field identifies the type of currency in the currency acceptor vault.

Field Name: RECEIPT GROUP
Screen: ATD Screen 16
DDL Name: ATDS1.[atds1-dev-info-diebold-10xx].MBC-RCPT-GRP

This field identifies which set of receipt templates from the RCPT file to use when formatting the receipt for MBC transactions.

Field Name: INELIGIBLE TRAN OVERRIDE
Screen: ATD Screen 16
DDL Name: ATDS1.[atds1-dev-info-diebold-10xx].INELIGIBLE-TXN-FLG

This field indicates if not-on-us MBC Money Exchange from Cash transactions from this terminal will be forced through the system as approved if a switch has denied the transaction as an ineligible transaction.

Field Name: MONEY XCHG COUNT
Screen: ATD Screen 16
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].NUM-MX

This field identifies the number of Money Exchange transactions that have been performed at this terminal during this balance period.

Field Name: MONEY XCHG AMOUNT
Screen: ATD Screen 16
DDL Name: ATDD1.[atdd1-mbc-data-diebold-10xx].AMT-MX

This field identifies the amount of Money Exchange transactions that have been performed at this terminal during this balance period.

BASE24-atm Transaction Log File (TLF)

The TLF AFT was modified to identify Money Exchange transactions in a detail record. Administrative detail records were enhanced to display the MBC rolled coin hopper information, currency acceptor information, and Money Exchange transaction activity.

Field Name: BEGIN CASH
Screen: TLF Screen 12
DDL Name: MBC-SETL-TKN.HOPR.BEG-CASH

This field identifies the amount of cash at the beginning of the balance period.

Field Name: INCREASE CASH
Screen: TLF Screen 12
DDL Name: MBC-SETL-TKN.HOPR.CASH-INCR

This field identifies the amount of cash added to the RCD hopper during the balance period.

Field Name: DECREASE CASH
Screen: TLF Screen 12
DDL Name: MBC-SETL-TKN.HOPR.CASH-DECR

This field identifies the amount of cash removed from the RCD hopper during the balance period.

Field Name: CASH OUT
Screen: TLF Screen 12
DDL Name: MBC-SETL-TKN.HOPR.CASH-OUT

This field identifies the amount of cash dispensed from the RCD hopper during the balance period.

Field Name: END CASH
Screen: TLF Screen 12
DDL Name: MBC-SETL-TKN.HOPR.END-CASH

This field identifies the amount of cash available from the RCD hopper at the end of the balance period.

Field Name: BEGIN CASH
Screen: TLF Screen 13
DDL Name: MBC-SETL-TKN.HOPR.BEG-CASH

This field identifies the amount of cash at the beginning of the balance period.

Field Name: INCREASE CASH
Screen: TLF Screen 13
DDL Name: MBC-SETL-TKN.HOPR.CASH-INCR

This field identifies the amount of cash added to the RCD hopper during the balance period.

Field Name: DECREASE CASH
Screen: TLF Screen 13
DDL Name: MBC-SETL-TKN.HOPR.CASH-DECR

This field identifies the amount of cash removed from the RCD hopper during the balance period.

Field Name: CASH OUT
Screen: TLF Screen 13
DDL Name: MBC-SETL-TKN.HOPR.CASH-OUT

This field identifies the amount of cash dispensed from the RCD hopper during the balance period.

Field Name: END CASH
Screen: TLF Screen 13
DDL Name: MBC-SETL-TKN.HOPR.END-CASH

This field identifies the amount of cash available from the RCD hopper at the end of the balance period.

Field Name: CURRENCY CODE
Screen: TLF Screen 14
DDL Name: MBC-SETL-TKN.CRNCY-ACCEPTOR.CRNCY-CDE

This field identifies the type of currency for the currency acceptor.

Field Name: AMOUNT
Screen: TLF Screen 14
DDL Name: MBC-SETL-TKN.CRNCY-ACCEPTOR.CASH-IN-VAULT

This field identifies the amount of cash accepted by the currency acceptor during the balance period.

Field Name: MONEY EXCHANGE NUMBER
Screen: TLF Screen 15
DDL Name: MBC-SETL-TKN.NUM-MX

This field identifies the number of Money Exchange transactions that were attempted during the balance period.

Field Name: MONEY EXCHANGE AMOUNT
Screen: TLF Screen 15
DDL Name: MBC-SETL-TKN.AMT-MX

This field identifies the amount of Money Exchange transactions that were attempted during the balance period.

Field Name: HOPPER #
Screen: TLF Screen 16
DDL Name:

This field identifies the hopper number of the cash increase/decrease transaction.

Field Name: AMT
Screen: TLF Screen 16
DDL Name:

This field identifies the amount of cash that was replaced/removed for the cash increase/decrease transaction.

Field Name: AMT DESCRIPTION
Screen: TLF Screen 16
DDL Name:

This field identifies the currency code of the cash that was replaced/removed for the cash increase/decrease transaction.

BASE24-atm Receipt File (RCPT)

The Receipt Templates for the MBC transactions Money Exchange from cash, Money Exchange from account and Bag Deposit must be added to the Receipt File. These transactions use existing transaction templates with the receipt group being the MBC receipt group on screen 16 of the ATD.

New receipt field type codes were added to print the MBC information on the receipts. The new codes are as follows:

Money Exchange codes:

AV	All denominations with monetary value received
AN	All denominations with number of rolls/bills received
RV	Requested denomination(s) with monetary value received
RN	Requested denomination(s) with number of rolls/bills received

N2 Amount of currency inserted for a Money Exchange from cash transaction

Bag Deposit code:

ID The bag ID for the Bag Deposit

Administrative codes:

VC Cash vaulted by the currency acceptor during the current balance period

MX Number of Money Exchange transactions attempted during the current balance period

AX Amount of Money Exchange transactions attempted during the current balance period.

An example receipt template for the Money Exchange from account receipt is as follows:

```
-----  
BASE24-ATM  RECEIPT  A41X      00/09/11  09:59  01 OF 01  
DEVICE TYPE: D1000 (DIEBOLD 1000) RECEIPT TYPE:WD (WITHDRAWAL)  
RECEIPT GROUP: ROLL           LANG CODE/MSG ID: 0  
FIID:**** FROM ACCT:** TO ACCT:** PRNTR TYPE:*(*****)  
@RECORD NO      #SQ>  
?IF NOFEES  
#TT #A1$>  
?ENDIF  
?IF FEES  
#TT #S1$>  
@FEE            #S2$>@TOTAL      #A1$>  
?ENDIF  
@ DENOMINATION  COUNT>#RN>  
#XL#LB$>  
#XA#AB$>>  
-----
```

A receipt for this template would appear as follows:

```
<Header Template Info>
:      :      :      :      :
RECORD NO          15
MONEY XCHG        $33.00
FEE                $1.00
TOTAL             $34.00
  DENOMINATION    COUNT
      0.01         2
      0.05         1
      0.25         2
      10.00        1

AVAILABLE BAL      $9906.57
LEDGER BAL         $9802.66
```

An example receipt template for the Money Exchange from cash receipt is as follows:

```
-----
BASE24-ATM  RECEIPT  A41X      00/09/11  09:59  01 OF 01
DEVICE TYPE: D1000(DIEBOLD 1000) RECEIPT TYPE: OL(LOG ONLY)
RECEIPT GROUP: ROLL              LANG CODE/MSG ID: 0
FIID:**** FROM ACCT:** TO ACCT:** PRNTR TYPE:* (*****)
@RECORD NO      #SQ>
?IF NOFEES
#TT #A1$>
?ENDIF
?IF FEES
#TT #S1$>@FEE      #S2$>@TOTAL      #A1$>
?ENDIF
@  DENOMINATION  AMOUNT>
#RV>
@NOTES INSERTED #N2>
-----
```

A receipt for this template would appear as follows:

```
<Header Template Info>
:      :      :      :      :
RECORD NO              15
MX FR CASH             $60.00
  DENOMINATION        AMOUNT
    0.05                4.00
    0.10               10.00
    0.25               40.00
    1.00                1.00
    5.00                5.00

NOTES INSERTED        60.00
```

An example receipt template for the Bag Deposit receipt is as follows:

```
-----
BASE24-ATM  RECEIPT      A41X      00/09/11  09:59  01 OF 01
DEVICE TYPE: D1000(DIEBOLD 1000) RECEIPT TYPE: DP(DEPOSIT)
RECEIPT GROUP: ROLL              LANG CODE/MSG ID:  0
FIID:****   FROM ACCT:** TO ACCT:** PRNTR TYPE:* (*****)
@RECORD NO.              #SQ>
#TT>
@TO  #TO00,06,X>>
@BAG ID  #ID>
@TOTAL AMOUNT  #A1 >
#XL#LB$>
#XA#AB$>>
-----
```

A receipt for this template would appear as follows:

```
<Header Template Info>
:      :      :      :      :
RECORD NO              15
DEPOSIT
TO  XXXX223455

BAG ID  33099842
TOTAL AMOUNT      $560.00
AVAILABLE BAL     $90093.50
LEGDER BAL        $90650.48
```

An example receipt template for the Administrative Deposit Totals receipt is as follows (note that this template does not require the receipt group to be the MBC receipt group):

```
-----  
BASE24-ATM  RECEIPT      A41X      00/09/11  09:59  01 OF 01  
DEVICE TYPE:D1000(DIEBOLD) RECEIPT TYPE:ADTO(ADMIN TOTALS)  
RECEIPT GROUP: ****                      LANG CODE/MSG ID: 0  
FIID:**** FROM ACCT:** TO ACCT:** PRNTR TYPE:* (*****)  
#E(@DEP0#D0 @DEP1#D1 @CHK #CK @MX #AX>  
@MSG #MG @PAY #PY @CSH ACPT #VC>  
@TOT #EN @CRD #CR #AT>  
-----
```

A receipt for this template would appear as follows:

```
<Header Template Info>  
<Admin Hopper Template Info>  
:      :      :      :      :  
DEP0 0  DEP1 1  CHK 2  MX 150.00  
MSG 1  PAY 0  CSH ACPT 55.00  
TOT 4  CRD 0  BAL
```

BASE24-atm Device Control Terminal (DCT)

The DCT has been modified to display the new status information for the MBC Rolled Coin Dispenser and Currency Acceptor devices. If one or more of the MBC Rolled Coin Dispenser cassettes are faulted or the Currency Acceptor is faulted, the terminal is flagged as faulted.

The DCT was not modified to display the Rolled Coin Hopper or Currency Acceptor information for a balance or subtotals function performed from the DCT. Nor was it modified to perform a rolled coin increase/decrease function. These functions are only supported via an administrative card at the terminal.

New Data Files

Not applicable.

Template Changes

COBNAMES

Token ids 'A8', 'A9' and 'AA' were added to the token table. The Merchant Banking Center Transaction Description table was added to change the display on the TLF detail screens for MBC Money Exchange transactions.

The Receipt Table was modified to contain the following new Merchant Banking Center field type codes:

AV	All dispensed denominations/value
AN	All dispensed denominations/number
RV	Requested dispensed denominations/value
RN	Requested dispensed denominations/number
N2	Total amount of notes inserted
ID	Bag ID for a bag deposit transactions

The Admin Receipt Table was modified to contain the following new Merchant Banking Center field type codes:

VC	Amount of cash vaulted by Currency Acceptor
MX	Number of Money Exchange transactions since the last balancing period
AX	Amount of Money Exchange transactions since the last balancing period

The MBC Hopper Currency table was added to support the six MBC roll coin hoppers needed for this enhancement. The default MBC currency code was added to determine the default currency code for the Diebold currency acceptor on screen 16 of the ATD requester.

X100NAMS

This is a new edit file which is used by the BASE24-atm Diebold 1000 series Merchant Banking Center Device Handler process. This file contains a table for the receipt verbiage for the MBC Money Exchange transactions. This table is referenced when the 'TT' field type code is used in a receipt template.

This file also contains the valid denominations table for the rolled coin cassettes. This table represents the Diebold denomination IDs A - H for the rolled coin cassette.

Device Configuration

The CNFGIX (Configuration File for the Diebold 10XX/478X) file was modified to contain a new load group for this enhancement.

Diebold 10XX/478X Configuration File

The Interbold Configuration File (CNFGIX) will require new states and screens to support the Merchant Banking Center transactions. A default set of screens will be provided, but these may require specific modifications depending upon the institutions requirements.

A new load group (600) has been added to the CNFGIX file for Merchant Banking Center support. This load group is built on top of the touch screen load group (006). This solution (utilizing touch screen functionality) was decided to be the preferred method to implement the Merchant Banking Center enhancement. If a customer desires to use MBC with a non-touch screen enabled terminal, the customer will need to modify load group 600 himself, but no changes will be required to the BASE24-atm source code. Please refer to the ***Diebold Merchant Banking Center Terminal Programming Guide (TP-820292)*** for details on formatting the MBC configuration information.

Mega Table (MEGATBL) Changes

Not applicable.

PI-Table Changes

Not applicable.

SEG-Table Changes

Not applicable.

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support screens 14, 15, and 16 in the ATD.

PATHCONF Changes

Not applicable.

CRT Access Security Changes

Access to the new ATD Diebold screens for MBC (screens 14, 15, 16) must be added through the Security screen for the appropriate users.

Routing Changes

Not applicable.

Network Configuration

Not applicable.

File Conversions

The ATD conversion program on the BA60UC04 subvolume has been updated to include the new MBC fields.

Host/Co-Network Impacts

Host code will be required to recognize and process the new Merchant Banking Center (MBC) tokens from the 'Withdrawal', 'Log Only' and 'Deposit' transactions. The Administrative transactions will also contain an MBC token. These tokens will be logged to the TLF and therefore will exist in the TLF Extract.

The Host must be aware of a new 'Log Only' reversal transaction (if these transactions are configured to be sent to the Host). These transactions are generated if there is an error processing a Money Exchange from cash transaction.

User/Operational Impacts

Surcharge Processing

The MBC Money Exchange transactions can be charged a surcharge. This surcharge is maintained via state flow in the CNFGIX file. This fee is maintained only for MBC Money Exchange transactions and is not to be confused with Withdrawal transaction surcharges accrued via settings in the SURF.

Transaction Limits

MBC Money Exchange transactions have limits associated with processing, which are supported via state flow in the CNFGIX file. These limits will not be stored on BASE24 or sent to the device handler. All limit checking is performed within the state flow processing.

There are three limits for the Money Exchange transactions; bill limit, roll limit and total change limit. The bill limit is the maximum number of bills allowed for each bill denomination. The roll limit is the maximum number of rolls allowed for each coin denomination. The total change limit is the maximum amount of change allowed. During the Money Selection process, if the current change mix exceeds the change limit, the bill limit, or the roll limit, the terminal exits from the Money Selection state and the transaction is cancelled.

Disk Impacts

The Transaction Log File (TLF) size will increase with the logging of the new Merchant Banking Center tokens.

Event Messages

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume or the BASE24 Documentation CD-ROM.

Vendor Impacts

This document assumes that the reader is aware of the hardware differences between a Merchant Banking Center terminal and a non-Merchant Banking Center terminal. Consult Diebold for more information.

Reference Documents

Not applicable.

ACI Worldwide Inc.

Dynamic Key Management

This section describes the changes associated with the Dynamic Key Management for BASE24-atm.

Section 2 contains a functional overview of the Dynamic Key Management enhancement.

Standard BASE24-atm modules have been modified to support this enhancement. Those modules are the Diebold 10XX/478X, NCR 5XXX and Triton device handlers.

Prerequisite: The ATD Infrastructure enhancement must be installed prior to implementing the Dynamic Key Management enhancement in the Diebold 10XX/478X or NCR 5XXX device handlers. In the ATD Infrastructure enhancement, the ATD records were expanded and new fields were added to the Dynamic Core data area to support Dynamic Key Management.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true to bind the DKM module to the Diebold 10XX/478X device handler.

- atm_dh_c1000_dkm_on

Set the following flag to true to bind the DKM module to the NCR 5XXX device handler.

- atm_dh_n50_dkm_on

Set the following flag to true to bind the DKM module to the Triton device handler.

- atm_dh_triton_dkm_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. The params are the same for all three device handlers. Different param records may be specified for each device type or device handler process, if desired.

Note that these modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-CONS-MAC-ERR-LMT

The number of consecutive message authentication code (MAC) errors that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32767. A value of 0 indicates that the device handler process does not reload keys regardless of the number of consecutive MAC errors.

ATM-DH-DKM-KEY-CHNG-INTRVL

This param specifies the maximum time allowed between key changes before the device handler downloads a new MAC and PIN key to the ATM. The value is in hours. Valid values are 0 through 32767. A value of 0 indicates that the device handler process does not reload keys regardless of the length of time since the last key change.

ATM-DH-DKM-MAC-ERR-LMT

This param specifies the number of message authentication code (MAC) errors that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32767. A value of 0 indicates that the device handler process does not reload keys regardless of the number of MAC errors.

ATM-DH-DKM-MAC-TXN-LMT

This param specifies the number of message authentication code (MAC) generations or verifications that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32767. A value of 0 indicates that the device handler process does not reload keys regardless of the number of MAC generations or verifications.

ATM-DH-DKM-PIN-ERR-LMT

This param specifies the number of personal identification number (PIN) verification errors that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32767. A value of 0 indicates that the device handler process does not reload keys regardless of the number of PIN verification errors.

ATM-DH-DKM-PIN-TXN-LMT

This param specifies the number of personal identification number (PIN) verifications that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32767. A value of 0 indicates that the

device handler process does not reload keys regardless of the number of PIN verifications performed.

Database Configuration

Not applicable.

New Data Files

Not applicable.

Template Changes

Not applicable.

Device Configuration

Not applicable.

Mega Table (MEGATBL) Changes

Not applicable.

PI-Table Changes

Not applicable.

SEG-Table Changes

Not applicable.

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable.

CRT Access Security Changes

Not applicable.

Routing Changes

Not applicable.

Network Configuration

Not applicable.

File Conversions

The new fields required in the ATD were added as a part of the ATD Infrastructure enhancement. As a part of that enhancement, the ATD records were expanded and the new fields needed for DKM were added to the Dynamic Core data area to support Dynamic Key Management in the ATD. The ATD Infrastructure enhancement must be installed prior to installing the Dynamic Key Management enhancement.

An inline conversion of the ATD and/or TDF will be performed to initialize the new data fields defined for Dynamic Key Management. If a release 5.3 TDF is being converted to release 6.0 version 3, the conversion program will initialize the new DKM data fields in the ATD and/or TDF.

Host/Co-Network Impacts

Not applicable.

User/Operational Impacts

This enhancement will automatically load new PIN and MAC keys to ATM terminals. Messages will be generated when the keys are loaded. Operators must be aware of this new functionality. Key load commands issued by operators will function as before. With this enhancement in place, the counters for transaction and error counts will be reset when an operator-initiated key load is performed.

The Diebold 10XX/478X and NCR 5XXX device handlers currently track consecutive MAC verification errors. Once the limit for the counter is reached, the ATM is shut down. Operator intervention is required to send a new MAC key and reopen the ATM. This enhancement will modify the functionality so the MAC key will automatically be sent (along with the PIN key) and the ATM reopened.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable.

Reference Documents

Not applicable.

Interactive Response Notification Fee Processing and Negative Surcharge Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Interactive Response Notification Fee Processing and Negative Surcharge Support enhancements.

Section 2 contains a functional overview of the Interactive Response Notification Fee Processing and Negative Surcharge Support enhancements.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

For the Interactive Response Notification Fee enhancement:

BASE24 Token File (TKN)

The following new BASE24 Base token has been added for this enhancement:

- Issuer Fee Rebate token (Token ID “30”) contains data relating to the transaction issuer fee or rebate amount assessed. This token is used if issuer fees or rebates are being processed.

If the Issuer Fee Rebate token is not logged to the Transaction Log File (TLF) or Interchange Log File (ILF), the appropriate token records must be configured in the TKN.

If the Issuer Fee Rebate token is to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface process, the appropriate token records must be configured in the TKN.

For the Negative Surcharge enhancement:

BASE24 Surcharge File (SURF)

The following existing fields in the Surcharge File (SURF) must be configured when performing negative surcharges:

Field Name: FLAT FEE
 PERCENT FEE
Screen: SURF screens 2 and 3
DDL Name: SURF.ATM.TRAN.FLAT-FEE
 SURF.ATM.TRAN.PCNT-FEE

Screens 2 and 3 of the SURF contain the surcharge information for transactions routed using the CPF and SPROUTE file, respectively. Either the flat fee, in whole and fractional currency units, can be used as a surcharge, or the percent fee, in whole and fractional units, can be used when calculating the surcharge. A flat fee, percent fee, or both must be defined for each transaction code. When using negative surcharges, a negative value can be entered in these fields. For additional information on these fields, refer to the ***BASE24 Base Files Maintenance Manual***.

The interactive response notification of fees can include both issuer fees and acquirer fees (i.e., surcharges). BASE24 maintains information in the Surcharge File (SURF) only for acquirer fees and does not maintain information in its database for issuer fees. The setting of issuer fees must be done at a host system or an interchange whose BASE24 interchange interface module has been enhanced to carry the issuer fee information in the new BASE24 Issuer Fee Rebate token.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

External Interfaces

The host or co-network must be modified to accept and process the Issuer Fee Rebate token in the ISO message from the BASE24-atm ISO Host Interface (HISO) or BASE24-atm BIC ISO process.

If negative surcharges are being used, the host or co-network must be modified to accept and process negative amounts in the existing Surcharge Data token in the TLF or ILF.

Super Extract Process

If the new Issuer Fee Rebate token is to be included in the TLF or ILF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

If negative surcharges are being used, the host must be modified to accept and process negative amounts in the existing Surcharge Data token in the TLF or ILF.

User/Operational Impacts

Disk Impacts

The TLF and ILF can require additional disk space because the new token for this enhancement (i.e., Issuer Fee Rebate token) can be included in the transaction records logged to these files.

If the TLF and/or ILF is extracted to a disk file, the extract record size is increased. This can increase the disk/tape space required for the extract file.

Operational Impacts

New event messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Multiple Currency Dispense

This section describes the changes, conversion requirements, and implementation requirements associated with the BASE24-atm Multiple Currency Dispense enhancement.

These changes, conversion requirements, and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the BASE24-atm Multiple Currency Dispense enhancement.

The Multiple Currency Dispense enhancement allows cardholders to select the currency to dispense cash in for standard BASE24-atm cash withdrawal transactions (withdrawal from checking account, withdrawal from savings account, withdrawal from credit account). A customer may configure from two to four different currencies that a cardholder can select to withdraw cash in, but only a single currency may be selected and dispensed per withdrawal transaction.

The Multiple Currency Dispense enhancement is supported only in the Diebold 10XX/478X and NCR 5XXX device handlers. It is compatible with the Multiple Currency Support enhancement, but the Multiple Currency Support enhancement is not required. In other words, the Multiple Currency extensions to the BASE24-atm Authorization module are not required in order for the Multiple Currency Dispense enhancement to function.

Prerequisite: The ATD Infrastructure enhancement must be installed prior to implementing the Multiple Currency Dispense enhancement in the Diebold 10XX/478X or NCR 5XXX device handlers. In the ATD Infrastructure enhancement, the ATD records were expanded and new fields were added to the Static Core data area to support Multiple Currency Dispense.

Refer to the **BASE24-atm Diebold 10XX/478X** or **BASE24-atm NCR 5XXX Device Support Manual** for additional information on the Multiple Currency Dispense enhancement.

Refer to the **BASE24-atm Multiple Currency Support Manual** for additional information on the Multiple Currency Support enhancement.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true to bind the Multiple Currency Dispense module to the Diebold 10XX/478X device handler.

- atm_dh_c1000_mcd_on

Set the following flag to true to bind the Multiple Currency Dispense module to the NCR 5XXX device handler.

- atm_dh_n50_mcd_on

LCONF Assign Changes

Refer to the Multiple Currency Support subsection in Section 4 of the **BASE24 Release 6.0 Implementation Guide (Version 2)** for information on the LCONF assigns required. The same assigns used in the Multiple Currency Support enhancement are required for the Multiple Currency Dispense enhancement. Note that these modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

LCONF Param Changes

Refer to the Multiple Currency Support subsection in Section 4 of the **BASE24 Release 6.0 Implementation Guide (Version 2)** for information on the LCONF params required. The same params used in the Multiple Currency enhancement are required for the Multiple Currency Dispense enhancement. Note that these modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

Database Configuration

BASE24 Terminal Data Files (ATD)

Field Name: CURRENCY CODE
Screen: ATD Screens 4 and 5
DDL Name: ATDD1.[atdd1-core].HOPR.CRNCY-CDE
TDF.HOPR.CRNCY-CDE

Screens 4 and 5 of the ATD contain the ISO currency code for each hopper. If the Multiple Currency Dispense enhancement is being used, each hopper's currency code may be set to a different value (with up to a maximum of four different currency codes). For additional information on these fields, refer to the **BASE24-atm Files Maintenance Manual**.

Field Name: CURRENCIES CONFIGURED FOR SELECTION BY THE
CARDHOLDER

Screen: ATD Screen 4

DDL Name: ATDS1.[atds1-core].SELCT-CRNCY-CDE

Screen 4 of the ATD contains four selection ISO currency codes for the Diebold 10XX/478X or NCR 5XXX device handlers. These fields define the selection currencies that are available to the cardholder at the ATM, and are only used on standard cash withdrawals if the Multiple Currency Dispense enhancement is being utilized. If some of the selection currencies are not used, for example, if only two or three different currencies are to be dispensed, the unused selection currency fields may be left blank. In order to properly dispense cash in a Multiple Currency Dispense environment, the selection currency codes must match currencies that are configured in the ATD hoppers. For additional information on these fields, refer to the **BASE24-atm Files Maintenance Manual**.

Field Name: AMT CURRENCY CODE

Screen: ATD Screen 6

DDL Name: ATDD1.[atdd1-core].AMT-CRNCY-CDE
TDF.AMT-CRNCY-CDE

Screen 6 of the ATD contains the terminal's currency code. This value may be set to a different value than the hopper currency codes if the Multiple Currency Dispense enhancement is being used. This value determines the currency that the terminal totals on screen 6 are accumulated in. For additional information on this field, refer to the **BASE24-atm Files Maintenance Manual**.

BASE24-atm Receipt File (RCPT)

The RCPT File can be used to display currency and exchange rate information on a cardholder's receipt. The following new field type codes can be used specifically for the Multiple Currency Dispense enhancement. Note that no default receipt templates have been provided for the Multiple Currency Dispense enhancement.

The CV field type code is used for the equivalent converted amount, i.e., the requested transaction amount converted using the exchange rate value.

The EV field type code is used for the exchange rate value.

The XC field type code is used for the converted currency description.

The XE field type code is used for the exchange rate description.

The XS field type code is used for the selected currency description.

Additional information on the Multiple Currency Dispense specific field type codes may be found in the **BASE24-atm Diebold 10XX/478X** or **BASE24-atm NCR 5XXX Device Support Manual**. Information on use of the RCPT file may be found in the **BASE24-atm Files Maintenance Manual**.

Token File (TKN)

No tokens were added for the Multiple Currency Dispense enhancement. However, the enhancement does use the existing Multiple Currency token, which is described in more detail in the Multiple Currency Support subsection in Section 4 of the **BASE24 Release 6.0 Implementation Guide (Version 2)** and the **BASE24-atm Multiple Currency Support Manual**.

New Data Files

Not applicable

Template Changes

C100NAMS

A table (in a section called `diebold_1000_mult_crncy_tbl`) was added to this file to contain multiple currency dispense verbiage for supporting multiple languages. Customers may need to modify this table to contain the proper verbiage for their language.

N50NAMS

A table (in a section called `ncr_mult_crncy_tbl`) was added to this file to contain multiple currency dispense verbiage for supporting multiple languages. Customers may need to modify this table to contain the proper verbiage for their language.

Device Configuration

The following files were modified to contain new load groups for this enhancement.

Diebold 10XX/478X Configuration File (CNFGIX)

The following Load Group was added for the Multiple Currency Dispense enhancement:

- 901 (Multiple Currency Dispense)

This load group contains states and screens specific to the Multiple Currency Dispense enhancement. It also contains generic currency verbiage screens that will

overlay ones in Load Group 000 that currently state "\$" or "DOLLAR". A customer can modify these screens with more specific currency verbiage if they choose.

A customer must modify the currency selection screen and information entry states within the 901 Load Group to include the specific currency verbiage to be displayed on the currency selection screen (screen 086).

The following Diebold 10XX/478X screens also contain currency specific verbiage (e.g., "\$" or "DOLLAR"), which a customer may wish to modify. These screens have not been modified for the Multiple Currency Dispense enhancement:

<u>Load Group #</u>	<u>Diebold 10XX/478X Screen #</u>
000	074
000	076
000	077
000	078
000	079
000	080
000	177
011	330
011	332
011	333
011	334
011	335
011	336
012	177
013	072
013	074
013	076
013	077
013	079
013	080
013	177

NCR 5XXX Configuration File (CNFG5070)

The following Load Group was added for the Multiple Currency Dispense enhancement:

- 901 (Multiple Currency Dispense)

This load group contains states and screens specific to the Multiple Currency Dispense enhancement. It also contains generic currency verbiage screens that will overlay ones in Load Group 000 that currently state "\$" or "DOLLAR". A customer can modify these screens with more specific currency verbiage if they choose.

A customer must modify the currency selection screen and information entry states within the 901 Load Group to include the specific currency verbiage to be displayed on the currency selection screen (screen 086).

General Device Configuration File Information

To contain information about the currency selected by the cardholder, the Multiple Currency Dispense enhancement makes use of General Purpose Buffer C in the native mode device message. Any other enhancements or CSMs that use General Purpose Buffer C will not be compatible with the Multiple Currency Dispense enhancement. This includes SSB Super Teller transactions and the use of the Dollar Selection Entry state in the Diebold 10XX/478X.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The new fields required in the ATD for the Multiple Currency Dispense enhancement were added as a part of the ATD Infrastructure enhancement. As a part of that enhancement, the ATD records were expanded and the new fields needed for Multiple Currency Dispense were added to the Static Core data area and initialized. The ATD Infrastructure enhancement must be installed prior to installing the Multiple Currency Dispense enhancement. An inline conversion of the ATD will be performed to initialize the new data fields defined for Multiple Currency Dispense.

If a release 5.3 TDF is being converted to release 6.0 version 3, the conversion program will initialize the new Multiple Currency Dispense data fields in the new ATD records.

For more information on file conversions, refer to the AAREADME file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Disk Impacts

The TLF and ILF require additional disk space because the new tokens for this enhancement (i.e., Multiple Currency token and optionally, BASE24-atm Balances token) can be included in the transaction records logged to these files.

If the Multiple Currency Dispense enhancement is used with a BASE24-atm Authorization module that doesn't support the Multiple Currency enhancement, the Multiple Currency token can be included in transaction records logged to the TLF/ILF only for withdrawal from checking, withdrawal from savings, or withdrawal from credit account transactions. These are the only transaction types that are supported for Multiple Currency Dispense.

If a Multiple Currency Authorization is being used, however, the Multiple Currency token can be included in transaction records logged to the TLF/ILF for any

transaction, since the Multiple Currency token is added for any transaction that Multiple Currency Authorization supports, not just standard withdrawals.

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Report Limitations

The set of BASE24-atm reports have not been enhanced to provide any type of multiple currency support. In a multiple currency dispense environment where a non-multiple currency authorization is used, some of the standard reports may still function correctly, with the exception of any reports (such as the terminal settlement reports) where there may be transaction totals in multiple currencies. If there are transactions in multiple currencies in this case, the totals would indicate hash totals only.

In a multiple currency dispense environment where a multiple currency authorization is used, reports are unlikely to function correctly. Additional information may be found in the **BASE24-atm Multiple Currency Support Manual**.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

NCR Bunch Note Acceptor / Check Processing Module Support

This section describes the changes, conversion requirements, and implementation requirements associated with the Bunch Note Acceptor / Check Processing Module Support for the NCR 5XXX Device Handler.

NCR's Bunch Note Acceptor / Check Processing Module (BNA/CPM) provides the terminal with hardware that recognizes documents inserted into the terminal without an envelope (both cash and checks). This support is implemented as part of the BASE24-atm NCR 5XXX SSB product. The BASE24-atm NCR 5XXX SSB product consists of three separate software extensions. The three modules are as follows:

- SSB Device Handler Extension
- SSB Enhanced ATM Application
- SSB Enhanced Check Processing Application

The absence of an envelope depository excludes the possibility of a Payment Enclosed or Message To Institution transaction.

The Force Cutover, Device Control Terminal (DCT), and Card initiated administrative functions were enhanced to include new Bunch Note Acceptor information on balance with standard cash, print subtotals, and balance with current cash transactions. The DCT was not modified to include any new screen information for administrative functions. The new Bunch Note Acceptor information will be displayed in the Transaction Log File (TLF) on a transaction detail screen for the associated administrative transaction.

DCT detailed terminal status screens contain BNA/CPM information in addition to the existing SSB information present.

New Receipt Conditionals were added to allow customers to configure their BNA/CPM Deposit Receipts with BNA/CPM descriptions.

Section 2 contains a functional overview of the NCR Bunch Note Acceptor / Check Processing Module Support Device Handler.

Prerequisite: The ATD Infrastructure enhancement must be installed prior to implementing the BNA/CPM enhancement in the NCR 5XXX device handler. In the ATD Infrastructure enhancement, the ATD records were expanded and new offsets were added to the Dynamic data area.

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable.

LCONF Param Changes

Not applicable.

Database Configuration

This section describes the fields in existing files that have been modified or added to support this enhancement. The setting of each field depends on the configuration desired by the institution.

BASE24 Card Prefix Files (CPF)

The CPF contains new cash deposit fields on BASE24-atm screen #4, [CASH] DEPOSIT CREDIT PERCENT and MAXIMUM CREDIT PER [CASH] DEPOSIT. By default the CPF processing utilizes zero for each of these fields.

BASE24-atm Terminal Data Files (ATD)

ATD configuration generally follows existing guidelines.

BNA/CPM specific configuration:

screen #8

DEPOSITORY (continues to use a value of '0')

screen #12

DEFAULT DEPOSITORY BIN (use '1' through '3')

IMAGE ARCHIVAL (N/A)

COURTESY AMT VERIFICATION (N/A), .

BASE24-atm Receipt Template Files (RCPT)

The Receipt Templates for BNA/CPM transactions use existing transaction templates. Customers can optionally add cash, check, envelope Receipt Conditionals to their templates.

CASH

CHECK

ENVLP

An example receipt template for the deposit receipt is as follows:

```
-----  
BASE24-ATM    RECEIPT    R60D          03/01/23  15:07  01 OF 01  
DEVICE TYPE: N50 (NCR 50XX)          RECEIPT TYPE: DP (DEPOSIT)  
RECEIPT GROUP: ****                  LANG CODE/MSG ID: 0  
FIID:****    FROM ACCT:**    TO ACCT:**    PRNTR TYPE:*(*****)  
#E^4@RECORD NO.#E^F    #SQ>  
?IF CASH  
#E^4@CSH #TT    #A1$>  
?ENDIF  
?IF CHECK  
#E^4@CHK #TT    #A1$>  
?ENDIF  
#E^4@TO#E^5#TO>>  
#E^4#XL#E^6#LB$>  
#E^4#XA#E^6#AB$>  
-----
```

A cash deposit receipt for this template would appear as follows:

```
      A C I    D E M O  
  
DATE          TIME    TERM  
MM/DD/YY    HH:MM    S1AN5070  
  
123456700000003-003  
  
330 SOUTH 108TH AVE  
  
RECORD NO.          572  
CSH DEPOSIT          $100.00  
TO          013333  
LEDGER BAL          $3512.00  
AVAIL BAL          $1760.00
```

BASE24-atm Device Control Terminal (DCT)

The DCT has been modified to display the new status information for the BNA/CPM devices.

The DCT was not modified to display the BNA/CPM information for a balance or subtotals function performed from the DCT. The terminal-maintained BNA counts are logged and viewable via the TLF Perusal.

New Data Files

Not applicable.

Template Changes

Not applicable.

Device Configuration

The CNFG5070 (Configuration File for the NCR 5XXX) file was modified to contain a new load group for this enhancement.

NCR 5XXX Configuration File

The NCR Configuration File (CNFG5070) will require new states and screens to support the BNA/CPM transactions. A default set of screens is provided; these are intended as examples and as such, require modifications depending upon the institutions requirements.

- A new load group (706) has been added to the CNFG5070 file for BNA/CPM support. This load group is built on top of the SSB load group (106). This load group provides an override for DPM, replacing it with CPM support as well as BNA support.
- With the absence of an envelope depository, the Payment Enclosed and Message To Institution transactions have been removed from the transaction menus.
- Additionally, this enhancement introduced new 'M' Screens, new State types and modifications to the Enhanced Configuration Parameters Load message.

Please refer to the ***APTRA Advance NDC Reference Manual (B006-6180)*** for details on the BNA/CPM configuration information.

Mega Table (MEGATBL) Changes

Not applicable.

PI-Table Changes

Not applicable.

SEG-Table Changes

Not applicable.

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable.

CRT Access Security Changes

Not applicable.

Routing Changes

Not applicable.

Network Configuration

Not applicable.

File Conversions

The ATD conversion program on the AT60UC04 subvolume has been updated to include the infrastructure for BNA/CPM ATDD1 device dependent data. The BASE24-atm Authorization utilizes a runtime initialization conversion for the CPF fields (the default values are zero).

Host/Co-Network Impacts

Host code will be required to recognize and process the new BNA/CPM (Cash Acceptor) token from 'Deposit' transactions. The Administrative transactions will also contain a BNA token. These tokens will be logged to the TLF and therefore will exist in the TLF Extract.

User/Operational Impacts

Transaction Limits

BNA Cash Deposit transactions have limits associated with processing, which are defined in the CPF file.

Disk Impacts

The Transaction Log File (TLF) size will increase with the logging of the new BNA/CPM tokens.

Event Messages

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume or the BASE24 Documentation CD-ROM.

Vendor Impacts

This document assumes that the reader is aware of the hardware differences between a BNA/CPM terminal and a non-BNA/CPM terminal. Consult hardware vendors for specifics.

Reference Documents

Not applicable.

Offline PIN Management Support

This section describes the changes associated with the Offline PIN Management Support enhancement for BASE24-atm.

Section 2 contains a functional overview of the Offline PIN Management Support enhancement.

For more detailed information on this enhancement see Section 4 (Base) or the ***BASE24-atm EMV Support Manual***.

Standard BASE24-atm modules have been modified to support this enhancement. Those modules are ATM Authorization and the NCR 5XXX Device Handler.

Prerequisite: The ATD Infrastructure enhancement must be installed prior to implementing the Offline PIN Management Support enhancement in the NCR 5XXX device handler. In the ATD Infrastructure enhancement, the ATD records were expanded and new offsets were added.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementing the Offline PIN Management Support enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Note that these modifications will be provided in the LCONFAT file located on the BA60MISC subvolume.

EMV-PIN-UNBLOCK-SCRIPT

Specifies the hexadecimal representation of binary data that is used to reset the offline bad PIN tries accumulator on a chip card. It includes both the Script header and the Script command. This param also provides the “template” used by BASE24-atm when generating a script to change the offline PIN on a chip card. This param is required if the EMV extensions are present and initialization fails if the param is missing.

Database Configuration

Token File (TKN)

The following BASE24 Base token has been added for this enhancement:

- The EMV Issuer Script Results Token (Token ID “BJ”) has been added and is used by the NCR 5XXX ATM Device Handler and the VisaNet and LINK Interchange Interfaces. The EMV Issuer Scripts Results Token contains information about the processing of EMV Script data. This token will be created by the acquirer process (e.g. device handler or interchange). It will contain information about the results of EMV script processing.

If the token for this enhancement is not to be logged to the TLF or ILF, the appropriate token records must be configured.

If the token for this enhancement is to be extracted with the TLF or ILF, sent to the host (via HISO), or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

Not applicable.

Template Changes

Not applicable.

Device Configuration

Where they are directly connected to BASE24, ATMs supporting offline PIN changes must be compliant with NCR’s CAM 2 implementation:

- Customers will require an NCR ATM type 56xx running NDC+ 06.04.00 and NDC CAM 2 FS Version 2.14 Software.
- In order to ensure that the N50 device handler receives ‘Last Transaction Status Information’ in transaction request messages, the enhanced configuration option ‘15’ should be set to ‘001’ at the ATM.

NCR 5XXX Configuration File (CNFG5070)

The NCR 5XXX Configuration File (CNFG5070) requires new states and screens to support the new EMV PIN Unblock transaction type. For more detailed information, see the CNN50DOC file on the AT60N50 subvolume. A default set of screens is provided, but these may require specific modifications depending upon the institution’s requirements.

Mega Table (MEGATBL) Changes

Not applicable.

PI-Table Changes

Not applicable.

SEG-Table Changes

Not applicable.

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable.

CRT Access Security Changes

Not applicable.

Routing Changes

Not applicable.

Network Configuration

Not applicable.

File Conversions

Not applicable.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interchange Interface, ISO Host Interface and Super Extract processes.

BIC ISO Interchange Interface Process

PIN Unblock transactions are supported with a new ISO processing code of "970000" in BASE24 external messages. Hosts should be analyzed for any impacts that may result from receiving the new transaction codes.

In approved PIN management transaction responses, the EMV Response Token and EMV Script token (containing a 'PIN Change' or 'PIN Unblock' script) may be included in the BASE24 external message. When the HISO and BICISO modules send a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token. When the HISO and BICISO modules receive a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token.

If the new EMV Issuer Script Results Token is configured to be included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

ISO Host Interface Process

PIN Unblock transactions are supported with a new ISO processing code of "970000" in BASE24 external messages. Hosts should be analyzed for any impacts that may result from receiving the new transaction codes.

In approved PIN management transaction responses, the EMV Response Token and EMV Script token (containing a 'PIN Change' or 'PIN Unblock' script) may be included in the BASE24 external message. When the HISO and BICISO modules send a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token. When the HISO and BICISO modules receive a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token.

If the new EMV Issuer Script Results Token is configured to be included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Super Extract Process

The new EMV PIN Unblock transactions are supported with an internal transaction code of "820000". Hosts should be analyzed for any impacts that may result from receiving TLF Extract records containing the new transaction code.

If the PIN Change Token (containing new PIN data) and/or new EMV Issuer Script Results Token is configured to be included in the TLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TLF and ILF require additional disk space if the PIN Change Token (containing new PIN data) and/or new EMV Issuer Script Results Token is configured to be included in the transaction records logged to these files. The total impact varies for each BASE24 user depending on the transaction volume in the BASE24 system.

User Impacts

The following configuration requirements apply when the PIN Change transaction is authorized on BASE24-atm:

- The PIN verification method must be IBM 3624-format or Visa PVV.
- The BASE24-atm authorization method must be Positive or Positive Balance, so the cardholder has a CAF record.
- PIN verification digits must be carried in the CAF.

Changes have been made to the TLF perusal and reporting modules to support the new transaction code of '820000' for EMV PIN Unblocks.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume or the BASE24 Documentation CD-ROM.

Performance Impacts

If BASE24 is configured to generate the 'PIN Change' and/or 'PIN Unblock' script command, then the additional HSM command will have an impact on system performance.

Vendor Impacts

Not applicable.

Reference Documents

- NCR: NDC+ CAM 2 Functional Specification, version 2.14, dated 14 November 2002.

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Transaction Security Services Multiple Logical Network Support

This section describes the changes associated with the Transaction Security Services Multiple Logical Network Support enhancement for BASE24-atm.

Section 2 contains a functional overview of the Transaction Security Services Multiple Logical Network Support enhancement.

For more detailed information on this enhancement see Section 4 (Base).

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement.

MULT-LN-PIN-KEY

This param points to the common PIN encryption key used to re-encrypt PINs on transactions passed between LNs in a multiple-LN BASE24 system. It must be set at the logical network level (i.e., with a 'used by' process name of all asterisks).

For LNs that use TSS, the param must contain a sixteen-digit hexadecimal key locator that points to a TSS ISECD (Interchange Security Data Source) record that in turn contains the actual encrypted PIN key.

For LNs that don't use TSS, the param must contain the actual sixteen-digit hexadecimal *encrypted* PIN key. The key has to be of single length because the BASE24 Security Utilities (SECUTILS) don't support longer keys for the non-TSS security device types.

This restriction means that, for BASE24 systems that contain one or more non-TSS LNs, the common key used by all LNs has to be of single length. If all LNs use TSS, however, either a single-, double-, or triple-length common key can be used.

Finally, if a Thales security device is in use, the common PIN key must be a Zone PIN Key (ZPK). A Terminal PIN Key (TPK) can't be used since no Thales PIN translation commands translate to TPKs.

Database Configuration

Interchange Security Data Source (ISECD)

A new TSS ISECD record must be added for each LN that uses TSS in a multiple-LN system. This record, which is pointed to by a corresponding MULT-LN-PIN-KEY LCONF param, should contain the common PIN key shared by all LNs.

If the system contains at least one LN that doesn't use TSS, this record should be configured with a single-length PIN encryption key. If all LNs use TSS, however, either a single-, double-, or triple-length key can be used.

If a Thales security device is in use, the common PIN key must be a Zone PIN Key (ZPK). A Terminal PIN Key (TPK) can't be used since no Thales PIN translation commands translate to TPKs.

When adding the ISECD common key record, only the following elements need to be entered. All other fields can be left empty or set to their defaults.

GUI Label: Interface Name
Element: INTF_NAM

Must contain the same value as the key locator specified in the MULT-LN-PIN-KEY LCONF param.

GUI Label: PIN Block Format
Element: PIN_BLK_TYP

Must be set to "ANSI PIN". The special PIN translation always re-encrypts the PIN as an ANSI PIN block.

GUI Label: Outbound Keys (Slot 1) Parts 1, 2, 3
Element: OUTBND_KPT1_KEYX

Must contain the common PIN encryption key. If an Atalla AKB security device is used, all three key parts must be filled. If a non-AKB Atalla security device is used, only the first part should be filled. If a Thales device is used, one, two, or three parts can be filled.

GUI Label: Outbound Current Index
Element: OUTBND_KPT1_IDX

Must contain a value of 1.

GUI Label: Inbound Keys (Slot 1) Parts 1, 2, 3
Element: INBND_KPT1_KEYX

Must contain the common PIN encryption key. If an Atalla AKB security device is used, all three key parts must be filled. If a non-AKB Atalla security device is used, only the first part should be filled. If a Thales device is used, one, two, or three parts can be filled.

GUI Label: Inbound Current Index
Element: INBND_KPT1_IDX

Must contain a value of 1.

GUI Label: Outbound Keys (Slot 1) Header
Element: OUTBND_KPT1_HDR

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB header. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Outbound Keys (Slot 1) MAC
Element: OUTBND_KPT1_KAM

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB MAC value. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Outbound Key Length
Element: OUTBND_KPT_LGTH

Must contain "Single", "Double", or "Triple" depending on the size of the common key entered in OUTBND_KPT1_KEYX.

GUI Label: Inbound Keys (Slot 1) Header
Element: INBND_KPT1_HDR

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB header. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Inbound Keys (Slot 1) MAC
Element: INBND_KPT1_KAM

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB MAC value. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Inbound Key Length
Element: INBND_KPT_LGTH

Must contain "Single", "Double", or "Triple" depending on the size of the common key entered in INBND_KPT1_KEYX.

GUI Label: External Key Block Format
Element: EXTRN_KEY_BLK_FRMT

If a Thales security device is used, this field should be set to "Variant method". If an Atalla security device is used (AKB or otherwise), the field should be set to "Standard Atalla".

New Data Files

Not applicable.

Template Changes

Not applicable.

Device Configuration

Not applicable.

Mega Table (MEGATBL) Changes

Not applicable.

PI-Table Changes

Not applicable.

SEG-Table Changes

Not applicable.

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable.

CRT Access Security Changes

Not applicable.

Routing Changes

Not applicable.

Network Configuration

Not applicable.

File Conversions

Not applicable.

Host/Co-Network Impacts

Not applicable.

User/Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume or the BASE24 Documentation CD-ROM.

Vendor Impacts

Not applicable.

Reference Documents

Not applicable.

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Section 6

BASE24-pos

Section 6 discusses the changes and enhancements made to BASE24-pos in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing BASE24-pos users must consider when upgrading from release 6.0 version 2.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-pos users, along with the associated conversion and implementation requirements that all BASE24-pos users must address.
- Optional enhancements describe the changes that affect only those BASE24-pos users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-pos users must address to implement the enhancements.

Section 6 must be used in conjunction with Section 4 to identify the changes that BASE24-pos users need to consider when upgrading to release 6.0 version 3.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-pos users must address in order to upgrade to release 6.0 version 3.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

1. When an optional enhancement is developed, it is usually developed as a bindable extension on its own subvolume (for example, the SPDH device handler's EMV extension on PS60ISPD). The main software module (PS60SPDH.ASPDHS) contains calls (hooks) to procedures that reside in the extension. Dummy procedures are used to allow the main module to compile cleanly when the optional enhancement is not purchased or is otherwise deactivated. In the past, these dummy procedures resided in a separate dummy file for each enhancement, or within the main module itself. (Note that these older dummy procedures will not be relocated at this time.) Beginning with version 3 of release 6.0 and going forward, certain modules will now employ a single "common" dummy file that will contain all dummy procedures for all new enhancements. The table below lists each module and its new dummy file.

Module	New Dummy File
ACI Standard POS Device Handler	ps60spdh.spdhd

2. The BASE24-pos ACI Standard POS Device Handler (SPDH) and the BASE24-pos ISO Host Interface have been enhanced to allow balance inquiry transactions to be reversed if they do not complete as intended. This is controlled via an LCONF param and is documented in the LCONF Param Changes section.
3. The BASE24-pos ACI Standard POS Device Handler (SPDH) can be configured to return the available balance to the terminal for all transactions if it is available. This is controlled via a new flag on the BASE24-pos Terminal Data files (PTD) and by the existence of the POS Balance token. This token was added with the Multiple Currency Support enhancement. Because of the use of the token, the functionality of returning the balance to the terminal cannot be processed with the Multiple Currency add-on product.

If the PTD flag is set to return the available balance to the terminal, the SPDH will add the POS Balance token to the internal request message. The issuer will update the token's TXN-AMT field with the available balance. When formatting the response message to the terminal, the SPDH will populate FID J with TXN-AMT-1 from the POS Balance token if the token exists, the FID is configured to be sent on the transaction response, and the PTD Balance Returned flag is set to Y.

4. A BASE24-pos ACI Standard POS Device Handler (SPDH) FID has been added to send the ISO response code to the terminal. This is documented in the Vendor Impacts section.

5. The BASE24-pos ACI Standard POS Device Handler (SPDH) has been enhanced to map BASE24 response codes to ISO response codes when BASE24 is the issuer. This is controlled via a table in a new file that is compiled into the executable object. The ISO response code is placed in the interchange response value field in the PSTM if it is not already populated. Until the table is updated with valid ISO response codes, the PSTM.ICHG-RESP will always be set to blanks. The table is called b24_pos_resp_tbl and is in the file SPDHNAMS.

6. A BASE24-pos file has been added called the Response Code Description File (RCDF). RCDF records define the response code description for each terminal group, FIID, language, record type (ISO or BASE24) and response code combination. The BASE24-pos ACI Standard POS Device Handler (SPDH) is the only process that uses the information in the file. The response code description can be configured to be sent to the terminal on a transaction response.

Information from the RCDF is used by the BASE24-pos SPDH process in the Response Code Description File Extended Memory Table file (RCDFEMT). Any time a change is made to the RCDF, the RCDFEMT should be rebuilt using the Extended Memory Table Build utility and reallocated (warmbooted) to processes that access the table using the EMT Control Commands screen or text commands entered from a network control facility.

The information in the RCDF is verified against the data in the Account Type Table File (ATT). The ATT contains the account types for the ISO account types displayed in the ACCOUNT TYPE field.

The key to RCDF records is a combination of the data entered in the TERMINAL GROUP, FIID, LANGUAGE INDICATOR, RECORD TYPE, ACCOUNT TYPE, and RESPONSE CODE fields.

7. The BASE24-pos Router/Authorization process has been enhanced to not send a response message to the originator when it receives a chargeback request (0402) message. This is controlled via a new LCONF param and is documented in the LCONF Param Changes section.
8. The BASE24-pos ISO Host Interface process has been enhanced to place all transactions in the Store-and-Forward file (SAF) prior to sending them to the host. This is controlled via an LCONF param and is documented in the LCONF Param Changes section.
9. The BASE24-pos Router/Authorization process has been enhanced to not update the PSTM.FRWD-INST-ID-NUM if the field is not equal to blanks or zeroes.

10. An administrative transaction has been added to the BASE24-pos ACI Standard POS Device Handler (SPDH) so the terminal can request new keys. This transaction is called the New Key transaction.
11. BASE24-pos has been enhanced to support generating chargeback transactions for pre-authorization completion transactions in which a hold was not found, the hold has expired, or the completion amount is greater than the hold amount. This is controlled via a new flag on the BASE24-pos Terminal Data files (PTD) and the existence of a new token.

If the PTD flag is set to generate chargebacks for pre-authorization completion transactions, the SPDH will add the POS Merchant token to the internal request message with a "Y" in the PRE-AUTH-CHRGBK field. When Router/Authorization processes a pre-authorization completion transaction, if the PRE-AUTH-CHRGBK field in the POS Merchant token is a "Y" and the hold cannot be found, the hold has expired or the completion amount is greater than the hold amount, a chargeback transaction is generated.

12. The BASE24-pos ACI Standard POS Device Handler process has been enhanced to send MAC verification failures as '9991' – Log Only messages to Router/Authorization to be logged to the PTLF. This is controlled via a new LCONF param and is documented in the LCONF Param Changes section.
13. New fields were added to the BASE24-pos Terminal Data (PTD) File DDL and screen that are used for Canadian-specific processing. These fields fall under the heading of Pass-Through Parameters.
14. The BASE24-pos Authorization process has been enhanced to support a cash back amount in preauthorization purchase and preauthorization purchase completion transactions. This transaction will use the same transaction code as the preauthorization purchase and preauthorization purchase completion codes. If the transaction is received then a check will be made to the IPCF to verify a purchase with cash back is allowed.

The entire transaction amount (purchase plus cash back) is to be carried in the PSTM.TRAN.AMT-1 field. The cash back amount is carried in the PSTM.TRAN.AMT-2 field. If a CAF is present only debit accounts/cards are allowed.

Preauthorization for a lesser amount has not been enhanced for preauthorization purchase with cash back transactions. This implies that preauthorization purchase with cash back transactions will be denied, instead of having the transaction amount reduced, if the transaction amount is greater than the applicable transaction limit value.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this release. These modifications will be provided in the LCONFPS file located on the BA60MISC subvolume.

RCDF

The fully qualified file name of the Response Code Description File. The RCDF contains one record for each combination of terminal group, FIID, language, record type and response code that is supported by a BASE24-pos ACI Standard POS Device Handler module in the system. This is a required assign for the RCDF Server object and the RCDFEMT Build utility object.

RCDFEMT

The fully qualified file name of the Response Code Description File Extended Memory Table file. This is a required assign for the RCDFEMT Build utility object and the BASE24-pos ACI Standard POS Device Handler modules.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this release. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-CHRGBCK-NO-RESP-ORIG

A param containing a value indicating an originator that chargeback transaction response messages will not be sent to. If the PSTM ORIG field is equal to the value in this LCONF parameter on a 0402 transaction then no response will be sent to the originator. The BASE24-pos Router/Authorization process reads this param. There is no default value.

POS-LOG-MAC-ERR

Indicates whether MAC verification errors will be logged to the PTLF. A value of Y indicates that MAC verification errors will be logged to the PTLF. The default is N. The BASE24-pos ACI Standard POS Device Handler process reads this param.

RCDFEMT-MAX-TBL-SIZE

The maximum size (in megabytes) of the RCDFEMT. This param is used to set the amount of memory allocated by the Extended Memory Table Build utility for the table. The maximum size allowed for the table is the default value of 127 MB.

RCDFEMT-WARM*Identifier*

A param containing the symbolic name of a process to be warmbooted for the RCDFEMT. Processes that can be warmbooted for the RCDFEMT are the BASE24-pos ACI Standard POS Device Handler processes.

This param indicates a process that receives an EMT warmboot command from the Device Control Terminal (DCT). The Device Control Terminal Server process retrieves all RCDFEMT-WARM*Identifier* params when a network control operator issues a command to reinitialize this specific EMT. A maximum of 500 processes can be EMT warmbooted from the DCT.

Each RCDFEMT-WARM param contains the name of one process. When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number, but can be any unique set of characters that has not already been declared by another RCDFEMT-WARM*Identifier* param. For example, RCDFEMT-WARM01 could contain the P1A^SPDH^01 process and RCDFEMT-WARM02 could contain the P1A^SPDH^02 process.

REVERSE-BAL-INQ

A code indicating whether balance inquiry transactions should be reversed if they do not complete as intended. A value of True, T, or 1 indicates that balance inquiry transactions can be reversed. A value of False, F, or 0 indicates that balance inquiry transactions cannot be reversed. The default is False. This param already exists in the BASE24 product. The BASE24-pos SPDH and BASE24-pos HISO processes were added to the list of processes that read the param.

SAF-PRIOR-TO-SEND

Indicates whether all transactions will be placed on the SAF prior to sending them to the Host. A value of Y indicates all transactions will be placed on the SAF prior to sending them to the Host. The default is N. This param is used by the BASE24-pos ISO Host Interface process.

Database Configuration

This section describes the new fields in the files that have been modified to support this release. The settings of each field depend on the configuration desired by the institution.

BASE24-pos Terminal Data files (PTD)

Field Name: BALANCE RETURNED
Screen: PTD Screen 2
DDL Name: PTDS1-CORE.BAL-RTRN-FLG

This field indicates whether the available balance will be returned to the terminal for all transactions. Valid values are Y (Yes, the available balance is returned to the terminal) and N (No, the available balance is not returned to the terminal). The default is N.

Field Name: PREAUTH CHARGEBACKS
Screen: PTD Screen 6
DDL Name: PTDS1-CORE. PRE-AUTH-CHRGBCK-FLG

This field indicates whether the chargeback transactions can be generated for pre-auth completion transactions from this terminal if the hold cannot be found, the hold has expired or the completion amount is greater than the hold amount. Valid values are Y (Yes, chargebacks will be generated) and N (No, chargebacks will not be generated). The default is N.

Field Name: DESTINATION DPC
Screen: PTD Screen 6
DDL Name: PTDS1-CORE. DPC-NUM

The DPC number of the accepting entity to route transactions to for POS pass-through processing. Valid values are "0000" - "9999". Default value is "0000".

Field Name: RECONCILIATION ENTITY
Screen: PTD Screen 6
DDL Name: PTDS1-CORE. DPC-NUM

The type of entity that will be used to reconcile for POS pass-through processing.

For additional information on the usage of these fields, refer to the **BASE24-pos Files Maintenance Manual**.

ACI Standard POS Device Handler Configuration File (ACNF)

Field Map records must be configured for the New Key administrative transaction supported by the SPDH for the terminals configured to send the New Key transaction. This transaction type does not have any required FIDs that need to be configured.

SPDH FID J – Available Balance must be added in the Field Map records for the transaction types that the available balance can be returned to the terminal for if returning the available balance is configured.

Refer to the ***BASE24-pos Standard POS Device Configuration Manual*** for additional information on setting up Field Map Records.

Token File (TKN)

The following BASE24-pos token has been added for the generation of chargeback transactions:

- The POS Merchant Token (Token ID “CI”) contains the data needed for transaction processing.

The functionality of returning the available balance to the terminal uses the existing POS Balance token to store the available balance.

If the BASE24-pos POS Merchant or the POS Balance token is not to be logged to the PTLF or ILF, the appropriate token records must be configured in the TKN.

If the BASE24-pos POS Merchant token or the POS Balance token is to be extracted with the PTLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured in the TKN.

New Data Files

This section describes the files that have been added to support this release.

Response Code Description File (RCDF)

The RCDF was created for storing the response code description for ISO and BASE24 response codes. Through configuration of RCDF records, customers define the response code descriptions that can be sent to terminals driven by the SPDH. Terminals can share common response code descriptions based on terminal group and FIID from the POS Terminal Data File (PTD). Refer to the ***BASE24-pos Files Maintenance Manual*** for additional information on configuring RCDF records.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The RCDF requester name was added to MEGATBL file, since RQMEGA communicates with the RCDF requester using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the screens for the RCDF requester/server.

PATHCONF Changes

The RCDF Server object is bound into the Classic Combined AFM Link Object (CCAFMLO).

CRT Access Security Changes

Access to the new RCDF requester screens must be added through the SEC screen for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

BASE24-pos Terminal Data files (PTD)

An inline conversion of the PTD will be performed to initialize the new data fields defined for this release to default values. If a release 5.3 file is being converted to release 6.0 version 3, the conversion programs on the PS60UC04 will initialize the new data fields in the PTD.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this release. Modifications to existing files and processing impact the Super Extract process.

Super Extract Process

Host code changes are required for all BASE24 users that utilize the OMF Extract process to accept and process information in the RCDF and PTD files.

If the POS Merchant token or the POS Balance token is to be included in the PTLF and ILF extracts, BASE24-pos users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The new RCDF and RCDFEMT impacts disk requirements. The total impact varies for each BASE24 user depending on the number of RCDF records defined in the BASE24 system.

The PTLF and ILF require additional disk space because the POS Merchant token and the POS Balance token can be included in the transaction records logged to these files.

The PTLF requires additional disk space if MAC verification failures are configured to be logged for the SPDH.

Operational Impacts

BASE24 operators must be aware that the new RCDF file will appear on the BASE24 POS Product Menu.

New event (log) messages were created as result of these general changes. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message

documentation files on \$<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

ACI Standard POS Device Handler (SPDH) Message Format

The following modifications have been made to the SPDH Message format for this release:

- SPDH optional data field X has been added to provide the ISO response code.

The following FID is available for standard processing:

FID	Picture	Length	Field Name	RQST	RESP
X	PIC 9(3)	3 bytes	ISO Response Code		✓

- SPDH optional data field g – response display has been enhanced to contain either the ISO response code display data or the BASE24 response code display data. If a response code exists in the PSTM ICHG-RESP field and an ISO RCDF record is found that matches the PSTM ICHG-RESP field, the ISO RCDF description is sent in the FID. If there is no value in the PSTM ICHG-RESP field or an ISO RCDF record is not found, the RCDF is searched for a BASE24 record that matches the BASE24 response code in the standard header. If a BASE24 RCDF record is found that matches the BASE24 response code, the BASE24 RCDF description is sent in the FID. If neither an ISO nor a BASE24 RCDF record is found that matches the response code for the transaction, the ARSP is used to retrieve the display data for the FID.

The following FID is available for standard processing:

FID	Picture	Length	Field Name	RQST	RESP
g	PIC X(48)	1 to 48 bytes	Response Display		✓

- The standard message header transaction type field must be set to a 96 for a New Key administrative transaction request message.

Refer to the ***ACI Standard POS Device Message Specifications Manual*** for additional information.

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24-pos enhancements that can be enabled or disabled with release 6.0 version 3, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- ACI Standard POS Device Handler Data Encryption
- ACI Standard POS Device Handler Dynamic Key Management
- CRT Authorization TQF Token Buffer Space Expansion
- Discretionary Data Suppression
- Enhanced Pre-Authorization Support
- Secure Internet Validation
- Transaction Security Services Multiple Logical Network Support

BASE24-pos users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements for future reference.

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ACI Standard POS Device Handler Data Encryption

This section describes the changes, conversion requirements, and implementation requirements associated with the ACI Standard POS Device Handler Data Encryption enhancement.

These changes, conversion requirements, and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the ACI Standard POS Device Handler Data Encryption enhancement.

This enhancement allows the available balance to be encrypted when sent to the terminal from the ACI Standard POS Device Handler (SPDH) on a transaction response. If the balance is to be returned to the terminal encrypted for any transaction type other than a balance inquiry, additional configuration is required. Refer to the General Changes listed in the General Enhancements section of this document for information on configuring available balances to be returned to the terminal for the SPDH.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the ACI Standard POS Device Handler Data Encryption enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

Set the following flag to true to bind the Data Encryption module to the SPDH.

- pos_dh_spdh_de_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field depend on the configuration desired by the institution.

BASE24-pos Terminal Data files (PTD)

Field Name: DATA ENCRYPTION TYPE

Screen: PTD Screen 7

DDL Name: PTDD1-CORE.DATA-ENCRYPT-TYP

A code indicating whether data encryption is used for the terminal and what form the encrypted data is sent to the terminal. Data encryption is only supported with Transaction Security Services (TSS). The TSS Channel Security File (CSECD) contains the actual encryption method used.

For additional information on the usage of this field, refer to the ***BASE24-pos Files Maintenance Manual***.

Note: The key locator for the data encryption key which is used as the key to access the CSECD will be maintained in the PIN MASTER KEY field on screen 7.

Channel Security File (CSECD)

For configuration information on the CSECD fields added for this enhancement, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Token File (TKN)

The following BASE24-pos token has been added for this enhancement:

- The Data Encryption Key token (token ID “BN”) contains the data encryption information needed for transaction processing.

If the token for this enhancement is not to be logged to the PTLF or ILF, the appropriate token records must be configured in the TKN.

If the token for this enhancement is to be extracted with the PTLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate TKN records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

To implement this enhancement, BASE24 must have access to a host security module. The enhancement supports the use of an Atalla security module. Refer to the Vendor Impacts subsection of this enhancement for further information.

File Conversions

Customers that have already implemented TSS will need to convert the CSECD before implementing this enhancement. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information on converting the file.

An inline conversion of the PTD will be performed to initialize the new data fields defined for Data Encryption. If a release 5.3 PTDF file is being converted to release 6.0 version 3, the conversion programs on the PS60UC04 will initialize the new Data Encryption fields in the PTD.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to the PTD records.

If the Data Encryption Key token is to be included in the PTLF and ILF Extracts, BASE24-pos users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The PTLF and ILF require additional disk space because the Data Encryption token can be included in the transaction records logged to these files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

ACI Standard POS Device Handler (SPDH) Message Formats

SPDH optional data field I has been added to provide the data encryption key.

The following FID is available for standard processing:

FID	Picture	Length	Field Name	RQST	RESP
I	PIC X(16)	16 bytes	Data Encryption Key		✓

Host Security Module Requirements

To implement this enhancement, BASE24 must have access to a host security module. The enhancement supports the use of the Atalla security module. The security module must have a version of software that is capable of supporting the Transaction Security Services enhancement.

- For Atalla HSMs, version 2.9 software and AKB version 1.0 are supported.

Reference Documents

Not applicable

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ACI Standard POS Device Handler Dynamic Key Management

This section describes the changes, conversion requirements, and implementation requirements associated with the ACI Standard POS Device Handler Dynamic Key Management enhancement.

These changes, conversion requirements, and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the ACI Standard POS Device Handler Dynamic Key Management enhancement.

Dynamic key management (DKM) can be performed by the ACI Standard POS Device Handler (SPDH) for the PIN encryption key, the MAC encryption key and the data encryption key. For information on data encryption with the SPDH, refer to the ACI Standard POS Device Handler Data Encryption enhancement.

CUSTMACS Changes

Set the following flag to true to bind the DKM module to the SPDH.

- pos_dh_spdh_dkm_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

ACI Standard Device Configuration File (ACNF)

Field Name: INCLUDE KMAC IN MAC
Screen: ACNF Screen 2
DDL Name: ACNF.PRO-DATA.FLAGS

This field indicates whether the MAC in messages exchanged between the SPDH module and the terminal includes the MAC encryption key (KMAC). Valid values are Y (Yes, include the KMAC when generating

the MAC) and N (No, do not include the KMAC when generating the MAC). The default is N.

Field Name: INCLUDE KME IN MAC
Screen: ACNF Screen 2
DDL Name: ACNF.PRO-DATA.FLAGS

This field indicates whether the MAC in messages exchanged between the SPDH module and the terminal includes the data encryption key (KME). Valid values are Y (Yes, include the KME when generating the MAC) and N (No, do not include the KME when generating the MAC). The default is N.

Field Name: INCLUDE KPE IN MAC
Screen: ACNF Screen 2
DDL Name: ACNF.PRO-DATA.FLAGS

This field indicates whether the MAC in messages exchanged between the SPDH module and the terminal includes the PIN encryption key (KPE). Valid values are Y (Yes, include the KPE when generating the MAC) and N (No, do not include the KPE when generating the MAC). The default is N.

Field Name: MAC KEY THRESHOLD
Screen: ACNF Screen 3
DDL Name: ACNF.PRO-DATA.MAC-TXN-LMT

This field defines the maximum number of transactions involving a MAC that can be performed before a MAC key exchange is required. Valid values are 0 - 9999. The default is 0.

Field Name: MAC KEY ERROR THRESHOLD
Screen: ACNF Screen 3
DDL Name: ACNF.PRO-DATA.MAC-ERR-LMT

This field defines the maximum number of transactions resulting in MAC errors that can be performed before a MAC key exchange is required. Valid values are 0 - 9999. The default is 0.

Field Name: CONSECUTIVE MAC KEY ERROR THRESHOLD
Screen: ACNF Screen 3
DDL Name: ACNF.PRO-DATA.CNSC-MAC-ERR-LMT

This field defines the maximum number of consecutive transactions resulting in MAC errors that can be performed before a MAC key exchange is required. Valid values are 0 - 9999. The default is 0.

Field Name: PIN KEY THRESHOLD
Screen: ACNF Screen 3
DDL Name: ACNF.PRO-DATA.PIN-TXN-LMT

This field defines the maximum number of transactions involving a PIN that can be performed before a PIN key exchange is required. Valid values are 0 - 9999. The default is 0.

Field Name: PIN KEY ERROR THRESHOLD
Screen: ACNF Screen 3
DDL Name: ACNF.PRO-DATA.PIN-ERR-LMT

This field defines the maximum number of transactions resulting in PIN key sync errors that can be performed before a PIN key exchange is required. Valid values are 0 - 9999. The default is 0.

Field Name: DATA KEY THRESHOLD
Screen: ACNF Screen 3
DDL Name: ACNF.PRO-DATA.DATA-TXN-LMT

This field defines the maximum number of transactions involving data encryption that can be performed before a data encryption key exchange is required. Valid values are 0 - 9999. The default is 0.

Field Name: DATA KEY ERROR THRESHOLD
Screen: ACNF Screen 3
DDL Name: ACNF.PRO-DATA.PIN-ERR-LMT

This field defines the maximum number of transactions resulting in data encryption key sync errors that can be performed before a data encryption key exchange is required. Valid values are 0 - 9999. The default is 0.

For additional information on the usage of these fields, refer to the **BASE24-pos Standard POS Device Configuration Manual**.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

An inline conversion of the PTD and ACNF will be performed to initialize the new data fields defined for Dynamic Key Management. If a release 5.3 file is being converted

to release 6.0 version 3, the conversion programs on the PS60UC04 will initialize the new DKM data fields in the PTD and ACNF.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

The LCONF param POS-DH-CONS-MAC-ERR-LMT can still be used to define the number of consecutive MAC errors that can occur before the device handler will load a new MAC key to the device. However, if this enhancement is enabled the value in the ACNF CONSECUTIVE MAC KEY ERROR THRESHOLD field will be used instead of the value from the LCONF param.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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CRT Authorization TQF Token Buffer Space Expansion

This section describes the changes, conversion requirements, and implementation requirements associated with the CRT Authorization TQF Token Buffer Space Expansion enhancement.

These changes, conversion requirements, and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the CRT Authorization TQF Token Buffer Space Expansion enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that can be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

CRT-AUTH-TQFT

The fully qualified file name of the Terminal Queue Token File (TQFT). This is an optional assign and is only needed if tokens are to be written to (and retrieved from if a reversal must be sent) the new TQFT instead of the existing Terminal Queue File (TQF).

LCONF Param Changes

This section describes the param records that can be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

CRT-AUTH-MAX-TQFTS

This param indicates the maximum number of TQFT file records that can be held in extended memory by the BASE24-pos CRT Authorization process. It is an optional parameter. If the TQFT is available for processing and this param is not present or has an invalid value (i.e., less than or equal to zero or greater than 32766) it will use a default value of 75.

CRT-AUTH-TKN-RETRV-OPT

This param indicates where the token data will be retrieved from if a reversal needs to be sent. Valid values are 0 (no tokens will be included on a reversal), 1 (token data will be retrieved from either the TQF or the TQFT on a reversal), 2 (tokens will be retrieved from the PTLF on a reversal). It is an optional param with a default value of 1. If it is set to 1 and the TQFT assign was successfully retrieved, the TQFT will be used. If it is set to 1 and the TQFT assign was not successfully retrieved (or does not exist) the TQF will be used.

Database Configuration

Not applicable

New Data Files

The TQF Token File (TQFT) contains one record for each Pathway terminal and transaction associated with the transaction in the transaction table of the TQF. The TQFT is used to store the token data that existed in the BASE24-pos Standard Internal Message (PSTM) for the specific transaction in the queue processed by CRT Authorization and acquired at this terminal. The use of the TQFT is optional, dependant upon customer configuration of where token data should be stored for CRT Authorization.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The record definition for the TQF was modified to define new fields needed for processing. While the record's length did not change, the length of some of existing fields were decreased in order to allow for the new fields. An optional file conversion program was developed to convert any existing records in the TQF to the new record format. The conversion program is needed only if the TQF contains records for transactions that need to be preserved.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

User Impacts

Users may optionally configure in the LNCF where token data will be retrieved on reversals. Users who have sufficient token buffer spaces in the TQF will not need to configure any additional LNCF assigns or params, as the new assign and params are all optional. Users who do not have sufficient token buffer space in the TQF, or those who wish to do so, will need to configure in the LNCF where they want the token information to be retrieved from the TQFT/TQF, the PTLF, or to not use token information on a reversal.

Users must be aware that if they configure their network to retrieve token data from the PTLF on reversals, any token data not logged to the PTLF will not exist on a reversal. This could adversely affect reversal processing.

Users who are configured to use the TQFT may have an additional I/O per transaction in order to write token information on each transaction response and an additional I/O to retrieve token information on a reversal if one is necessary.

Users who are configured to retrieve token data from the PTLF on reversals will have an additional I/O per reversal transaction in order to retrieve the token information from the PTLF.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Discretionary Data Suppression

This section describes the changes, conversion requirements, and implementation requirements associated with the Discretionary Data Suppression enhancement.

These changes, conversion requirements, and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Discretionary Data Suppression enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-ACQ-AVS-DATA-SUPPRS

This param indicates whether address verification data in acquired transactions is logged to the PTLF. Valid values are Y (suppress address verification data) and N (do not suppress) with a default of N.

POS-ACQ-CVD2-DATA-SUPPRS

This param indicates whether card verification data in acquired transactions is logged to the PTLF. Valid values are Y (suppress card verification data) and N (do not suppress) with a default of N.

POS-ACQ-DISCR-DATA-SUPPRS

This param indicates whether discretionary data from Track 1 or Track 2 in acquired transactions is logged to the PTLF. All data past the expiration date in the track is considered discretionary data. Valid values are Y (suppress discretionary data) and N (do not suppress) with a default of N.

Database Configuration

Address verification data in the PSTM may also be carried in the IAVS-DATA-TKN (tkn-id = "CF"). This data may be suppressed by using the Token screen (TKN) to create a token record with the "TRAN LOG" field of the POS IAVS DATA token set to "N".

Users may also choose to suppress track 1 data by using the Token screen to create a token record with the "TRAN LOG" field of the TRACK1 token set to "N", or by using the LCONF param provided by this enhancement to truncate the discretionary data in the token.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Customers who use discretionary data suppression will no longer have discretionary data available on the PTLF for some transactions. Any host or other network entity that may use data from the PTLF should be analyzed for impacts.

User/Operational Impacts

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Enhanced Pre-Authorization Support

This section describes the changes, conversion requirements, and implementation requirements associated with the Enhanced Preauthorization Support enhancement.

These changes, conversion requirements, and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Enhanced Preauthorization Support enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

PRE-AUTH-HOLD-STORE-TIME

This param indicates the amount of time, in minutes, the terminal has to send a preauthorization hold completion transaction to the acquirer for enhanced preauthorization hold processing. This param is used by the BASE24-pos Device Handler/Router/Authorization process. Valid values are 0 through 32767. The default value is 120.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

Cardholder Authorization File (CAF)

Field Name: HOLD STATUS

Screen: CAF Screen 6

DDL Name: CAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.PR-TIMESTAMP

The status of each enhanced preauthorization hold in this CAF record. The transaction hold status is cleared when the hold expires, when a completion comes in on the hold account, or when the hold is canceled by a CRT operator. A file refresh can clear a hold or change the time it is to expire. Valid values are as follows:

EXPIRED Enhanced Preauthorization hold is no longer considered.

ON HOLD Enhanced Preauthorization hold is still in effect.

The length of a hold depends on the transaction originator. If the transaction originates at a BASE24-pos terminal, the hold time length can be specified by the terminal or by the PRE-AUTH HOLD TIME field on BASE24-pos Terminal Data files (PTD) screen 3. If the transaction originates from an ISO host, the hold time length is included in the message. If the transaction originates at an interchange, the hold time length can be specified in the PRE-AUTH HOLD TIME field on screen 11 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

Field Name: APPRV CODE

Screen: CAF Screen 6

DDL Name: CAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.APPRV-CDE

The approval code for the transaction. This value is used to associate a preauthorized purchase completion transaction with the proper preauthorized purchase transaction.

Field Name: ACCT TYPE

Screen: CAF Screen 6

DDL Name: CAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.ACCT-TYP

The type of account that has funds on hold. It is used to uniquely associate the preauthorized purchase completion transaction with the proper preauthorized purchase transaction. The account type listed in this field should match the value in the TYPE field on CAF screen 4 for the same account. Not all values are valid for all products. However, the values recognized by all BASE24 products are as follows:

01–09	Checking accounts
11–19	Savings accounts
31–39	Credit accounts

Field Name: ACCOUNT NUMBER
Screen: CAF Screen 6
DDL Name: CAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.ACCT

The account number associated with the hold.

Field Name: SEQUENCE NUMBER
Screen: CAF Screen 6
DDL Name: CAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.SEQ-NUM

The sequence number of the transaction. This value is used with the terminal ID to match the completion to the hold if a match isn't found using the approval code.

Field Name: HOLD AMOUNT
Screen: CAF Screen 6
DDL Name: CAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.HOLD-AMT

The transaction amount that is associated with this hold. Transaction amounts can be entered at the POS terminal. However, if the transaction amount is not provided, BASE24-pos device handler processes obtain the transaction amount from in the DEFAULT PRE-AUTH AMOUNT field on BASE24-pos Terminal Data files (PTD) screen 3. If the transaction originates at an interchange and the transaction amount is not provided, some Interchange Interface processes obtain the transaction amount from the DEFAULT PRE-AUTH AMOUNT field on screen 11 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

For additional information on the usage of these fields, refer to the **BASE24 Base Files Maintenance Manual**.

Institution Definition File (IDF)

A FIID Auth File Segment Indicator on screen 5 of the IDF has been added for the Enhanced Pre-Auth segment added to the CAF and UAF for this enhancement. The file segment indicator must be updated to a "Y" for any institution that processes enhanced preauthorization transactions with the CAF or UAF.

The HOLDS LVL field on screen 16 of the IDF has been enhanced to allow holds to be stored on the UAF for transactions authorized online to a host.

For additional information on the usage of this field, refer to the **BASE24 Base Files Maintenance Manual**.

Token File (TKN)

The following BASE24-pos token has been added for the Enhanced Pre-Authorization Support enhancement:

- The POS Merchant Token (Token ID "CI") contains the enhanced pre-authorization data needed for transaction processing.

If the BASE24-pos token for this enhancement is not to be logged to the PTLF or ILF, the appropriate token record must be configured in the TKN.

If the token for this enhancement is to be extracted with the PTLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate TKN records must be configured.

Usage Accumulation File

Field Name: HOLD STATUS

Screen: UAF Screen 6

DDL Name: UAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.PR-TIMESTAMP

The status of each enhanced preauthorization hold in this UAF record. The transaction hold status is cleared when the hold expires, when a completion comes in on the hold account, or when the hold is canceled by a CRT operator. A file refresh can clear a hold or change the time it is to expire. Valid values are as follows:

EXPIRED Enhanced Preauthorization hold is no longer considered.

ON HOLD Enhanced Preauthorization hold is still in effect.

The length of a hold depends on the transaction originator. If the transaction originates at a BASE24-pos terminal, the hold time length can be specified by the terminal or by the PRE-AUTH HOLD TIME field on BASE24-pos Terminal Data files (PTD) screen 3. If the transaction originates from an ISO host, the hold time length is included in the message. If the transaction originates at an interchange, the hold time length can be specified in the PRE-AUTH HOLD TIME field on screen 11 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

Field Name: APPROVAL CODE
Screen: UAF Screen 6
DDL Name: UAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.APPRV-CDE

The approval code for the transaction. This value is used to associate a preauthorized purchase completion transaction with the proper preauthorized purchase transaction.

Field Name: ACCOUNT TYPE
Screen: UAF Screen 6
DDL Name: UAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.ACCT-TYP

The type of account that has funds on hold. It is used to uniquely associate the preauthorized purchase completion transaction with the proper preauthorized purchase transaction. The values recognized by all BASE24 products are as follows:

01–09	Checking accounts
11–19	Savings accounts
31–39	Credit accounts

Field Name: SEQUENCE NUMBER
Screen: UAF Screen 6
DDL Name: UAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.SEQ-NUM

The sequence number of the transaction. This value is used with the terminal ID to match the completion to the hold if a match isn't found using the approval code.

Field Name: HOLD AMOUNT
Screen: UAF Screen 6
DDL Name: UAF.ENHNC-PREAUTH.ENHNC-PRE-AUTH.HOLD-AMT

The transaction amount that is associated with this hold. Transaction amounts can be entered at the POS terminal. However, if the transaction amount is not provided, BASE24-pos device handler processes obtain the transaction amount from in the DEFAULT PRE-AUTH AMOUNT field on BASE24-pos Terminal Data files (PTD) screen 3. If the transaction originates at an interchange and the transaction amount is not provided, some Interchange Interface processes obtain the transaction amount from the DEFAULT PRE-AUTH AMOUNT field on screen 11 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

For additional information on the usage of these fields, refer to the **BASE24 Base Files Maintenance Manual**.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

An entry was added to the SEG-TABLE and the SEG-DESCR-TABLE for the enhanced pre-auth segment.

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the new Enhanced Pre-Auth screens for the CAF and UAF.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new CAF and UAF screens must be added through the SEC screen for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

If the token (i.e., POS Merchant Token) used for this enhancement is to be included in the PTLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to the CAF and UAF records.

Refresh Process

Host code changes are required for all BASE24 users that utilize the release 6.0 CAF Refresh file message format to accept and process the new Enhanced Preauthorization segment information in the CAF file. Users utilizing the release 5.x CAF Refresh file message format will not be able to refresh this new segment.

User/Operational Impacts

Disk Impacts

The PTLF and ILF require additional disk space because the POS Merchant token can be included in the transaction records logged to these files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Secure Internet Validation

This section describes the changes, conversion requirements, and implementation requirements associated with the Secure Internet Validation enhancement.

These changes, conversion requirements, and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Secure Internet Validation enhancement.

ACI's E-Commerce products support the ability for a cardholder to register for either the Visa (3-D Secure) or MasterCard (SPA UCAF) Internet security products and to perform cardholder authentication during an online purchase. These are supported through cardholder registration and cardholder authentication during a purchase with an Access Control Server (ACS) that is compliant with Visa or MasterCard's rules.

The Secure Internet Validation (SIV) enhancement extends secure internet processing to BASE24-pos. BASE24-pos will participate in the initial authentication via a card verification transaction and in the actual purchase via a purchase, preauthorization request, or preauthorization completion transaction.

Note: Preauthorization processing for a lesser amount is not supported in the Secure Internet Validation enhancement.

If Virtual Primary Account Numbers (VPANs) are being used, the currency code and transaction amount in the card verification are based on the issuer's currency. In addition, the currency code and amount in the actual purchase are based on the issuer's currency. Thus, the BASE24-pos router/authorization process uses the currency code and transaction amount fields in the PSTM (i.e., PSTM.ORIG-CRNCY-CDE and PSTM.TRAN.AMT-1) for SIV processing.

If VPANs are not being used, the currency code and transaction amount in the card verification are based on the merchant's currency. In addition, the currency code and amount in the actual purchase are based on the merchant's currency. Thus, the BASE24-pos router/authorization process uses the currency code and transaction amount fields in the Original Currency 6.0 Token (i.e., ORIG-CRNCY-60-TKN.CRNCY-CDE and ORIG-CRNCY-60-TKN.AMT-1) for SIV processing. If the token is not available, the currency code and transaction amount fields in the PSTM are used.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the Assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-EAF

The fully qualified file name of the E-Commerce Authentication File. This is required for Secure Internet Validation customers using VPANs, validating CAVVs without a host security module, or validating AAVs.

POS-EAF-PREV

The fully qualified file name of the previous E-Commerce Authentication File. This is required for Secure Internet Validation customers using VPANs, validating CAVVs without a host security module, or validating AAVs.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-EAF-NOTIFY-INTERVAL

The number of writes made to the E-Commerce Authentication File (EAF) between calculating the percentage of file full. Each time a write is made to the EAF, a counter is increased. When the counter is equal to the value in this param, the Router module calculates how full the EAF is. The counter is then reset to zero. When the file reaches the percent full specified in the POS-EAF-NOTIFY-PERCENTAGE param, the Router module generates an event message indicating that the file is becoming full. Until more file space is made available, the Router module continues to generate event messages indicating that the file is becoming full at the interval specified in this param. Valid values are 5 to 32766, and the default is 5.

POS-EAF-NOTIFY-PERCENTAGE

The percentage of file full at which the Router module of the BASE24-pos Device Handler/Router/Authorization process generates an event message indicating that the E-Commerce Authentication File (EAF) is becoming full. The Router module calculates the percentage of file full and continues to calculate the percentage at the interval indicated in the POS-EAF-NOTIFY-INTERVAL param as described in the following paragraphs until more file space is made available. If the percentage has increased by one percent or more since the preceding calculation, the Router module

generates an event message indicating that the file is becoming full. Valid values are 10 through 99, and the default is 70.

POS-SETL-EAF-NOTIFY^{identifier}

The symbolic name of a BASE24-pos Device Handler/Router/Authorization process that the Main Settlement process must notify after the new EAF file has been created. The Main Settlement process uses this param. A maximum of 500 processes can be notified.

Each POS-SETL-EAF-NOTIFY param contains the name of one process. When multiple processes must be notified, this param can include an optional *identifier* to specify params used for different processes in the same logical network. The optional *identifier* is usually a number but can be any unique set of characters that has not already been declared by another POS-SETL-EAF-NOTIFY^{identifier} param. For example, the POS-SETL-EAF-NOTIFY01 param could contain the P1A^RTAU01 process and the POS-SETL-EAF-NOTIFY02 param could contain the P1A^SPDH01 process.

POS-SETL-EAF-RETENTION-PERIOD

The number of days the EAF and the previous EAF should be retained before the BASE24-pos settlement process cuts over the file.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

Card Prefix File (CPF)

Field Name: SIV CHECK TYPE
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-CHK-TYP

Indicates whether SIV card verification is to be attempted.

Field Name: SIV KEYA GROUP
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-KEYA-GRP

An identifier used to show the card verification group to which this record belongs on the Key Authorization file (KEYA).

Field Name: CHECK IF ONLINE SIV
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-CHK

Indicates whether a pre-screen SIV card verification check should be performed by the Router/Authorization processes before sending a transaction to the host.

Field Name: BAD SIV ACTION
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-BAD-DISP

The action that the Router/Authorization processes will take when a cardholder uses a card that contains invalid verification data.

Field Name: SIV NOT PRESENT ACTION
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-NOT-PRSN-DISP

The action that the Router/Authorization processes will take when a Cardholder Authentication Verification Value (CAVV) or Accountholder Authentication Value (AAV) was not present in the Secure Internet Validation transaction.

External Message File (EMF)

Data elements P-12 and P-13 must be included in the ISO message that is sent to the E-Commerce product on a BASE24-pos 0430 response. Thus, the EMF record used on outbound BASE24-pos 0430 messages must be updated or added to include data elements P-12 and P-13.

Host Configuration File (HCF)

When the E-Commerce product sends a reversal for BASE24-pos card verification transactions, E-Commerce requires a response. Thus, the ACK TO DPC field on screen 1 of the HCF must be sent to Y for the ISO Host Interface configured for the E-Commerce product.

Key Authorization file (KEYA)

KEYA records may be added for a SIV KEYA group, since the SIV KEYA group may be different than the CARD VERIFICATION KEYA group. In addition, BASE24 user may not have performed card verification, but the user is configuring SIV card verification.

BASE24-pos Transaction Log File (PTLF)

Field Name: VIRTUAL PAN
Screen: PTLF Screen 4
DDL Name: PSEUDO-CRD-NUM-TKN.PSEUDO-CRD-NUM

The Virtual PAN data from the Pseudo Card Number Token. This field is only displayed when the Pseudo Card Number Token is present and the transaction is purchase, preauthorization purchase, preauthorization completion, mail phone order, or card verification. Otherwise, this new screen field is not visible.

Field Name: EXP DAT
Screen: PTLF Screen 4
DDL Name: PSEUDO-CRD-NUM-TKN.EXP-DAT

The expiration date of the Virtual PAN from the Pseudo Card Number Token. This field is only displayed when the Pseudo Card Number Token is present and the transaction is purchase, preauthorization purchase, preauthorization completion, mail phone order, or card verification. Otherwise, this new screen field is not visible.

Token File (TKN)

The following BASE24-pos token has been added for the Secure Internet Validation enhancement:

- The Pseudo Card Number Token (Token ID “BL”) contains the Virtual PAN data needed for transaction processing.

If the BASE24-pos token for this enhancement is not to be logged to the PTLF or ILF, the appropriate token record must be configured in the TKN.

If the token for this enhancement is to be extracted with the PTLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate TKN records must be configured.

New Data Files

E-Commerce Authentication File (EAF)

The E-Commerce Authentication File (EAF) is used to store E-Commerce transactions for transaction validation. An EAF file (EAF) and a previous EAF file (EAFPREV) will exist for each logical network. These two files together must be kept for the maximum life of a transaction. Due to potential recurring payments, it is

recommended that the files must be kept for 300 days, and therefore the EAF file must be kept for at least 150 days as well as the EAFPREV file.

A new record will be written to the EAF file or another transaction entry will be added to an EAF record on an E-Commerce card verification transaction. A transaction entry in either the EAF and EAFPREV files will be updated on a purchase, preauthorization purchase, preauthorization purchase completion, or mail phone order transaction.

The EAF and EAFPREV records are referenced by the Virtual PAN (VPAN) and real PAN, so that a VPAN can be linked to a real PAN. There is not BASE24 AFT subsystem access to the EAF or EAFPREV files.

Template Changes

CUSTCNST

The constant CUST-EAF-REC-TXN-MAX-L from the CUSTCNST file on the PS60DDL subvolume needs to be modified if the Secure Internet Validation enhancement is being used. The value of this constant must be set according to the customer's configuration.

After the modification of this constant, the BASE24-pos DDLs and Secure Internet Validation enhancement code must be recompiled. The value in this constant dictates the maximum number of E-commerce authentication transactions outstanding that a customer's given logical network will contain in the EAF and EAFPREV files. It is imperative that a customer consider the number of transactions outstanding for one cardholder based on their VPAN or possibly their real PAN in the EAF and EAFREV files.

The use of this constant allows a customer to tailor the size of the EAF and EAFPREV files to its actual customer usage, thereby eliminating some possibly wasted space in the EAF and EAFPREV files. The value in this constant should never be set to more than 46, as that is the maximum number of E-commerce authentication transactions outstanding.

EAF

There is one new template for the EAF and EAFPREV files. The template must be created before starting BASE24-pos settlement. When settlement is started, the EAF and EAFPREV files are created using the new EAF template file.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

The BASE24-pos user must consider the new pre-screening checks defined in the POS segment of the CPF for this enhancement.

Network Configuration

An ISO Host Interface (Acquiring) process may be needed to be configured and maintained between the issuer's ACS or SPA Wallet Server, since the issuer ACS or SPA Wallet Server must send a card verification transaction into BASE24 using the ISO Host Interface process. This must be received and approved prior to allowing the cardholder to purchase goods such that the EAF can be populated prior to receiving the purchase transaction.

File Conversions

Card Prefix File (CPF)

If the BASE24-pos user wishes to enable SIV processing, the user must update the new fields in the POS segment of the CPF with appropriate values. Otherwise, these new fields in the POS segment of the CPF indicate that SIV processing is disabled.

Host/Co-Network Impacts

Super Extract Process

If the token (i.e., Pseudo Card Number Token) for this enhancement is to be included in the PTLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to the CPF records.

The PTLF record for the card verification portion of a SIV transaction will contain a non-zero transaction amount. Host code changes may be required for this situation.

ISO Host Interface Process

If the token for this enhancement (i.e., Pseudo Card Number Token) is included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Host code changes may be required for those BASE24-pos users that wish to specify an expiration date other than the YYMM format for a Virtual PAN during the card verification portion of a SIV transaction. Data element 126 of the ISO message is used to specify whether the SIV transaction expires in x minutes, y hours, or z days.

User/Operational Impacts

Disk Impacts

The new EAF and EAFPREV impacts disk requirements. The total impact varies for each BASE24 user depending on the number of EAF and EAFPREV records defined in the BASE24 system.

The KEYA may require additional disk space when new KEYA records are added for a SIV KEYA group. The same KEYA record may be used when the SIV KEYA group matches the CARD VERIFICATION KEYA group.

The PTLF and ILF require additional disk space as the new token for this enhancement (i.e., Pseudo Card Number Token) can be included in the transaction records logged to these files.

Operational Impacts

PTLF Perusal support the data defined in the new Pseudo Card Number Token. Operators should be trained to recognize the new information on the screen.

BASE24-pos response code 077 will be returned by the Router/Authorization processes when the cardholder already had the maximum number of E-Commerce authentication transactions outstanding. Additional SIV card verification transactions will continue to be denied until at least one of the authentication transactions expires or is removed from the EAF. Previously, this response code was only returned when the cardholder already had the maximum number of preauthorization transactions outstanding. The maximum number of preauthorizations outstanding is specified on CPF screen 7.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Performance Impacts

E-Commerce transactions that are written to the EAF require additional disk I/O.

For those BASE24-pos users that configure the CAVV validation with a host security module (i.e., HSM) an Inter-Process Communication (IPC) message is added for each call to the HSM.

User Impacts

BASE24 users utilizing the Secure Internet Validation enhancement must enter the appropriate information in the PS60DDL.CUSTCNST file.

Vendor Impacts

Not applicable

Reference Documents

- Electronic-Commerce Initiatives BASE24 and ACI Enterprise Payment System Support, December 31, 2001

- Europay “Maestro Implementation Guide for e-Commerce, Dual Message”, January 2002
- MasterCard International “SPA and UCAF Functional Specification”, Version, 1.0, April 3, 2001
- MasterCard International “SPA and UCAF Detailed Specification”, Version, 1.00, June 30, 2001
- Visa “3-D Secure Functional Requirements Access Control Server”, Version 1.0, August 3, 2001, (Ref. No. 70002-01)
- Visa “3-D Secure Functional Requirements Access Control Server”, Version 1.02, July 16, 2002, (Ref. No. 70002-01)
- Visa “3-D Secure Introduction”, Version 1.01, September 28, 2001 (Ref. No. 70001-01)
- Visa “3-D Secure Protocol Specification Core Functions”, Version 1.02, July 16, 2002, (Ref. No. 70000-01)
- Visa “April 2002 VisaNet Business Enhancements Technical Letter, (80007-01)
- Visa “April 2002 VisaNet Business Enhancements Technical Letter Update Bulletin #1, (Ref. No. 80007-02)
- Visa “October 2002 VisaNet Business Enhancements Technical Letter Update Bulletin #2, (Ref. No. 80011-03)
- Visa “Visa Payer Authentication Service Technical Requirements”, Version 1.14, January 2, 2001
- Visa “V.I.P. System SMS POS Technical Specifications (U.S.),” January 30, 1999

Transaction Security Services Multiple Logical Network Support

This section describes the changes associated with the Transaction Security Services Multiple Logical Network Support enhancement for BASE24-pos.

Section 2 contains a functional overview of the Transaction Security Services Multiple Logical Network Support enhancement.

For more detailed information on this enhancement see Section 4 (Base).

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement.

MULT-LN-PIN-KEY

This param points to the common PIN encryption key used to re-encrypt PINs on transactions passed between logical networks (i.e., LN) in a multiple logical network BASE24 system. It must be set at the logical network level (i.e., with a “used by” process name of all asterisks).

For LNs that use TSS, the param must contain a sixteen-digit hexadecimal key locator that points to a TSS ISECD (Interchange Security Data Source) record that in turn contains the actual encrypted PIN key.

For LNs that do not use TSS, the param must contain the actual sixteen-digit hexadecimal *encrypted* PIN key. The key has to be of single length because the BASE24 Security Utilities (SECUTILS) don't support longer keys for the non-TSS security device types.

This restriction means that, for BASE24 systems that contain one or more non-TSS LNs, the common key used by all LNs has to be of single length. If all LNs use TSS, however, either a single-length, double-length, or triple-length common key can be used.

Finally, if a Thales security device is in use, the common PIN key must be a Zone PIN Key (ZPK). A Terminal PIN Key (TPK) can't be used since no Thales PIN translation commands translate to TPKs.

Database Configuration

Interchange Security Data Source (ISECD)

A new TSS ISECD record must be added for each LN that uses TSS in a multiple logical network system. This record, which is pointed to by a corresponding MULT-LN-PIN-KEY LCONF param, should contain the common PIN key shared by all LNs.

If the system contains at least one LN that doesn't use TSS, this record should be configured with a single-length PIN encryption key. If all LNs use TSS, however, either a single-length, double-length, or triple-length key can be used.

If a Thales security device is in use, the common PIN key must be a Zone PIN Key (ZPK). A Terminal PIN Key (TPK) can't be used since no Thales PIN translation commands translate to TPKs.

When adding the ISECD common key record, only the following elements need to be entered. All other fields can be left empty or set to their defaults.

GUI Label: Interface Name
Element: INTF_NAM

Must contain the same value as the key locator specified in the MULT-LN-PIN-KEY LCONF param.

GUI Label: PIN Block Format
Element: PIN_BLK_TYP

Must be set to "ANSI PIN". The special PIN translation always re-encrypts the PIN as an ANSI PIN block.

GUI Label: Outbound Keys (Slot 1) Parts 1, 2, 3
Element: OUTBND_KPT1_KEYX

Must contain the common PIN encryption key. If an Atalla AKB security device is used, all three key parts must be filled. If a non-AKB Atalla security device is used, only the first part should be filled. If a Thales device is used, one, two, or three parts can be filled.

GUI Label: Outbound Current Index
Element: OUTBND_KPT1_IDX

Must contain a value of 1.

GUI Label: Inbound Keys (Slot 1) Parts 1, 2, 3
Element: INBND_KPT1_KEYX

Must contain the common PIN encryption key. If an Atalla AKB security device is used, all three key parts must be filled. If a non-AKB Atalla security device is used, only the first part should be filled. If a Thales device is used, one, two, or three parts can be filled.

GUI Label: Inbound Current Index
Element: INBND_KPT1_IDX

Must contain a value of 1.

GUI Label: Outbound Keys (Slot 1) Header
Element: OUTBND_KPT1_HDR

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB header. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Outbound Keys (Slot 1) MAC
Element: OUTBND_KPT1_KAM

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB MAC value. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Outbound Key Length
Element: OUTBND_KPT_LGTH

Must contain "Single", "Double", or "Triple" depending on the size of the common key entered in OUTBND_KPT1_KEYX.

GUI Label: Inbound Keys (Slot 1) Header
Element: INBND_KPT1_HDR

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB header. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Inbound Keys (Slot 1) MAC
Element: INBND_KPT1_KAM

If an Atalla AKB security device is used, this field must be filled with the common PIN encryption key's AKB MAC value. If an AKB device is not used, this field is not displayed and is unused.

GUI Label: Inbound Key Length
Element: INBND_KPT_LGTH

Must contain "Single", "Double", or "Triple" depending on the size of the common key entered in INBND_KPT1_KEYX.

GUI Label: External Key Block Format
Element: EXTRN_KEY_BLK_FRMT

If a Thales security device is used, this field should be set to "Variant method". If an Atalla security device is used (AKB or otherwise), the field should be set to "Standard Atalla".

New Data Files

Not applicable.

Template Changes

Not applicable.

Device Configuration

Not applicable.

Mega Table (MEGATBL) Changes

Not applicable.

PI-Table Changes

Not applicable.

SEG-Table Changes

Not applicable.

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable.

CRT Access Security Changes

Not applicable.

Routing Changes

Not applicable.

Network Configuration

Not applicable.

File Conversions

Not applicable.

Host/Co-Network Impacts

Not applicable.

User/Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable.

Reference Documents

Not applicable.

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Section 7

BASE24-teller

Section 7 discusses the changes and enhancements made to BASE24-teller in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing BASE24-teller users must consider when upgrading from release 6.0 version 2.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-teller users, along with the associated conversion and implementation requirements that all BASE24-teller users must address.
- Optional enhancements describe the changes that affect only those BASE24-teller users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-teller users must address to implement the enhancements.

Section 7 must be used in conjunction with Section 4 to identify the changes that BASE24-teller users need to consider when upgrading to release 6.0 version 3.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-teller users must address in order to upgrade to release 6.0 version 3.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

The Extract Configuration File (ECF) has been modified to support the configuration of reading past the initially determined end-of-file of the BASE24-teller Transaction Log File (TTLF) or reading up to the initially determined end-of-file of the TTLF.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

User Impacts

BASE24-teller users that extract their TTLF files must set the ECF Screen 1 field READ PAST INITIAL EOF to a value of Y for the ECF record that is used on the last daily extract of a TTLF file.

The ECF conversion program sets the ECF base segment field READ-PAST-INITIAL-EOF to a value of N. The ECF record used for the last daily extract of the TTLF must be updated to have the value of Y for this field.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the BASE24-teller enhancements that can be enabled or disabled with release 6.0 version 3, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Multiple Currency Support
- Transaction Security Services

BASE24-teller users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements for future reference.

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Multiple Currency Support

The changes, conversion requirements, and implementation requirements associated with the Multiple Currency Support enhancement are described in Section 4.

BASE24-teller has not been enhanced to provide full multiple currency support. However, it has been modified to deny transactions that are received with a transaction currency code that does not match the currency code configured for the corresponding Positive Balance File (PBF) record(s).

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Transaction Security Services

This section describes the changes, conversion requirements, and implementation requirements associated with the Transaction Security Services enhancement for BASE24-teller.

Section 2 contains a functional overview of the Transaction Security Services enhancement.

For more detailed information on this enhancement see Section 4 (Base).

Standard BASE24-teller modules have been modified where required to support Transaction Security Services. The BASE24-teller Authorization processes have been modified to support triple DES encryption/decryption.

Note: Customers using Atalla Identikey PIN verification, Diebold PIN verification methods, or non-ANSI PIN blocks cannot implement Transaction Security Services at this time.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

TTDF

All key fields in the TTDF records must be updated to contain key locators. The key locators are used by Transaction Security Services to read CSECD records to retrieve the actual encryption/decryption keys used by the teller device.

Refer to the File Conversions subsection of this enhancement to see how the key fields in the TTDF can be updated programmatically, using the terminal ID as the default key locator for each TTDF record, and corresponding CSECD records created.

The TTDF requester/server have been modified to read the SECURE-DEV-TYP param in the LCONF. If the SECURE-DEV-TYP param indicates Transaction

Security Services, the key fields maintained in the TTDF files are automatically modified to contain the terminal ID when a new record is added. The terminal ID is the key locator for teller device records in the CSECD.

If the Transaction Security Services database is partitioned across multiple logical networks, a two-character hexadecimal partition identifier, set by the LCONF param LN-IND, is used as the first two characters of the key locator. For key locators in the terminal data files, the first fourteen characters of the terminal ID is appended to the partition identifier to create the key locator.

CSECD

A CSECD (Channel Security) record must exist for every set of encryption/decryption keys used by teller devices in the BASE24 network. Normally, every teller device uses unique encryption/decryption keys so there would be one CSECD record for every TTDF record.

Refer to the File Conversions subsection of this enhancement to see how CSECD records for each TTDF record can be created programmatically.

If a teller device is configured in the BASE24-teller Terminal Data File (TTDF) to send PINs in the clear to any interchanges on the BASE24 system, a Channel Security record (CSECD) must be manually added for the encrypted intermediate key with a PIN block type of PIN/PAD.

If an interchange interface is configured to support clear PINs, a Channel Security record must be manually added for the encrypted intermediate key with a PIN block type of ANSI.

New Data Files

A new data file is required that contains encryption/decryption keys referenced by key locators during transaction processing. This file can be created by running conversion programs or records can be added, deleted or modified manually using the ACI Desktop Graphical User Interface (GUI).

- CSECD Channel Security

The following table shows the relationship of existing BASE24-teller database files to the corresponding Transaction Security Services database files. The first column lists the existing BASE24 files and the second column lists the files accessed by the Transaction Security Services process to get the real encryption/decryption keys used during transaction processing.

BASE24 Database File	Transaction Security Services Database File
TTDF	CSECD

Template Changes

Not applicable

Device Configuration

Teller devices supporting Triple DES encryption must have new software/hardware configured to use the Triple DES encryption algorithm and double-length or triple-length keys.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the keys stored in the CSECD records is through the new ACI Desktop Graphical User Interface, which is set up separately from the BASE24 Pathway CRT access system. Refer to the ***BASE24 Classic 6.0 Installation Guide*** for further information.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

File conversions for the following BASE24-teller files are supplied with the Transaction Security Services enhancement. There are no structural changes in any files, but Transaction Security Services files can be created and data populated from BASE24-teller files. BASE24-teller files are updated with key locators to point BASE24-teller processes to the correct Transaction Security Services records. Transaction Security Services data can be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs.

- TTDF

BASE24-teller Terminal Data File (TTDF)

The TTDF conversion program can be run to create CSECD records while updating TTDF records with key locators that point to the appropriate CSECD records. Running the conversion program is not required. CSECD records can be entered manually using the ACI Desktop Graphical User Interface instead. Key fields in corresponding TTDF records would have to be updated manually as well.

The TTDF conversion program on the TR60UC04 subvolume can be used to create or update the CSECD. If modifications to the conversion program are necessary, the program can be modified by editing the TSSTTDFS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ttdfstss.t,name $ttdfm,nowait/  
tssttdfm -force -list -listreplace
```

Users must update the TSSTTDFR obey file on the TR60UC04 subvolume with the appropriate information. TSSTTDFR must then be obeyed to execute the conversion program.

Both a new and old CSECD file must exist before the conversion is run. The new CSECD file must be created using the FUP utility. The old CSECD file will be the output of a previous TTDF conversion for Transaction Security Systems or will be an empty file the first time a terminal data file conversion is run.

For more information on this file conversion, refer to the AAREAD05 on the BA60UC05 subvolume and TSSTTDFR files on the TR60UC04 subvolume.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- Host Security Module RG7000 Programmers Manual (1270A514 Issue 3)
- Banking Command Reference Manual (424645-002 May 2001)
- Atalla Key Block Banking Command Reference Manual (TP522901-001, October 2001)

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Section 8

BASE24 Kernel/IEP

Section 8 discusses the changes and enhancements made to BASE24 Kernel/IEP in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing BASE24 Kernel/IEP users must consider when upgrading from release 6.0 version 2.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Kernel/IEP users, along with the associated conversion and implementation requirements that all BASE24 Kernel/IEP users must address.
- Optional enhancements describe the changes that affect only those BASE24 Kernel/IEP users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 8 must be used in conjunction with Section 4 to identify the changes that BASE24 Kernel/IEP users need to consider when upgrading to release 6.0 version 3.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Kernel/IEP users must address in order to upgrade to release 6.0 version 3.

There are no general enhancements inherent to BASE24 Kernel/IEP.

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Optional Enhancements

This section identifies the BASE24 Kernel/IEP enhancements that can be enabled or disabled with release 6.0 version 3, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24 Kernel/IEP.

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Section 9

BASE24-telebanking

Section 9 discusses the changes and enhancements made to BASE24-telebanking in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing BASE24-telebanking users must consider when upgrading from release 6.0 version 2.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-telebanking users, along with the associated conversion and implementation requirements that all BASE24-telebanking users must address.
- Optional enhancements describe the changes that affect only those BASE24-telebanking users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-telebanking users must address to implement the enhancements.

Section 9 must be used in conjunction with Sections 4 and 8 to identify the changes that BASE24-telebanking users need to consider when upgrading to release 6.0 version 3. Enhanced Telebanking users must also use Section 10, BASE24-billpay, to identify all changes, as Enhanced Telebanking shares modules with BASE24-billpay.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-telebanking users must address in order to upgrade to release 6.0 version 3.

There are no general enhancements inherent to BASE24-telebanking.

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Optional Enhancements

This section identifies the BASE24-telebanking enhancements that can be enabled or disabled with release 6.0 version 3, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24-telebanking.

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Section 10

BASE24-billpay

Section 10 discusses the changes and enhancements made to BASE24-billpay in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing BASE24-billpay users must consider when upgrading from release 6.0 version 2.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-billpay users, along with the associated conversion and implementation requirements that all BASE24-billpay users must address.
- Optional enhancements describe the changes that affect only those BASE24-billpay users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-billpay users must address to implement the enhancements.

Section 10 must be used in conjunction with Sections 4, 8 and 9 to identify the changes that BASE24-billpay users need to consider when upgrading to release 6.0 version 3.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-billpay users must address in order to upgrade to release 6.0 version 3.

There are no general enhancements inherent to BASE24-billpay.

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Optional Enhancements

This section identifies the BASE24-billpay enhancements that can be enabled or disabled with release 6.0 version 3, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24-billpay.

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Section 11

BASE24-customer service

Section 11 discusses the changes and enhancements made to BASE24-customer service in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing BASE24-customer service users must consider when upgrading from release 6.0 version 2.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-customer service users, along with the associated conversion and implementation requirements that all BASE24-customer service users must address.
- Optional enhancements describe the changes that affect only those BASE24-customer service users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-customer service users must address to implement the enhancements.

Section 11 must be used in conjunction with Sections 4 and 8 to identify the changes that BASE24-customer service users need to consider when upgrading to release 6.0 version 3.

ACI Worldwide Inc.

General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-customer service users must address in order to upgrade to release 6.0 version 3.

There are no general enhancements inherent to BASE24-customer service.

ACI Worldwide Inc.

Optional Enhancements

This section identifies the BASE24-customer service enhancements that can be enabled or disabled with release 6.0 version 3, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24-customer service.

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Section 12

BASE24 Interchange Interfaces

Section 12 discusses the changes and enhancements made to BASE24 Interchange Interfaces in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing BASE24 Interchange Interface users must consider when upgrading from release 6.0 version 2.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Interchange Interface users, along with the associated conversion and implementation requirements that all BASE24 Interchange Interface users must address.
- Optional enhancements describe the changes that affect only those BASE24 Interchange Interface users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 Interchange Interface users must address to implement the enhancements.

Section 12 must be used in conjunction with Section 4 to identify the changes that BASE24 Interchange Interface users need to consider when upgrading to release 6.0 version 3.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Interchange Interface users must address in order to upgrade to release 6.0 version 3.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here. Further details may be found in the update guides and manuals on the individual Interchange Interface subvolumes.

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General Changes

The VisaNet, Star ISO and MAC MASM Interchange Interfaces have been modified to support the Transaction Subtype Token (Token ID “BM”). This token contains information relating to transactions that are a subtype of another transaction. The token includes the transaction subtype, the issuer processing code and the acquirer processing code. The transaction subtype field from the token is used in conjunction with the SUBTPTBL file to fill the description of the token on the PTLF/TLF perusal screen.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following BASE24 Base token has been added for this enhancement:

- The Transaction Subtype Token (Token ID “BM”) has been added and is used by the VisaNet, STAR ISO and MAC MASM Interchange Interfaces. This token contains information relating to transactions that are a subtype of another transaction. The token includes the transaction subtype, the issuer processing code and the acquirer processing code. The transaction subtype field from the token is used in conjunction with the SUBTPTBL file to fill the description of the token on the PTLF/TLF perusal screen.

If the token for this enhancement is not to be logged to the PTLF, TLF or ILF, the appropriate token records must be configured.

If the token for this enhancement is to be extracted with the PTLF, TLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interchange Interface, ISO Host Interface, and Super Extract processes.

BIC ISO Interchange Interface Process

If the new Transaction Subtype Token is configured to be included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

ISO Host Interface Process

If the new Transaction Subtype Token is configured to be included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Super Extract Process

If the new Transaction Subtype Token is configured to be included in the TLF, PTLF, and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TLF, PTLF, and ILF require additional disk space if the new Transaction Subtype Token is configured to be included in the transaction records logged to these files. The total impact varies for each BASE24 user depending on the transaction volume in the BASE24 system.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF, PTLF, and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens may cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the BASE24 Interchange Interface enhancements that can be enabled or disabled with release 6.0 version 3, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Discretionary Data Suppression
- Format 2 File Support
- Offline PIN Management Enhancement

BASE24 Interchange Interface users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements for future reference.

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Discretionary Data Suppression

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Discretionary Data Suppression enhancement. The VisaNet, Banknet, MDS, EPS-Net, and BIC ISO Interfaces have been modified to support Discretionary Data Suppression.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Discretionary Data Suppression enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement.

SWI-ACQ-AVS-DATA-SUPPRS

This param indicates whether address verification data in acquired transactions is logged to the ILF. Valid values are Y (suppress address verification data) and N (do not suppress) with a default of N.

SWI-ACQ-CVD2-DATA-SUPPRS

This param indicates whether card verification data in acquired transactions is logged to the ILF. Valid values are Y (suppress card verification data) and N (do not suppress) with a default of N.

SWI-ACQ-DISCR-DATA-SUPPRS

This param indicates whether discretionary data from Track 1 or Track 2 in acquired transactions is logged to the ILF. All data past the expiration date in the track is considered discretionary data. Valid values are Y (suppress discretionary data) and N (do not suppress) with a default of N.

Database Configuration

Address verification data in the PSTM may also be carried in the IAVS-DATA-TKN (tkn-id = "CF"). This data may be suppressed by using the Token screen (TKN) to create a token record with the "TRAN LOG" field of the POS IAVS DATA token set to "N".

Users may also choose to suppress track 1 data by using the Token screen to create a token record with the "TRAN LOG" field of the TRACK1 token set to "N", or by using the LCONF param provided by this enhancement to truncate the discretionary data in the token.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The BIC ISO Interface compares the terminal logical network from the internal message and the logical network in the PCT (from ICFE screen 1 Interchange Logical Net) to determine if a transaction is eligible for discretionary data suppression. Customers who have interfaces operating in a switch to switch environment should ensure that the Interchange Logical Net values from the ICFE are unique to allow proper identification of transactions eligible for discretionary data suppression.

File Conversions

Not applicable

Host/Co-Network Impacts

Customers who use discretionary data suppression may no longer have discretionary data available on the ILF for some transactions (depending on the configuration parameters for any given interface). Any host or other network entity that may use data from the ILF should be analyzed for impacts.

User/Operational Impacts

Not applicable.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Format 2 File Support

The Format 2 File Support enhancement has been implemented in the following BASE24 Interchange Interface:

- BIC ISO Interface

Section 2 of the **BASE24 Release 6.0 Implementation Guide (Version 2)** contains a functional overview of the Format 2 File Support enhancement.

For more detailed information on this enhancement, refer to Section 4 of the **BASE24 Release 6.0 Implementation Guide (Version 2)**.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

BASE24 users using Format 2 ILF files must start at cutover. Before cutover, rename the existing template ILF files (i.e., the primary file and all of its alternate key files), and create the Format 2 template files (i.e., the primary file and all of its alternate key files). At cutover, the ILF is created using the new Format 2 template files.

Create the Format 2 ILF template file in the following manner using the FUP utility where the existing ILF template file name is VAYYMMDD, the existing ILF template alternate key file name is VBYMMDD. The maximum values for the primary extent size and secondary extent size has changed with Format 2 files, and therefore the user can use the SET EXT command in the following creation of the ILF template

files (i.e., both the primary ILF template file and the alternate key ILF template file) using FUP.

```
-SET LIKE VAYYMMDD  
-RENAME VAYYMMDD,OPYYMMDD  
-RENAME VBYYMMDD,OAYYMMDD  
-SET FORMAT 2  
-SET REC 4048  
-CREATE VAYYMMDD
```

The Format 2 ILF template file in the create command must have the same value as specified in the LCONF for this interchange interface's ILF. When the interchange interface cuts over, the next day's ILF will be created using the Format 2 ILF template file. The previous ILF files can be format 1 files and other ILF files can be Format 2 files.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

For those BASE24 users implementing this enhancement and extracting the ILF to a Format 2 file, the host code must process the new block format for the extract Format 2 disk file. In addition, the host code may utilize the new file format indicator that is specified in the Extract File Header to decipher the file format of the extract disk file. The value 2 in this field denotes a Format 2 file. The fourth byte of the user field in the Extract File Header was used to store this information.

User/Operational Impacts

Disk Impacts

Any file converted to Format 2 with alternate key(s) requires more disk space for the alternate key file(s), as the key length for Format 2 files is increased. This is true when the primary file is Format 2, regardless of whether the alternate key file(s) is Format 2.

Since the maximum data area per block in Format 2 files is less than what is available in format 1 files, users may require more disk space for any file that is converted to a Format 2 file. Users should check the length of their ILF records (for those Interchange Interfaces modified to support Format 2 File Support) to see if fewer ILF records will fit in the new Format 2 maximum data area blocks. The Format 2 maximum data area is 4048 for entry sequenced files, like the ILF. If fewer ILF records will fit in the new Format 2 maximum data area per block, users will require more disk space for the same number of records.

The user should do the same check for entry sequenced files against the ILFX file, if a Format 2 version of this file is to be used when extracting to disk.

User Impacts

BASE24 users using Format 2 ILF files must modify their template files to be Format 2 files. Refer to the Template Changes subsection of this enhancement for further information about ILF Template files.

BASE24 users extracting their ILF files to Format 2 disk files must configure the appropriate information in the ECF.

BASE24 users using Format 2 ILF files must start at cutover. Before cutover, rename the existing template (i.e., the primary file and all of its alternate key files), and create the Format 2 template files (i.e., the primary file and all of its alternate key files). At cutover, the new ILF is created using the new Format 2 template files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the following event message documentation files or the BASE24 Documentation CD-ROM:

- \$<volume>.SW60BABI.BABILOGM for the BIC ISO Interchange Interface process
- \$<volume>.BA60LOGM subvolume. This subvolume contains event message documentation files for BASE24 Classic modules, except interchange interface processes.

System Limitations

BASE24 users must calculate the length of the records that are written to the ILF. Format 2 Interchange log files allow a maximum logical record length of 4048 bytes. Depending on which, if any, tokens are written to the interchange log files, the record length of an interface log file record could exceed the limit the 4048 bytes.

Before you change ILFs to Format 2 files, ensure these records do not exceed the 4048 limit. BASE24 users utilizing format 1 are not affected, since the DDLs for the interchange log files were not changed to set the maximum record length to 4048.

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

Offline PIN Management Enhancement

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Offline PIN Management enhancement. The VisaNet, EPS-Net, LINK, and BIC ISO Interfaces have been modified to support the Offline PIN Management enhancement.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Offline PIN Management enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following new BASE24 Base token has been added for this enhancement:

- The EMV Issuer Script Results Token (Token ID “BJ”) has been added and is used by the VisaNet, and LINK Interchange Interfaces. The EMV Issuer Scripts Results Token contains information about the processing of EMV Script data. This token will be created by the acquirer process (e.g. device handler or interchange). It will contain information about the results of EMV script processing.

If the new token for this enhancement is not to be logged to the TLF or ILF, the appropriate token records must be configured.

If the new token for this enhancement is to be extracted with the TLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interchange Interface, ISO Host Interface, and Super Extract processes.

If a host does not wish to use the EMV Issuer Script Results token data, no modifications are necessary. Otherwise, hosts will have to create back-end programs to utilize the token data received from the Host Interface and Extract processes.

If a Host or BIC ISO Interface passes a transaction with the EMV Issuer Script Results token and fixed format is being utilized, 82 bytes of additional data will be passed. The customer may want to consider not passing this token out of HISO or BIC ISO links configured to use fixed formats.

The new PIN Unblock transactions are supported with an internal transaction code of "820000". PIN Unblock transactions are supported with a new ISO processing code "970000" in BASE24 external messages (used by HISO and BIC ISO). Hosts should be analyzed for any impacts that may result from receiving the new transaction codes.

BIC ISO Interchange Interface Process

If the new EMV Issuer Script Results Token is configured to be included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

ISO Host Interface Process

If the new EMV Issuer Script Results Token is configured to be included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Super Extract Process

If the new EMV Issuer Script Results Token is configured to be included in the TLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TLF and ILF require additional disk space if the new EMV Issuer Script Results Token is configured to be included in the transaction records logged to these files.

The total impact varies for each BASE24 user depending on the transaction volume in the BASE24 system.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Section 13

BASE24 Value-Added Products

Section 13 discusses the changes and enhancements made to the BASE24 value-added products in release 6.0 version 3, along with the resulting conversion and implementation requirements that existing users of these products must consider when upgrading from release 6.0 version 2.

The value-added products discussed in this section include the following:

- BASE24-InfoBase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Settlement Manager
- ACI Payments Manager
- ACI Smart Chip Manager

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 value-added product users, along with the associated conversion and implementation requirements that all BASE24 value-added product users must address.
- Optional enhancements describe the changes that affect only those BASE24 value-added product users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 value-added product users must address to implement the enhancements.

Section 13 must be used in conjunction with Sections 4 through 12 to identify the changes that BASE24 value-added product users need to consider when upgrading to release 6.0 version 3.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 value-added product users must address in order to upgrade to release 6.0 version 3.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the general enhancements described here.

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BASE24-InfoBase

This section describes the enhancements that impact the BASE24-InfoBase value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting BASE24-InfoBase users.

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BASE24 Retail Products

This section describes the enhancements that impact the value-added BASE24 Retail Products and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the users of the BASE24 Retail Products.

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BASE24-card

This section describes the enhancements that impact the BASE24-card value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting BASE24-card users.

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ACI Host Link

This section describes the enhancements that impact the ACI Host Link value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Host Link users.

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ACI Plastic Card Control System

This section describes the enhancements that impact the ACI Plastic Card Control System value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Plastic Card Control System users.

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ACI Claims Manager

This section describes the enhancements that impact the ACI Claims Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Claims Manager users.

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ACI Proactive Risk Manager

This section describes the enhancements that impact the ACI Proactive Risk Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Proactive Risk Manager users.

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ACI Settlement Manager

This section describes the enhancements that impact the ACI Settlement Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Settlement Manager users.

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ACI Payments Manager

This section describes the enhancements that impact the ACI Payments Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Payments Manager users.

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ACI Smart Chip Manager

This section describes the enhancements that impact the ACI Smart Chip Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Smart Chip Manager users.

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Optional Enhancements

This section identifies the BASE24 enhancements that can be enabled or disabled with release 6.0 version 3 that have an impact on the following BASE24 value-added products, along with the changes, conversion requirements, and implementation requirements associated with each.

- BASE24-InfoBase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Payments Manager
- ACI Settlement Manager
- ACI Smart Chip Manager

BASE24 value-added product users can choose whether or not to implement the enhancements described in this section. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 3 changes, conversion requirements, and implementation requirements for future reference.

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BASE24-InfoBase

This section describes the enhancements that impact the BASE24-InfoBase value-added product.

There are no optional enhancements inherent to BASE24-InfoBase.

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BASE24 Retail Products

This section describes the enhancements that impact the value-added BASE24 Retail Products.

There are no optional enhancements inherent to BASE24 Retail Products.

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BASE24-card

This section describes the enhancements that impact the BASE24-card value-added product.

There are no optional enhancements inherent to BASE24-card.

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ACI Host Link

This section describes the enhancements that impact the ACI Host Link value-added product.

There are no optional enhancements inherent to ACI Host Link.

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ACI Plastic Card Control System

This section describes the enhancements that impact the ACI Plastic Card Control System value-added product.

There are no optional enhancements inherent to ACI Plastic Card Control System.

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ACI Claims Manager

This section describes the enhancements that impact the ACI Claims Manager value-added product.

There are no optional enhancements inherent to ACI Claims Manager.

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ACI Proactive Risk Manager

This section describes the enhancements that impact the ACI Proactive Risk Manager value-added product.

There are no optional enhancements inherent to ACI Proactive Risk Manager.

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ACI Payments Manager

This section describes the enhancements that impact the ACI Payments Manager value-added product.

There are no optional enhancements inherent to ACI Payments Manager.

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ACI Settlement Manager

This section describes the enhancements that impact the ACI Settlement Manager value-added product.

There are no optional enhancements inherent to ACI Settlement Manager.

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ACI Smart Chip Manager

This section describes the enhancements that impact the ACI Smart Chip Manager value-added product.

There are no optional enhancements inherent to the ACI Smart Chip Manager.

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Appendix A

Logical Network Configuration File Changes

Appendix A contains the Logical Network Configuration File (LCONF) changes.

This information includes assigns and params that have been added, deleted or modified with release 6.0 version 3 along with their associated page numbers. Refer to the specified page numbers for explanations of how these assigns and params can or must be set.

Refer to the ***BASE24 Logical Network Configuration File (LCONF) Manual*** for instructions on adding them to the LCONF.

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ASSIGNS

The following assigns were added or modified for release 6.0 version 3. Refer to the specified page numbers for explanations of how these assigns can or must be set.

BASE24-pos General Changes:	
RCDF	6-8
RCDFEMT	6-8
 CRT Authorization TQF Token Buffer Space Expansion:	
CRT-AUTH-TQFT	6-31
 Secure Internet Validation:	
POS-EAF	6-48
POS-EAF-PREV	6-48
 Transaction Security Services Diebold and Identikey PIN Verification Support:	
KEYA	4-29

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PARAMS

The following params were added or modified for release 6.0 version 3. Refer to the specified page numbers for explanations of how these params can or must be set.

Automated Key Distribution System:

ATM-DH-AKDS-TSS-TIMER	5-17
ATM-DH-N50-AKDS-SERIAL-NUM-VRFY	5-17

BASE24-pos General Changes:

POS-CHRGBCK-NO-RESP-ORIG	6-8
POS-LOG-MAC-ERR.....	6-8
RCDFEMT-MAX-TBL-SIZE	6-9
RCDFEMT-WARMIdentifier	6-9
REVERSE-BAL-INQ	6-9
SAF-PRIOR-TO-SEND	6-9

CRT Authorization TQF Token Buffer Space Expansion:

CRT-AUTH-MAX-TQFTS.....	6-31
CRT-AUTH-TKN-RETRV-OPT	6-32

Discretionary Data Suppression:

POS-ACQ-AVS-DATA-SUPPRS	6-35
POS-ACQ-CVD2-DATA-SUPPRS	6-35
POS-ACQ-DISCR-DATA-SUPPRS	6-35
SWI-ACQ-AVS-DATA-SUPPRS	12-11
SWI-ACQ-CVD2-DATA-SUPPRS.....	12-11
SWI-ACQ-DISCR-DATA-SUPPRS	12-11

Dynamic Key Management:

ATM-DH-CONS-MAC-ERR-LMT	5-44
ATM-DH-DKM-KEY-CHNG-INTRVL.....	5-44
ATM-DH-DKM-MAC-ERR-LMT	5-44
ATM-DH-DKM-MAC-TXN-LMT	5-44
ATM-DH-DKM-PIN-ERR-LMT	5-44
ATM-DH-DKM-PIN-TXN-LMT	5-44

Enhanced Pre-Authorization Support:

PRE-AUTH-HOLD-STORE-TIME	6-39
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Offline PIN Management Support:

EMV-PIN-UNBLOCK-SCRIPT	4-14
	5-65

Secure Internet Validation:

POS-EAF-NOTIFY-INTERVAL	6-48
POS-EAF-NOTIFY-PERCENTAGE	6-48
POS-SETL-EAF-NOTIFYidentifier	6-49
POS-SETL-EAF-RETENTION-PERIOD	6-49

Transaction Security Services Multiple Logical Network Support:

MULT-LN-PIN-KEY	4-35
	5-71
	6-57

Appendix B

Host/CO-Network Impacts

Appendix B contains a list of release 6.0 version 3 enhancements that have host/co-network impacts, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

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Enhancements

The following enhancements have host/co-network impacts. Refer to the specified page numbers for more information about these enhancements.

ACI Standard POS Device Handler Data Encryption	6-22
ATD Infrastructure	5-13
BASE24 Interchange Interfaces General Changes	12-7
BASE24-pos General Changes	6-13
Diebold Merchant Banking Center	5-40
Discretionary Data Suppression	6-37
	12-13
Enhanced Pre-Authorization Support	6-45
Format 2 File Support	12-17
Interactive Response Notification Fee Processing and Negative Surcharge Support	5-49
NCR Bunch Note Acceptor / Check Processing Module Support	5-63
Offline PIN Management Support	4-18
	5-67
	12-21
Persistent UAF Support	4-26
Secure Internet Validation	6-54

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Appendix C

CUSTMACS File Changes

Appendix C contains a list of release 6.0 version 3 enhancements that have CUSTMACS changes, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

BASE24 users should review these changes to determine how the CUSTMACS file should be set for their BASE24 environment before running MAKE to compile their system. More information on all MAKE flags used within BASE24 can be found in the FLGSDOC file located on the BA60MAKE subvolume delivered with BASE24 Release 6.0 Version 3 code.

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Enhancements

The following enhancements have CUSTMACS file changes:

ACI Standard POS Device Handler Data Encryption	6-19
ACI Standard POS Device Handler Dynamic Key Management	6-25
Automated Key Distribution System	5-17
Diebold Merchant Banking Center	5-25
Dynamic Key Management.....	5-43
Multiple Currency Dispense	5-52

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Appendix D

Re-Sequenced Files and New Files

Appendix D consists of the following tables:

- A table illustrating the new files for the optional BASE24 Base Persistent UAF Support enhancement.
- A table illustrating the new files for the BASE24-atm General Changes.
- A table illustrating the new files for the optional BASE24-atm Automated Key Distribution System enhancement.
- A table illustrating the new files for the optional BASE24-atm Diebold Merchant Banking Center enhancement.
- A table illustrating the new files for the optional BASE24-atm Dynamic Key Management enhancement.
- A table illustrating the new files for the optional BASE24-atm Multiple Currency Dispense enhancement.
- A table illustrating the new files for the BASE24-pos General Changes.
- A table illustrating the new files for the optional BASE24-pos ACI Standard POS Device Handler Data Encryption enhancement.
- A table illustrating the new files for the optional BASE24-pos ACI Standard POS Device Handler Dynamic Key Management enhancement.
- A table illustrating the new files for the optional BASE24-pos CRT Authorization TQF Token Buffer Space Expansion enhancement.
- A table illustrating the new files for the optional BASE24-pos Secure Internet Validation enhancement.

No files were re-sequenced in release 6.0 version 3.

New Files for the Optional BASE24 Base Persistent UAF Support Enhancement
Table

Persistent UAF Support New File Location and File Name	File Language Type
ba60ext.uafo	Generated during a compile
ba60ext.uafoe	Generated during a compile
ba60ext.uafof	TAL
ba60ext.uafof	MAKE
ba60ext.uafof	TAL

New Files for the BASE24-atm General Changes Table

General Changes	
New File Location and File Name	File Language Type
at60n50.n50d	TAL
at601000.c1000d	TAL
at60trit.triton	TAL

New Files for the Optional BASE24-atm Automated Key Distribution System
Enhancement Table

Automated Key Distribution System New File Location and File Name	File Language Type
at60an50.atan50fm	MAKE
at60an50.atan50mm	MAKE
at60an50.n50akde	Generated during a compile
at60an50.n50akdg	TAL
at60an50.n50akdm	MAKE
at60an50.n50akdo	Generated during a compile
at60an50.n50akds	TAL

New Files for the Optional BASE24-atm Diebold Merchant Banking Center
Enhancement Table

Diebold Merchant Banking Center New File Location and File Name	File Language Type
at60x100.atx100fm	MAKE
at60x100.atx100mm	MAKE
at60x100.c100mbce	Generated during a compile
at60x100.c100mbcg	TAL
at60x100.c100mbcm	MAKE
at60x100.c100mbco	Generated during a compile
at60x100.c100mbcs	TAL
at60x100.x100nams	TAL

New Files for the Optional BASE24-atm Dynamic Key Management Enhancement Table

Dynamic Key Management New File Location and File Name	File Language Type
Diebold 10xx/478x Device Handler	
at60k100.atk100fm	MAKE
at60k100.atk100mm	MAKE
at60k100.c100dkme	Generated during a compile
at60k100.c100dkmg	TAL
at60k100.c100dkmm	MAKE
at60k100.c100dkmo	Generated during a compile
at60k100.c100dkms	TAL
NCR 5xxx Device Handler	
at60kn50.atkn50fm	MAKE
at60kn50.atkn50mm	MAKE
at60kn50.n50dkme	Generated during a compile
at60kn50.n50dkmg	TAL
at60kn50.n50dkmm	MAKE
at60kn50.n50dkmo	Generated during a compile
at60kn50n50dkms	TAL
Triton Device Handler	
at60ktrt.atkn50fm	MAKE
at60ktrt.atkn50mm	MAKE
at60ktrt.tritdkme	Generated during a compile
at60ktrt.tritdkmg	TAL
at60ktrt.tritdkmm	MAKE
at60ktrt.tritdkmo	Generated during a compile
at60ktrt.tritdkms	TAL

New Files for the Optional BASE24-atm Multiple Currency Dispense Enhancement Table

Multiple Currency Dispense New File Location and File Name	File Language Type
Diebold 10XX/478X Device Handler	
at60u100.atu100fm	MAKE
at60u100.atu100mm	MAKE
at60u100.c100mcue	Generated during a compile
at60u100.c100mcug	TAL
at60u100.c100mcum	MAKE
at60u100.c100mcuo	Generated during a compile
at60u100.c100mcus	TAL
DH Multiple Currency Utilities	
at60umcd.atmcdhl	Generated during a compile
at60umcd.atmcdhle	Generated during a compile
at60umcd.atmcdhlg	TAL
at60umcd.atmcdhlm	MAKE
at60umcd.atmcdhls	TAL
at60umcd.atumcdfm	MAKE
at60umcd.atumcdm	MAKE
at60umcd.atumcdmm	MAKE
ba60logm.atatmcdh	DOC
NCR 5XXX Device Handler	
at60un50.atun50fm	MAKE
at60un50.atun50mm	MAKE
at60un50.n50mcue	Generated during a compile
at60un50.n50mcug	TAL
at60un50.n50mcum	MAKE
at60un50.n50mcuo	Generated during a compile
at60un50.n50mcus	TAL

New Files for the BASE24-pos General Changes Table

General Changes	
New File Location and File Name	File Language Type
ps60aft.rqrcdfss	ACIAID
ps60aft.rqrcdfxm	MAKE
ps60aft.rqrcdfxs	COBOL
ps60aft.svrcdfg	TAL
ps60aft.svrcdfm	TAL
ps60aft.svrcdfts	MAKE
ps60emtb.pemtbfm	MAKE
ps60emtb.pemtbn	MAKE
ps60emtb.pemtbnm	MAKE
ps60emtb.pemtdm	MAKE
ps60emtb.pemtds	DDL
ps60emtb.rcdemttg	TAL
ps60emtb.rcdemttm	MAKE
ps60emtb.rcdemtts	TAL
ps60ddl.ddlfrcdf	DDL
ps60spdh.spdhd	TAL
ps60spdh.spdhnams	TAL

New Files for the Optional BASE24-pos ACI Standard POS Device Handler Data Encryption Enhancement Table

SPDH Data Encryption	
New File Location and File Name	File Language Type
ps60dspd.psdspdfm	MAKE
ps60dspd.psdspdm	MAKE
ps60dspd.psdspdmm	MAKE
ps60dspd.spdhdee	Generated during a compile
ps60dspd.spdhdeg	TAL
ps60dspd.spdhdem	MAKE
ps60dspd.spdhdeo	Generated during a compile
ps60dspd.spdhdes	TAL

New Files for the Optional BASE24-pos ACI Standard POS Device Handler Dynamic Key Management Enhancement Table

SPDH Dynamic Key Management New File Location and File Name	File Language Type
ps60kspd.pskspdfm	MAKE
ps60kspd.pskspdm	MAKE
ps60kspd.pskspdmm	MAKE
ps60kspd.spdhdkme	Generated during a compile
ps60kspd.spdhdkmg	TAL
ps60kspd.spdhdkmm	MAKE
ps60kspd.spdhdkmo	Generated during a compile
ps60kspd.spdhdkms	TAL
ps60uc04.cnvancf	Generated during a compile
ps60uc04.cnvacnfe	Generated during a compile
ps06uc04.cnvacnfm	MAKE
ps60uc04.cnvacnfo	Generated during a compile
ps60uc04.cnvacnfr	Run file
ps60uc04.cnvacnfs	TAL

New Files for the Optional BASE24-pos CRT Authorization TQF Token Buffer Space
Expansion Enhancement Table

CRT Authorization	
New File Location and File Name	File Language Type
bp60crta.ddlftqft	DDL
bp60crta.cnvtqf	Generated during a compile
bp60crta.cnvtqfe	Generated during a compile
bp60crta.cnvtqfm	MAKE
bp60crta.cnvtqfo	Generated during a compile
bp60crta.cnvtqfr	Run file
bp60crta.cnvtqfs	TAL

New Files for the Optional BASE24-pos Secure Internet Validation Enhancement Table

Secure Internet Validation New File Location and File Name	File Language Type
ps60ddl.custcnst	DDL
ps60ddl.ddlfeaf	DDL
ps60rtau.rtaud	TAL
ps60vrta.psvrtafm	MAKE
ps60vrta.psvrtamm	MAKE
ps60vrta.rtausiv	Generated during a compile
ps60vrta.rtausive	Generated during a compile
ps60vrta.rtausivg	TAL
ps60vrta.rtausivm	MAKE
ps60vrta.rtausivo	Generated during a compile
ps60vrta.rtausivs	TAL

Appendix E

Moved, Renamed, Deleted Files

Appendix E contains a list of files that were Moved, Renamed, or Deleted in BASE24 Release 6.0 Version 3.

File Location and File Name	Action Taken	Additional Action Taken or Note
BP60CRTA.DDLS	Deleted	No longer used
BP60ADMN.DDLS	Deleted	No longer used

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Appendix F

Regional Specific Information

Appendix F contains the information that is required to implement the changes for release 6.0 version 3 that are specific to each region.

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